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Newton-Raphson-based optimizer: A new population-based metaheuristic algorithm for continuous optimization problems

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ABSTRACT

The Newton-Raphson-Based Optimizer (NRBO), a new metaheuristic algorithm, is suggested and developed in this paper. The NRBO is inspired by Newton-Raphson's approach, and it explores the entire search process using two rules: the Newton-Raphson Search Rule (NRSR) and the Trap Avoidance Operator (TAO) and a few groups of matrices to explore the best results further. The NRSR uses a Newton-Raphson method to improve the exploration ability of NRBO and increase the convergence rate to reach improved search space positions. The TAO helps the NRBO to avoid the local optima trap. The performance of NRBO was assessed using 64 benchmark numerical functions, including 23 standard benchmarks, 29 CEC2017 constrained benchmarks, and 12 CEC2022 benchmarks. In addition, the NRBO was employed to optimize 12 CEC2020 real-world constrained engineering optimization problems. The proposed NRBO was compared to seven state-of-the-art optimization algorithms, and the findings showed that the NRBO produced promising results due to its features, such as high exploration and exploitation balance, high convergence rate, and effective avoidance of local optima capabilities. In addition, the NRBO also validated on challenging wireless communication problem called the internet of vehicle problem, and the NRBO was able to find the optimal path for data transmission. Also, the performance of NRBO in training the deep reinforcement learning agents is also studied by considering the mountain car problem. The obtained results also proved the NRBO's excellent performance in handling challenging real-world engineering problems.

1. Introduction

The past several years have seen a flurry of interest in real-world optimization problems that are difficult to solve in various fields, together with but not restricted to computer science-associated challenges and data analytics, as well as wireless communication and engineering design challenges (Abdel-Basset et al., 2018). Common features of such optimization problems include the presence of constraints, decision variables, objectives, and other similar elements. Today's modern research, particularly in engineering and manufacturing control problems, heavily relies on applying optimization algorithms (Halim et al., 2020; Sreedhar et al., 2020). Analytical or numerical programming and heuristics or metaheuristic techniques are two categories used to solve real-world engineering problems (Osaba et al., 2021). The initial classification of the Metaheuristic Algorithm (MA) separates it into two categories: single-solution and population-based algorithms. Single-Solution-Based (SSB) algorithms are more efficient than population-based algorithms. Depending on their complexity, numerous real-world problems could be characterized as single-objective, multi--objective, or multimodal optimization problems. Because of such complicated features, conventional mathematical methods such as gradient descent and conjugate gradient cannot handle these situations efficiently and effectively (Okwu and Tartibu, 2021; Kumar et al., 2022).

On the other hand, the MA of the latter categories is not gradient-based and, as a result of their randomized behaviour, has a high likelihood of avoiding local optima (Gong et al., 2020; Abualigah et al., 2021). The MAs provide many advantages over traditional techniques, including ease of implementation, reliability, and efficiency. Because of such characteristics, MAs can overcome the limitations of classic methodologies, such as early convergence and local optima stalling (Dréo et al., 2006). When handling optimization problems, because MAs are members of the randomized optimization class, they gain from using stochastic operators to prevent locally optimum solutions as much as possible (Oliva et al., 2021). In MAs, the two important search stages are exploration and exploitation. These two phases must be balanced to get

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Nomenclature

x_0 - Initial or starting point
x_{n+1} - Next approximation
$f'(x), f''(x_0), f'''(x_0)$ - First-order, second-order, and third-order derivatives
lb and ub - Lower and upper bounds
dim - Problem dimension
x_j - Decision vector/variable
N_p - Population size
X_n - Population matrix
X_b and X_w - Best and worst population position, respectively
δ - Adaptive coefficient for balancing exploration and exploitation
IT - Current iteration
Max_IT - Maximum number of iterations
ρ - Adaptive coefficient for improving exploitation
r_1 and r_2 - Random integers
r_3, a and b - Uniform random numbers between (0,1)
θ_1 and θ_2 - Uniform random numbers between (−1,1) and (−0.5,0.5)
β - Binary number, either 1 or 0
x_n^{IT} - Current vector position
$X_{r_1}^{IT}$ and $X_{r_2}^{IT}$ - Random position vectors
$X1_n^{IT}$ - New vector position by updating x_n^{IT}
$X2_n^{IT}$ - New vector position by considering X_b, $X_{r_1}^{IT}$ and $X_{r_2}^{IT}$
$X3_n^{IT}$ New vector position by considering $X1_n^{IT}$, $X2_n^{IT}$, and x_n^{IT}
div^{IT} - Average population diversity
div_{max} - Maximum population diversity
P_c - Probability of route link
P_r - Probability of packet reception
D - Route delay
$D_t(e_j)$ - Threshold value for the delay
T - Time variation of the graphical model
$G(i,s)$ - Network map

the best optimal solutions. Consequently, MAs are suitable for various situations in both research and industrial applications due to their specific qualities (Costa and Oliveira, 2003). A typical disadvantage of most MAs, on the other hand, is that they are often resistant to parameter tuning that the inventor chooses. On the other hand, the MAs may not necessarily lead to global optimum solutions (Okwu and Tartibu, 2021; Dréo et al., 2006). Aside from that, they are typically more computationally expensive than traditional algorithms because they use a population of solutions. The MAs can be divided into Swarm Intelligence (SI) and Evolutionary Algorithms (EA) (Hussain et al., 2018). Nature's evolutionary principles inspire the EAs. For instance, Biogeography-Based Optimization (BBO) (Simon, 2008), Differential Evolution (DE) (Kumar et al., 2022; Qin et al., 2009), Genetic Algorithms (GA) (Goldberg and Holland, 1988), Evolutionary Strategy (ES) (Rechenberg, 1989), etc., are a few algorithms in the EA category. However, their parameters must be ideally adjusted before starting the optimization procedure to prevent the local optima trap. As a result, numerous studies have been conducted to determine the optimal parameters for such EAs.

The SI algorithms are a form of artificial intelligence used to replicate the communal characteristics of social swarms in the environment. It is possible to achieve this by having agents follow local rules without centralized control. The aggregate capabilities of the swarm are significantly more extensive than the sum of individual swarm behaviours. The SI algorithms generate more than one randomized solution and refine those solutions over the optimization procedure. SI comprises Particle Swarm Optimization (PSO) (Eberhart and Kennedy, 1995; Shi and Eberhart, 1998) and Ant Colony Optimization (ACO) (Dorigo et al., 1996), which are probably the most common methods. For instance, the ACO simulates ants' social actions to identify the shortest distance between the food source and the nest. The PSO mimics the movement of a flock of birds and their foraging activity. Recently, numerous SI algorithms have been reported for solving various optimization problems. For instance, Artificial Bee Colony (ABC) (Karaboga and Basturk, 2007), Bat Algorithm (BA) (Bora et al., 2012), Cuckoo Search (CS) (Yang and Deb, 2010), Glow-worm swarm optimization (Li and Huang, 2016), Grey Wolf Optimizer (GWO) (Mirjalili et al., 2014; Premkumar et al., 2022), Dragon Fly Optimizer (DFO) (Khishe and Safari, 2019), Firefly algorithm (Johari et al., 2013), Ant Lion Optimizer (ALO) (Mirjalili, 2015a), Moth Flame Optimizer (MFO) (Mirjalili, 2015b), Emperor Penguin Optimizer (EPO) (Dhiman and Kumar, 2018), Elephant Herding Optimizer (EHO) (Jadoun et al., 2021), butterfly optimization (Arora and Singh, 2019), Salp Swarm Algorithm (SaSA) (Mirjalili et al., 2017), Whale Optimization Algorithm (WOA) (Mirjalili and Lewis, 2016), chicken swarm optimizer (Meng et al., 2014), Crow Search Algorithm (CrSA) (Zamani et al., 2019), elephant clan optimizer (Jafari et al., 2021), Mante-Ray Foraging Optimizer (MRFO) (Zhao et al., 2020), Red Deer Algorithm (RDA) (Fathollahi-Fard et al., 2020), Harris Hawks Optimizer (HHO) (Heidari et al., 2019), Spotted Hyena Optimizer (SHO) (Dhiman and Kumar, 2017), Slime Mould Optimizer (SMO) (Li et al., 2020), Squirrel Search Algorithm (SSA) (Jain et al., 2019), Marine Predator Algorithm (MPA) (Faramarzi et al., 2020a), tunicate swarm algorithm (Kaur et al., 2020), Hunger Games Search Optimizer (HGSO) (Manjula Devi et al., 2022), Sooty Tern Optimization Algorithm (STOA) (Dhiman and Kaur, 2019), Aquila Optimizer (AO) (Abualigah et al., 2021), Artificial Gorilla-Troop Optimizer (AGTO) (Abdollahzadeh et al., 2021a), African Vultures Optimization Algorithm (AVOA) (Abdollahzadeh et al., 2021b), COOT Algorithm (Naruei and Keynia, 2021), Remora Optimization Algorithm (ROA) (Jia et al., 2021), etc. are a few algorithms in SI category. The following are mostly preferred Population-Based (PB) optimization algorithms: Simulated Annealing (SA) (Kirkpatrick et al., 1983), harmony search (Geem et al., 2001), imperialist competitive algorithm (Shahrazad and Alikhanzadeh, 2015), gravitational search algorithm (Rashedi et al., 2009), Teaching–Learning-Based Optimizer (TLBO) (Crawford et al., 2020), Krill Herd (Gandomi and Alavi, 2012), Interior Search Algorithm (ISA) (Trivedi et al., 2016), Multi-Verse Optimizer (MVO) (Mirjalili et al., 2016), Inclined Planes system Optimization (IPO) (Mozaffari et al., 2016), Henry Gas Solubility Optimization (HGSO) (Hashim et al., 2019), Atom Search Optimization (ASO) (Irudayaraj et al., 2022), Political Optimizer (PO) (Askari et al., 2020), Equilibrium Optimizer (EO) (Faramarzi et al., 2020b), Gradient-Based Optimizer (GBO) (Ahmadianfar et al., 2020; Premkumar et al., 2021a), Runge-Kutta Optimization (RKO) (Ahmadianfar et al., 2021), Crystal Structure Algorithm (CSA) (Talatahari et al., 2021), etc. Further, the algorithms are classified as EAs, swarm-based, physics-inspired, human-inspired, and bio-inspired. Fig. 1 illustrates the broad classification of metaheuristic algorithms.

As seen in Fig. 1, most MAs created are based on population information. Algorithms preserve and enhance several candidate solutions by using various operators. On the other hand, an SSB applies a solution to discover the optimal solution. When a population of solutions is utilized to search for good solutions in a region, the likelihood of finding viable solutions among the candidate solutions is greater than one. However, compared to SSB algorithms, PB strategies often need more time, and, in addition, the majority do not have a decent or greedy intensification phase. Nevertheless, it is clear that the number of SSB approaches is far lower than the PB algorithms (Fathollahi-Fard et al., 2018).

Studies have extensively concentrated on PB algorithms during the

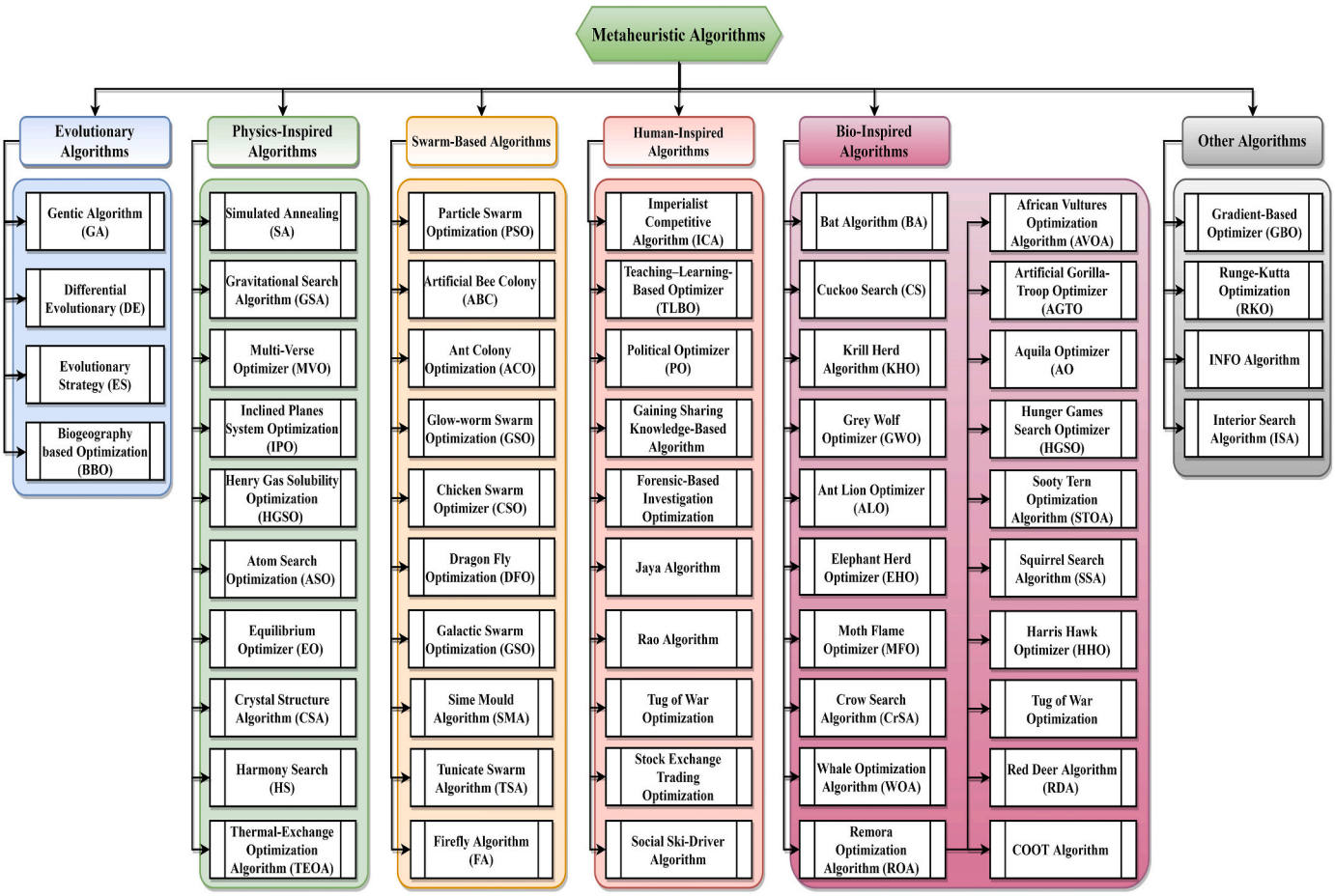


Fig. 1. Broad classification of metaheuristic algorithms.

last three decades, with only a few SSB algorithms. The SSB algorithms are less complicated than PB algorithms. Generally, the PB methods can be categorized into gradient-based (GB) methods and MAs. The gradient-based methods have been broadly employed to solve optimization problems. An extreme point at which the gradient is equal to zero must be identified to determine an optimal solution using the gradient-based methods. Exploring the search directions in these methods requires determining the derivatives of the objective function and the constraints. The two main disadvantages of this type of optimization are: (1) the convergence speed is very slow, and (2) there is no guarantee of achieving the optimal solution. Some initial points (i.e., population) are randomly generated in the second category. Each point has a search direction determined by the information acquired from previous results. The optimization process is continued by updating the search directions until the convergence criterion is met. Such optimization techniques (i.e., MAs) have been widely utilized to optimize engineering problems. The MAs provide great robustness to finding the global optima, while the gradient-based methods tend to converge into local optima. However, the non-gradient-based methods require higher computational capacities, especially for problems with high-dimensional search spaces. Hence, it will be very worthwhile to develop an optimization method that uses a gradient method to skip the unfeasible points and move towards the feasible area and also takes advantage of the capabilities of the population-based optimization methods.

Despite the variety in such algorithms, there seems to be one thing they all have in prevalent: MAs split the search procedure into two stages: exploitation and exploration. The exploratory phase occurs when an MA seeks to locate the best regions of a particular search space using a variety of search parameters. An MA should utilize its random operators to comprehensively investigate diverse portions of the search region. On the other hand, the exploitation stage enables the optimization to focus on the neighbourhood of the entire search space that contains higher-quality solutions rather than the entire search area (Premkumar et al., 2021b). The contradictory nature of exploitation and exploration necessitates using a robust and reliable optimizer to balance the exploitative and exploratory abilities. The local optima trap significantly increases if the balance between exploration and exploitation is not properly balanced (Li et al., 2021).

In addition to the MAs, Reinforcement Learning (RL) is also getting more attention to solve optimization problems. The RL is a machine learning paradigm where an agent learns to make decisions by interacting with an environment. The agent takes action in the environment and receives rewards based on the results, aiming to maximize the cumulative reward over time (Mazyavkina et al., 2021). Deep Reinforcement Learning (DRL) combines RL with Deep Learning (DL) techniques, using neural networks to approximate the policies or value functions. DRL can be applied to optimization problems where the decision-making process can be modelled as a sequence of decisions or actions (Dell'Aversana and Dell'Aversana, 2022; Dussauge et al., 2023). For instance, Deep Q-Networks (DQN) combines Q-learning with deep neural networks. It was one of the pioneering methods that showed the potential of combining RL with DL, initially used to play Atari games but later extended to optimization problems (Mnih et al., 2015). Some other applications are as follows: (i) Deep Deterministic Policy Gradients (DDPG) algorithm that adapts the concepts of DQN for continuous action spaces. It is valuable for optimization problems where actions are not discrete (Silver et al., 2014); (ii) Proximal Policy Optimization (PPO) tries to balance exploiting known good actions and exploring new ones. It is known for its stability in training, which can be crucial for many optimization tasks (Schulman et al., 2017); (iii) Actor-critic (AC)

methods combine the benefits of value-based and policy-based methods. An actor learns the policy, and a critic evaluates it (Mnih et al., 2016; Haarnoja et al., 2018). This combination can speed up the learning process for optimization problems. Silver et al. (2014) demonstrated using DRL for continuous control tasks with DDPG (Silver et al., 2014). Bello et al. (2016) used DRL to tackle the travelling salesman problem, showing that DRL methods could produce near-optimal solutions and sometimes outperform traditional heuristics (Bello et al., 2016). The authors of (Sánchez et al., 2022) have reviewed DRL for efficient resource allocation in computer systems. Lillicrap et al. (2015) implemented DDPG for various robotic control tasks, pushing the boundaries of what DRL could achieve in real-world settings (Lillicrap et al., 2015a). Schulman et al. (2017) introduced PPO, enhancing training stability, which is particularly crucial for challenging optimization tasks (Schulman et al., 2017). Andrychowicz et al. (2017) developed hindsight experience replay to help agents learn from failures, proving useful for certain optimization scenarios where failure experiences are abundant (Andrychowicz et al., 2017).

DRL algorithms often require a significant amount of data to achieve good performance. Balancing exploration and exploitation remains a challenge. Novel techniques like entropy regularization have been proposed to address this (Haarnoja et al., 2018). Training DRL agents for one optimization problem does not guarantee performance on another. Work on meta-learning and transfer learning, like that by (Finn et al., 2017), provides a promising avenue. The authors of (Grumbach et al., 2022a, 2022b) have compared the performance of DRL methods against traditional metaheuristic algorithms, finding mixed results dependent on problem specifics. Researchers have proposed hybrid models, integrating DRL with metaheuristics, to benefit from the strengths of both methods (Kosanoglu et al., 2022; Chen et al., 2020). While DRL has shown promise for various optimization problems, the field is still evolving. Challenges remain, especially regarding training stability, sample efficiency, and transferability. Future research may delve deeper into hybrid models, adaptive exploration techniques, and improved sample efficiency. Though the DRL methods are advantageous over the metaheuristic algorithms in solving some complex optimization problems, it has several drawbacks. (i) Training deep neural networks can be time-consuming and computationally expensive; (ii) DRL algorithms can sometimes be unstable during training, particularly if not correctly tuned; (iii) DRL usually requires a significant amount of data or interactions with the environment; however, the metaheuristic algorithms do not necessarily have this demand; (iv) Metaheuristic algorithms often have intuitive steps and can be easier to interpret than DRL models; (v) DRL models can overfit to their training environments, leading to poor performance on new, unseen instances of the optimization problem; (vi) Setting up and tuning DRL models usually demands a deeper understanding of both RL and DL, which might not be the case for some metaheuristic algorithms.

Deep reinforcement learning has shown significant potential in solving optimization problems, offering the ability to generalize across problem instances and handle large-scale and complex problems. However, its adoption comes with challenges, including long training times and potential stability issues. Traditional metaheuristic algorithms still find relevance due to their simplicity, transparency, and suitability for specific problems. The choice between DRL and metaheuristics depends on the optimization problem, available computational resources, and the practitioner's expertise.

On the other hand, even though most PB algorithms are preferred for optimization problems and effectively balance the exploration and exploitation stages, algorithms require several stages and numerous parameters to be tuned. Furthermore, according to the No-Free-Lunch (NFL) theory, none of the previously listed metaheuristics can solve all optimization problems (Wolpert and Macready, 1997). According to this theorem, a new algorithm that outperforms present metaheuristics on current or future problems is always possible as long as it is based on current or future optimization problems. The metaheuristic algorithms can not always promise to find the global optima for every optimization problem because of the random elements in the search mechanisms. A difficult and unsolved challenge with metaheuristic algorithms is finding a good balance between exploration and exploitation. Additionally, according to the NFL theory, no single optimization technique can handle all possible optimization problems. NFL also suggests that this field of research is quite dynamic, encouraging the years-long invention of numerous new metaheuristic optimization techniques. This work proposes a new metaheuristic algorithm with population-based features named "Newton-Raphson-Based Optimizer (NRBO)" to close the research gap.

The metaheuristic algorithms provide great robustness to find the global optima, while the gradient-based methods tend to converge into local optima. However, the non-gradient-based methods require higher computational capacities, especially for problems with high-dimensional search spaces. Hence, it will be very worthwhile to develop an optimization method that uses a gradient method to skip the unfeasible points and move towards the feasible area and also takes advantage of the capabilities of the population-based optimization methods. Thus, one of the unique features of this study is to combine the concept of the gradient-based method with the population-based method to create a powerful and efficient NRBO algorithm to overcome the drawbacks of gradient and non-gradient algorithms. As a result, the above-said limitations motivate our attempt to build a novel population-based metaheuristic algorithm called NRBO that is simple and efficient. The proposed NRBO is inspired by Newton-Raphson's approach, and it explores the entire search process using two rules: the Newton-Raphson Search Rule (NRSR) and the Trap Avoidance Operator (TAO) and a few groups of matrices to explore the best results further. The main contributions of the paper are listed as follows.

- A new population-based metaheuristic algorithm called NRBO inspired by the Newton-Raphson method is proposed and analyzed.
- Two operators called NRSR and TAO are proposed to improve the exploration and exploitation of NRBO.
- The performance of NRBO is validated using 23 classical benchmark functions, 29 CEC2017 benchmark functions, and 12 CEC2020 real-world engineering problems. In addition, 12 CEC2022 benchmarks are utilized and compared with CEC2022 winners.
- The performance comparison is made among seven newly reported algorithms using statistical data analysis, convergence analysis, runtime analysis, balance analysis, and Friedman's rank statistical test.
- The capability of NRBO is further investigated on one of the challenging wireless communication problems called the Internet of Vehicles (IoV) problem to enhance the data broadcasting over a reliable path.
- The performance of NRBO in training the DRL agents is also briefly studied by considering the mountain car problem.

The paper is structured as follows: Section 2 discusses the background study of the MAs and the basic concepts of the Newton-Raphson method. In addition, the inspiration and mathematical modelling of the proposed NRBO are comprehensively discussed, along with the formulation procedure of NRSR and TAO. Section 3 discusses the results obtained on classical benchmark problems, CEC2017, and CEC2022 problems and the performance comparison based on several metrics. In addition, the performance comparison between NRBO and the CEC2022 winners is also discussed. Section 4 presents the results obtained by the NRBO and other state-of-the-art algorithms on 12 CEC2020 real-world engineering problems. Section 5 comprehensively discusses the results of the NRBO and other algorithms on the IoV problem. Section 6 discusses the performance study of the DRL agents training in mountain car problems and the summary of the proposed work, limitations, and challenges, and Section 7 concludes the paper.

2. Background and methodology

This section comprehensively discusses the theoretical background and the concepts of Newton's method. In addition, the inspiration and the formulation procedures of the proposed population-based meta-heuristic algorithm called Newton-Raphson-Based Optimizer (NRBO) are explained comprehensively.

2.1. Background

Mostly, the optimization algorithms are classified into two categories: Gradient-Based (GB) algorithms, such as Newton's Method (NM) (Amrein and Wihler, 2014), Gradient-Descent Algorithm (GDA) (Madgwick et al., 2011), Levenberg Marquardt Algorithm (LMA) (Moré, 1978), and Quasi-Newton's Method (QNM) (Weerakoon and Fernando, 2000), and non-gradient-based algorithms called MAs, such as GA, DE, PSO, ABC, ACO, SA, JAYA algorithm, GWO, SSA, EO, MPA, GBO, RKO, HGSO, SMO, AGTO, etc., as discussed in earlier. The GB approaches have been widely used in solving optimization problems in different applications. Identifying an absolute extreme, where the gradient is equivalent to 0, is required to discover an optimum solution employing GB techniques. The NM and conjugate direction methods are GB methods based on this principle. The GB methods and most other optimization approaches require selecting a search path, after which the search proceeds in that direction in pursuit of the best solution. In order to investigate the search directions, it is necessary first to establish the derivative of the fitness function and the constraints. The two most significant drawbacks of GB methods are slow convergence speed and no assurance that the best solution will be obtained.

In MAs, the selected algorithm produces certain initial population positions randomly. With respect to each position, a search path is established by the intelligence gathered from earlier findings. The optimization is carried out until the maximum number of iterations or other convergence conditions by updating the search patterns. The application of MAs to various real-world engineering problems has become extremely prevalent. The MAs offer excellent robustness in searching for optimal solutions, whereas the GB approaches get trapped by local optimal solutions. On the other hand, the MAs necessitate more CPU core utilization, which is particularly important for problems with large search spaces. To avoid impracticable locations and advance towards the viable region, it would be extremely beneficial to build an algorithm that employs a GB and takes advantage of the availability of population-based algorithms. As a result, one of the unique aspects of this research is combining Newton's method with population-based methods to develop an advanced and effective algorithm that overcomes the shortcomings of traditional algorithms.

2.2. Newton-Raphson Method (NRM)

Newton's method, commonly known as the Newton-Raphson Method (NRM), is a root-finding procedure that uses the first few components of the Taylor Series (TS) of a function $f(x)$ in the region of a presumed root to locate the root of the function (Chapter 5 Newton, 2007; Bazaraa et al., 2006; Kalantari, 2000). For $f(x)$ a polynomial, NRM is fundamentally similar to Horner's method (He and J-S Shiue). The NRM initiates with a single point (x_0) and then employs the TS evaluated at x_0 for identifying one more point nearby the previous solution. This step is repeated until the appropriate solution has been found. The TS of $f(x)$ about the point ($x = x_0 + \epsilon$) is represented as follows.

$$f(x_0+\epsilon)=f(x_0)+f^{'}(x_0)\epsilon+\frac{f^{''}(x_0)(\epsilon)^2}{2!}+\frac{f^{'''}(x_0)(\epsilon)^3}{3!}+\cdots \tag{1}$$

Keeping the second-order terms,

$$f(x_0+\epsilon)\approx f^{'}(x_0)\epsilon+\frac{f^{''}(x_0)(\epsilon)^2}{2} \tag{2}$$

Equation (2) can be utilized to find the offset ϵ required to locate closer to the root originating from x_0. Assume $f(x_0+\epsilon)=0$ and Eq. (2) solved for $\epsilon \equiv \epsilon_0$ gives,

$$\epsilon_0=-\frac{f^{'}(x_0)}{f^{''}(x_0)} \tag{3}$$

Eq. (3) is the second-order tuning to the position of the root. By allowing $x_1 = x_0 + \epsilon_0$, determine the next position of the root, and the procedure can be repeated until it converges to the optimal root using Eq. (4).

$$\epsilon_n=-\frac{f^{'}(x_n)}{f^{''}(x_n)} \tag{4}$$

Regrettably, this process can be unbalanced near a local peak or a horizontal asymptote. Still, the method can be applied iteratively to find the next approximation with a proper initial position.

$$x_{n+1}=x_n-\frac{f^{'}(x_n)}{f^{''}(x_n)},\ n=1,2,3,\ldots \tag{5}$$

In NRM, an approximate zero is a starting point x_0 that ensures safe convergence of the algorithm.

2.3. Newton-raphson-based optimizer (NRBO)

The proposed NRBO defines the search path by applying the NRM to discover the search region using a few vector sets and two operators, such as NRSR and TAO to explore the search domain.

2.3.1. Initialization

Imagine the following scenario: The optimization is performed on an unconstrained single-objective optimization problem with the following definition.

$$\begin{aligned}&\text{Minimize}: f(x_1,x_2,\ldots,x_n)\\ &lb\le x_j\le \text{ub},\ j=1,2,\ldots,dim\end{aligned} \tag{6}$$

where $f(x)$ is the fitness function to be minimized, x_j denotes the decision vector, *dim* represents the problem dimension, *lb* denotes the lower bounds, and u *b* denotes the upper bounds. Like other MH algorithms, NRBO initiates its search for optimal solutions by producing initial random populations inside the boundaries of the candidate solutions. Based on the fact that there are N_p number of populations, and each population consists of *dim* decision variables/vectors. Therefore, the random population is generated using Eq. (7).

$$x_j^n=lb+rand\times(ub-lb),\ n=1,2,\ldots,N_p \text{ and } j=1,2,\ldots,dim \tag{7}$$

where x_j^i denotes the position of j th dimension of n th population, and *rand* denotes the random number between (0,1). Eq. (8) presents the population matrix that can portray the populations with all dimensions.

$$\boldsymbol{X_n}=\begin{bmatrix} x_1^1 & x_2^1 & \cdots & x_{dim}^1\\ x_1^2 & x_2^2 & \cdots & x_{dim}^2\\ \vdots & \vdots & \ddots & \vdots\\ x_1^{N_p} & x_2^{N_p} & \cdots & x_{dim}^{N_p}\end{bmatrix}_{N_p\times dim} \tag{8}$$

2.3.2. Newton-Raphson Search Rule (NRSR)

The vectors are controlled by the NRSR, allowing for a more accurate exploration of the viable region and obtaining better positions. The NRSR is based on the notion that the NRM is presented to boost the exploration tendency and speed up the convergence. Since numerous optimization techniques are not differentiable, a mathematical NRM is used to replace the explicit formulation of the function in such situations. The NRM begins with a presumed initial solution and proceeds to the next location in a definite direction. In order to get the NRSR from

Eq. (5), the second-order derivative must be determined using the TS. The TS for $f(x-\Delta x)$ and $f(x+\Delta x)$ is written as follows.

$$f(x+\Delta x)=f(x)+f^{'}(x_0)\Delta x+\frac{1}{2!}f^{''}(x_0)\Delta x^2+\frac{1}{3!}f^{'''}(x_0)\Delta x^3+\cdots \tag{9}$$

$$f(x-\Delta x)=f(x)-f^{'}(x_0)\Delta x+\frac{1}{2!}f^{''}(x_0)\Delta x^2-\frac{1}{3!}f^{'''}(x_0)\Delta x^3+\cdots \tag{10}$$

By subtracting/adding Eq. (9) and Eq. (10), the expressions for $f^{'}(x)$ and $f^{''}(x)$ are derived as follows:

$$f^{'}(x)=\frac{f(x+\Delta x)-f(x-\Delta x)}{2\Delta x} \tag{11}$$

$$f^{''}(x)=\frac{f(x+\Delta x)+f(x-\Delta x)-2\times f(x)}{\Delta x^2} \tag{12}$$

Substitute Eq. (11) and Eq. (12) in Eq. (5), and the updated root position is rewritten as follows.

$$x_{n+1}=x_n-\frac{(f(x_n+\Delta x)-f(x_n-\Delta x))\times\Delta x}{2\times(f(x_n+\Delta x)+f(x_n-\Delta x)-2\times f(x_n))} \tag{13}$$

Considering that the NRSR is supposed to be the primary component of the NRBO, it is necessary to make certain adjustments in order to manage the population-based search. As a result of Eq. (13), the positions of x adjacent to each other are denoted by $x_n+\Delta x$ and $x_n-\Delta x$, respectively, as shown in Fig. 2. This pair of neighbouring positions is transformed into two other vectors in the population by the NRBO. Because $f(x_n)$ is a minimization problem, as illustrated in Fig. 2, position $x_n+\Delta x$ has a poorer fitness value than position x, whereas the position $x_n-\Delta x$ has greater fitness than position x_n. So, the NRBO replaces the position $x_n-\Delta x$ for the position X_b, which has a better position in its neighbourhood than the position x_n, while position $x_n+\Delta x$ is replaced by the position of X_w, which has a worse position in its neighbourhood than the position x_n. A further advantage of the suggested approach is that it uses the position (x_n) rather than its fitness ($f(x_n)$), which saves time in the computation. Following that, the suggested NRSR is expressed as follows:

$$\text{NRSR}=randn\times\frac{(X_w-X_b)\times\Delta x}{2\times(X_w+X_b-2\times x_n)} \tag{14}$$

where *randn* denotes the normal-distributed random number with mean 0 and variance 1, X_w denotes the worst position and X_b denotes the best position. Eq. (14) can enhance the present solution by updating its location by assisting it with the current solution. Eq. (14) has a random parameter to increase the search capabilities of the NRBO and better balance exploitation and exploration abilities.

As a rule of thumb, the proposed algorithm must be capable of reaching a balance between diversification and intensification in order to discover optimal solutions in the search space and ultimately converge to the global solution. It is possible to enhance the algorithm by applying an adaptable coefficient called δ. The expression for δ is presented in Eq. (15).

$$\delta=\left(1-\left(\frac{2\times IT}{Max_IT}\right)\right)^5 \tag{15}$$

where *IT* denotes the current iteration, and *Max_IT* denotes the maximum number of iterations. In order to maintain a balance between the exploration and exploitation phases, the parameter δ adapts itself during the iterative process. Fig. 3 illustrates the variation of δ during each iteration. The value of δ varies from 1 to −1 as per Eq. (15).

Together with the adaptive parameter δ, the proposed NRSR improves the NRBO by considering random actions during the optimization procedure, increasing diversification, and avoiding local optima while significantly reducing the number of iterations. The expression for Δx in Eq. (14) is presented in Eq. (16).

$$\Delta x=rand(1,dim)\times\left|X_b-X_n^{IT}\right| \tag{16}$$

where X_b denotes the best solution obtained so far, and $rand(1:dim)$ is a random number with *dim* decision variables. Now, Eq. (13) is modified by considering NRSR and rewritten as follows.

$$x_{n+1}=x_n-NRSR \tag{17}$$

The exploitation of the proposed NRBO is improved by including another parameter called ρ, which directs the population in the right direction. The expression for ρ is given as follows.

$$\rho=a\times\left(X_b-X_n^{IT}\right)+b\times\left(X_{r_1}^{IT}-X_{r_2}^{IT}\right) \tag{18}$$

where a and b are random numbers between (0,1), r_1 and r_2 are different integers randomly selected from the population. However, the values of r_1 and r_2 are not equal. The current position of the vector (X_n^{IT}) has been updated by Eq. (19).

$$X1_n^{IT}=x_n^{IT}-\left(randn\times\frac{(X_w-X_b)\times\Delta x}{2\times(X_w+X_b-2\times X_n)}\right)+\left(a\times\left(X_b-X_n^{IT}\right)+b\times\left(X_{r_1}^{IT}-X_{r_2}^{IT}\right)\right) \tag{19}$$

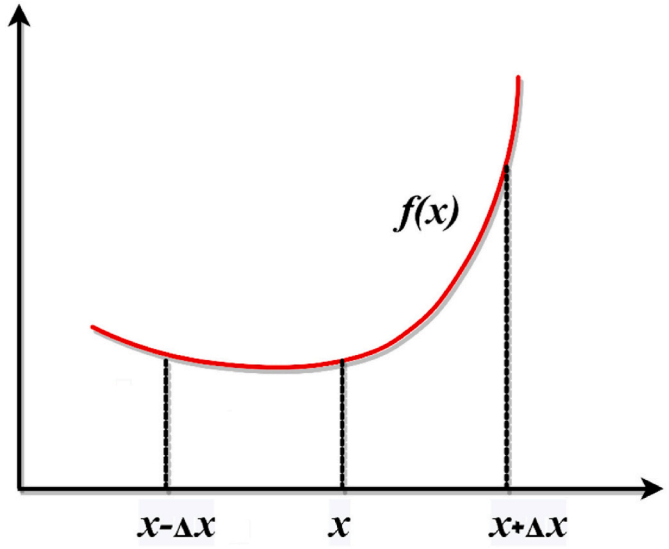


Fig. 2. Location of the position x.

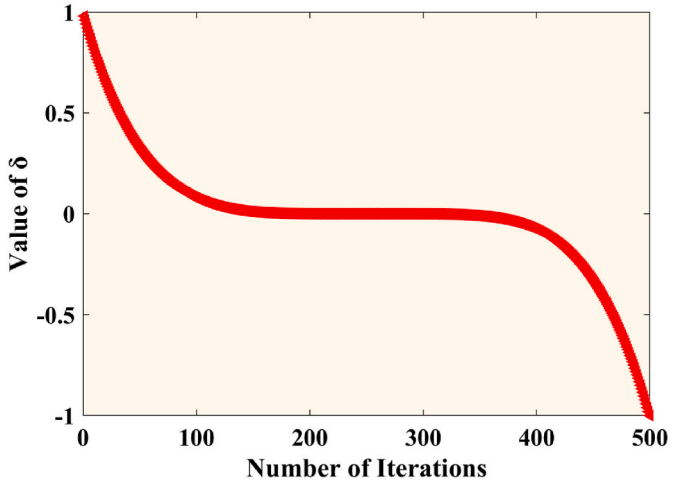


Fig. 3. Variation of δ during the iterative process.

where $X1_n^{IT}$ is a new vector position by updating x_n^{IT}. The NRSR is further improved by NRM proposed by (Weerakoon and Fernando, 2000; Magreñán and Argyros, 2018), and Eq. (13) is altered and rewritten as follows.

$$\mathrm{NRSR} = randn \times \frac{(y_w - y_b)\times\Delta x}{2 \times (y_w + y_b - 2\times x_n)} \tag{20}$$

$$y_w = r_1 \times (\mathrm{Mean}(Z_{n+1} + x_n) + r_1 \times \Delta x) \tag{21}$$

$$y_b = r_1 \times (\mathrm{Mean}(Z_{n+1} + x_n) - r_1 \times \Delta x) \tag{22}$$

$$Z_{n+1} = x_n - randn \times \frac{(X_w - X_b)\times\Delta x}{2 \times (X_w + X_b - 2\times x_n)} \tag{23}$$

where y_w and y_b are the location of two vectors generated using Z_{n+1} and x_n, and r_1 denotes the random number between (0,1). The enhanced version of NRSR is presented in Eq. (20). Using Eq. (20), Eq. (19) is updated as follows.

$$X1_n^{IT} = x_n^{IT} - \left(randn \times \frac{(y_w - y_b)\times\Delta x}{2 \times (y_w + y_b - 2\times x_n)}\right) + \left(a \times \left(X_b - {X_n}^{IT}\right) + b \times \left(X_{r_1}^{IT} - X_{r_2}^{IT}\right)\right) \tag{24}$$

It is necessary to construct the new vector $X2_n^{IT}$ by substituting the location of the best vector X_b with the position of the current vector x_n^{IT} in Eq. (24).

$$X2_n^{IT} = X_b - \left(randn \times \frac{(y_w - y_b)\times\Delta x}{2 \times (y_w + y_b - 2\times x_n)}\right) + \left(a \times \left(X_b - {X_n}^{IT}\right) + b \times \left(X_{r_1}^{IT} - X_{r_2}^{IT}\right)\right) \tag{25}$$

The exploitation phase is the primary focus of this search direction strategy. When it comes to local search, the search approach proposed by Eq. (25) is virtuous, but it has limitations when it comes to global search, whereas the search strategy presented by Eq. (24) is virtuous for global search but has limitations when it comes to local search. However, the NRBO uses both Eq. (24) and (25) to improve both the diversification and intensification phases. The new position vector during the next iteration is expressed in Eq. (26).

$$x_n^{IT+1} = r_2 \times \left(r_2 \times X1_n^{IT} + (1 - r_2) \times X2_n^{IT}\right) + (1 - r_2) \times X3_n^{IT} \tag{26}$$

$$X3_n^{IT} = X_n^{IT} - \delta \times \left(X2_n^{IT} - X1_n^{IT}\right) \tag{27}$$

where r_2 denotes the random number between (0,1).

2.3.3. Trap Avoidance Operator (TAO)

The TAO has been included to improve the effectiveness of the suggested NRBO for handling real-world problems. The TAO is a modified and enhanced operator adopted from (Ahmadianfar et al., 2020). The position of x_n^{IT+1} can be dramatically altered by using TAO. It produces a solution with enhanced quality X_{TAO}^{IT} by combining the best position X_b and the current vector position X_n^{IT}. The solution X_{TAO}^{IT} is produced if the value of *rand* is less than *DF* using Eq. (28).

$$\begin{cases} X_{TAO}^{IT} = X_n^{IT+1} + \theta_1 \times \left(\mu_1 \times x_b - \mu_2 \times X_n^{IT}\right) + \theta_2 \times \delta \times \left(\mu_1 \times \mathrm{Mean}\left(X^{IT}\right) - \mu_2 \times X_n^{IT}\right), \textbf{ if } \mu_1 < 0.5 \\ X_{TAO}^{IT} = x_b + \theta_1 \times \left(\mu_1 \times x_b - \mu_2 \times X_n^{IT}\right) + \theta_2 \times \delta \times \left(\mu_1 \times \mathrm{Mean}\left(X^{IT}\right) - \mu_2 \times X_n^{IT}\right), \text{ Otherwise} \end{cases} \tag{28a}$$

$$X_n^{IT+1} = X_{TAO}^{IT} \tag{28b}$$

where *rand* denotes the uniform random number between (0,1), θ_1 and θ_2 are uniform random numbers between (−1,1) and (−0.5,0.5), respectively, *DF* denotes the deciding factor that controls the NRBO performance, and μ_1 and μ_2 are random numbers, and it is generated using Eq. (29) and Eq. (30), respectively.

$$\mu_1 = \begin{cases} 3 \times rand, & \text{if}\Delta < 0.5 \\ 1, & \text{Otherwise} \end{cases} \tag{29}$$

$$\mu_2 = \begin{cases} rand, & \text{if}\Delta < 0.5 \\ 1, & \text{Otherwise} \end{cases} \tag{30}$$

where *rand* signifies a random number between (0,1) and Δ denotes a number between (0,1). Eq. (29) and Eq. (30) are further simplified as follows.

$$\mu_1 = \beta \times 3 \times rand + (1 - \beta) \tag{31}$$

$$\mu_2 = \beta \times rand + (1 - \beta) \tag{32}$$

where β denotes a binary number, either 1 or 0, and *rand* denotes the random number. If the value of Δ is greater than or equal to 0.5, then the value of β is 0; otherwise, the value is 1. Because of the randomness in the selection of the parameters μ_1 and μ_2, the population becomes more diverse and escapes from the local optimum solutions, which helps to improve its diversification. As similar to NRBO, the GBO is also inspired by Newton's method. Therefore, the NRBO concept may looks similar to GBO but still, the performance is much better than GBO due to the unique features of NRBO. The pseudocode of NRBO is presented in the ***Algorithm***. The flowchart of the proposed NRBO is shown in Fig. 4.

2.3.4. Computational complexity

The computational complexity of initialization is $O(N)$, the complexity of objective function evaluation and sorting is $O(N + NlogN)$, the complexity of updating control parameters is $O(N \times D)$, and the complexity of updating solutions is $O(N \times D)$. Accordingly, the computation complexity of NRBO is equal to $O(N\times(1 + it \times N \times (1 + logN + 2 \times N)$ very less compared to the original Newton-Raphson method that is equal to $O(N \times N \times N)$.

3. Results and discussions on numerical optimization problems

This section first covers the research approach used in this study, followed by a discussion of the findings. The research approach used in this research is quantitative, similar to that used in other metaheuristic studies. The proposed technique is tested on many test functions, i.e., 23 classical benchmark functions, 29 CEC2017 benchmark functions (Ahmadianfar et al., 2021), and 12 CEC2022 benchmark functions (Luo et al., 2022), which are discussed in detail in the following section. Due to the stochastic nature of the algorithm, the algorithm is executed 30 times individually, and the average results are considered to be a near-optimal solution. The metaheuristic algorithms provided near-optimal solutions during each algorithm run due to the random

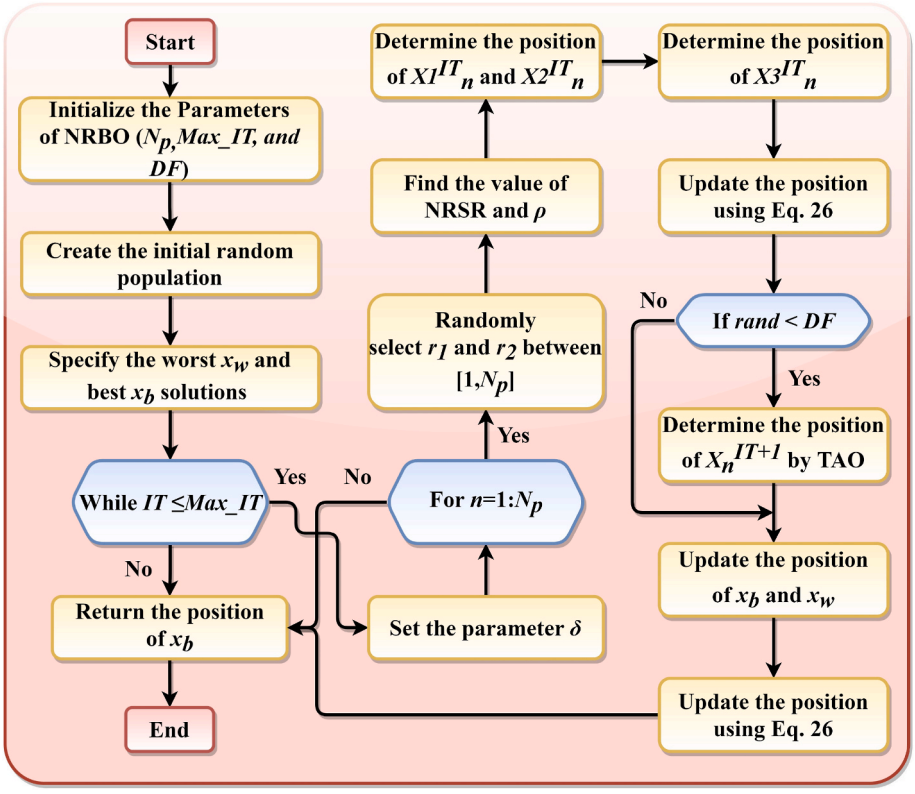


Fig. 4. Flowchart of NRBO algorithm.

numbers in the search mechanism. And this is true for all metaheuristic algorithms, including the proposed algorithm. Therefore, it is necessary to run the algorithm at least 30 times individually before its effectiveness is reported using qualitative evaluation metrics to get fair results, as discussed in the literature. The proposed NRBO is compared to seven well-known recent population-based algorithms, such as GBO, GWO, SCA, RKO, HHO, CSA, and EO, for results validation. The comparison is carried out utilizing statistical measurements (Min, Mean, and Standard Deviation (STD)) and Friedman's Ranking Test (FRT) to confirm the superiority of the proposed NRBO over other algorithms. Aside from that, the scalability analysis, sensitivity analysis, and convergence analysis of NRBO are also explored to determine their impact on its overall performance. To be clear, the datasets employed in this study are groups of benchmark test functions that have been widely used in research to analyze the performance of optimization algorithms.

3.1. Benchmark test functions

This sub-section aims to illustrate the effectiveness of the proposed NRBO in conjunction with the conceptual observations through several simulation experiments. An established and well-known set of benchmark test functions is selected based on a thorough literature study (Yao et al., 1999). This set includes 23 test functions with unimodal, multiple-dimensional, and fixed-dimensional multimodal characteristics and 29 CEC2017 benchmark test functions with separable, non-separable, and rotated characteristics, all of which have been widely employed to assess the effectiveness of NRBO and other selected algorithms. The classical benchmark test functions are listed in Supplementary Table S1. F1–F7 are unimodal functions (i.e., they have a single optimal result) that are purposely used to evaluate the algorithm's ability to exploit. The middle group encompasses multimodal functions (F8–F13) with more than one optimal outcome to be found. In such functions, local optimal results affect the exploration behaviour of the NRBO, whereas an algorithm must be able to locate the space globally and escape getting stuck in locally optimal solutions in order to locate the global optimum in such functions. F14–F23 are multimodal fixed dimensional functions similar to multimodal functions but instead have low and fixed dimensions. The benchmark functions of CEC2017 are listed in Supplementary Table S2. Like conventional functions, F24–F25 are unimodal functions, F26–F32 are simple multimodal problems, F33–F42 are hybrid functions, and F43–F52 are composite functions. The benchmark functions of CEC2022 are listed in Supplementary Table S3. Like CEC2017 functions, F53 are unimodal functions, F54–F57 are basic problems, F58–F60 are hybrid functions, and F61–F64 are composite functions. These functions have been constructed by combining the most intricate sorts of numerical optimization problems described in the previous work, which have been separable, non-separable, shifted, rotated, and enlarged in various ways.

3.2. Simulation setup

In order to examine the effectiveness and feasibility of the results obtained by the proposed NRBO, the authors have contrasted its results with various state-of-the-art algorithms from the literature, such as GBO, GWO, SCA, RKO, HHO, CSA, and EO. The parameters for such algorithms are listed in Table 1. The authors have utilized the default settings for the parameters set by the authors or developers of each selected algorithm. The proposed and all other selected algorithms are developed in MATLAB R2020b, installed on a machine running Windows 11 64-bit and having 16 GB of memory. Furthermore, the populations and iterations of all optimization algorithms are set to 30 and 750, respectively, for comparability.

***Algorithm*: Pseudocode of Newton-Raphson-Based Optimizer (NRBO)**

Initialization Phase

1. Select the proper values of population size (N_p), maximum number of iterations (Max_IT), and the deciding factor (DF)
2. Initialize the initial random population position
3. Assess the initial fitness value and the specify X_b and X_w

Main Loop

while ($IT < Max_IT$)

for n =1:N_p

for j=1:dim

Choose randomly $r_1 \neq r_2 \neq n$ between [1, N_p]

Determine the solution x_n^{IT+1} by Eq. 26

end ***for***

----------%%% **Trap Avoidance Operator (TAO)** %%%----------

if $rand < DF$

Find the solution X_{TAO}^{IT} by Eq. 28

$X_n^{IT+1} = X_{TAO}^{IT}$

else

Determine the solution x_n^{IT+1} by Eq. 26

end ***if***

Update X_b and X_w

end ***for***

IT=IT+1

end ***while***

Termination

Return the best solution X_b if termination criteria met

Table 1
Parameter settings of all algorithms.

Algorithm	Parameters	Value
NRBO (Proposed)	Deciding Factor (DF)	0.6
GBO (Ahmadianfar et al., 2020)	Probability rate (pr)	0.5
GWO (Mirjalili et al., 2014)	Constant a	[2 0]
SCA (Mirjalili, 2016)	Constant a	[2 0]
RKO (Ahmadianfar et al., 2021)	Constant a and b	20, 12
	Probability rate (pr)	0.5
HHO (Heidari et al., 2019)	Rabbit Energy	[2 0]
	β	1.5
CSA (Talatahari et al., 2021)	Parameter-free framework	
EO (Faramarzi et al., 2020b)	Constant a_1	2
	Constant a_2	1
	Generation probability (GP)	0.5

3.3. Results visualization

This sub-section reports and discusses how the findings of the qualitative data investigation of the NRBO when handling multimodal and unimodal functions compare to those obtained by other methods. During the optimization process, qualitative analysis is used to intuitively inspect location updates as well as fitness modifications in the populations. Fig. 5 shows four metrics: parameter space, the search history curve, the trajectory curve, and average fitness. In the trajectory curve, we can see how the first population's initial variable progresses over iterations. The mean objective/fitness value curves depict the changes in the population's mean fitness as a function of time.

There is evidence in the search history, as shown in Fig. 5, where the proposed NRBO demonstrates strong search space coverage while concentrating on possible locations. A review of the trajectory curve reveals that the solutions in NRBO have been subjected to dramatic shifts in the beginning phases, which then progressively diminish in the later stages. The NRBO eventually converges on a position and utilizes the feasible region. All trajectories illustrate the algorithm's core exploratory activity, characterized by abrupt movements. In addition, large values during initial iterations and small values in further iterations can ensure the fast convergence of NRBO and an effective quest near the global best and the transition from exploratory to exploitative structures (van den Bergh and Engelbrecht, 2006). Furthermore, this research can reveal the attitude of NRBO toward exploration. From the mean objective value curve, the variations in the NRBO fitness throughout the iterations have shown the positive influence of employing TAO and NRSR on fitness. Although the mean fitness of NRBO is distorted in certain situations, the curve displays a slow descent, which indicates that the overall performance of the population is strengthening over the period.

In addition to the case presented earlier, Fig. 6 provides the fitness function assessment and convergence behaviour of the NRBO on unimodal function. NRBO was initialized with 100 search agents for this investigation and ran for 750 iterations. Thereafter, the initial search agents and the search agents from various iteration levels are distributed to show how the search agents' convergences changed while NRBO was carried out. With the help of this graphic, one can see how the NRBO behaves when it comes to convergence and decide whether or not the global optimum is reached. As the number of iterations on the unimodal function of F1 increases, Fig. 6 shows that NRBO has a reasonable level of convergence speed and the capacity to search around the globally optimal solution point actively. A difficult problem that algorithms usually should address is the possibility that the algorithm could become stuck in local optima or stagnate as a result of the quicker rates of

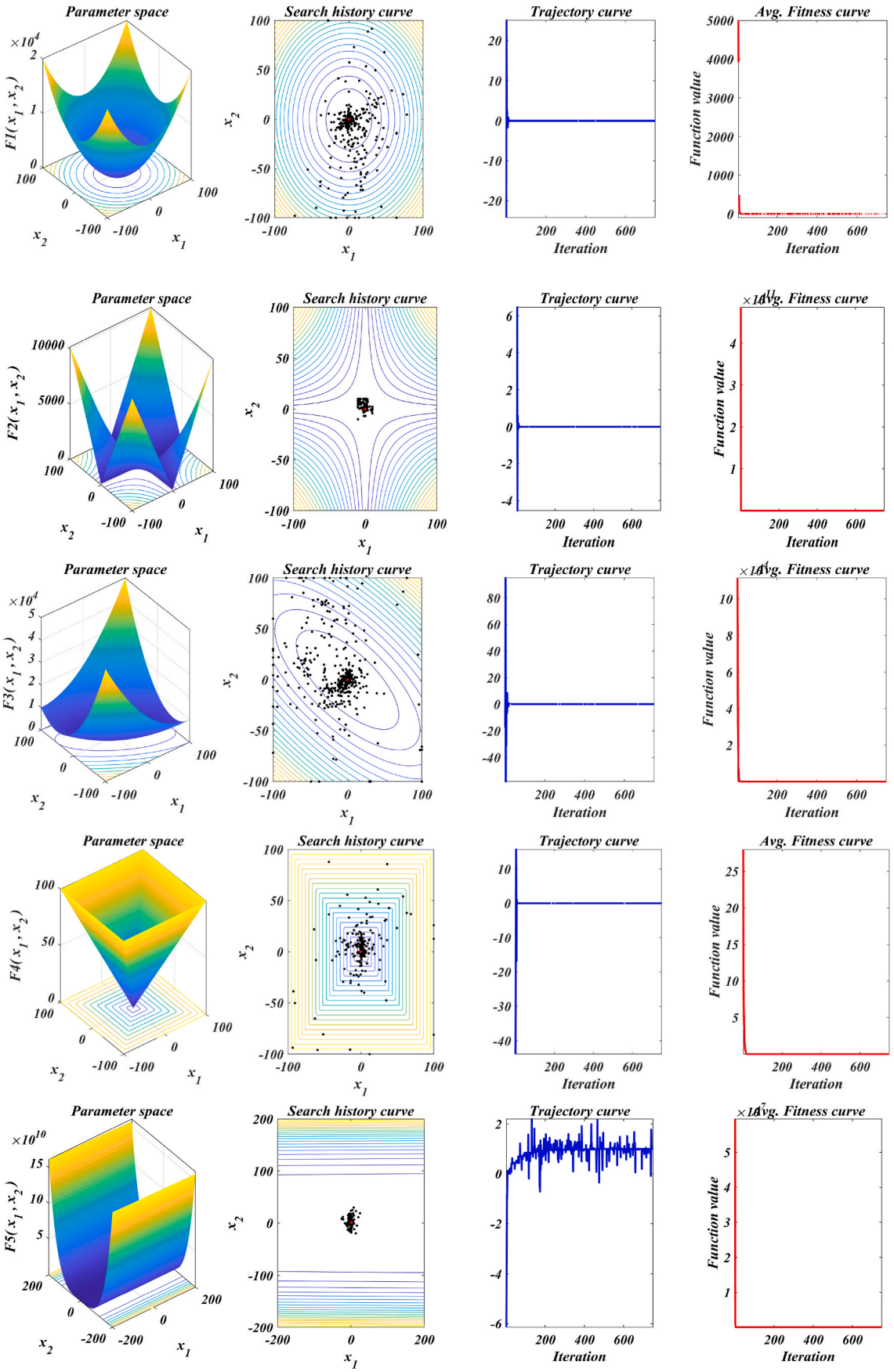


Fig. 5. Qualitative analysis of all 23 standard benchmark test functions.

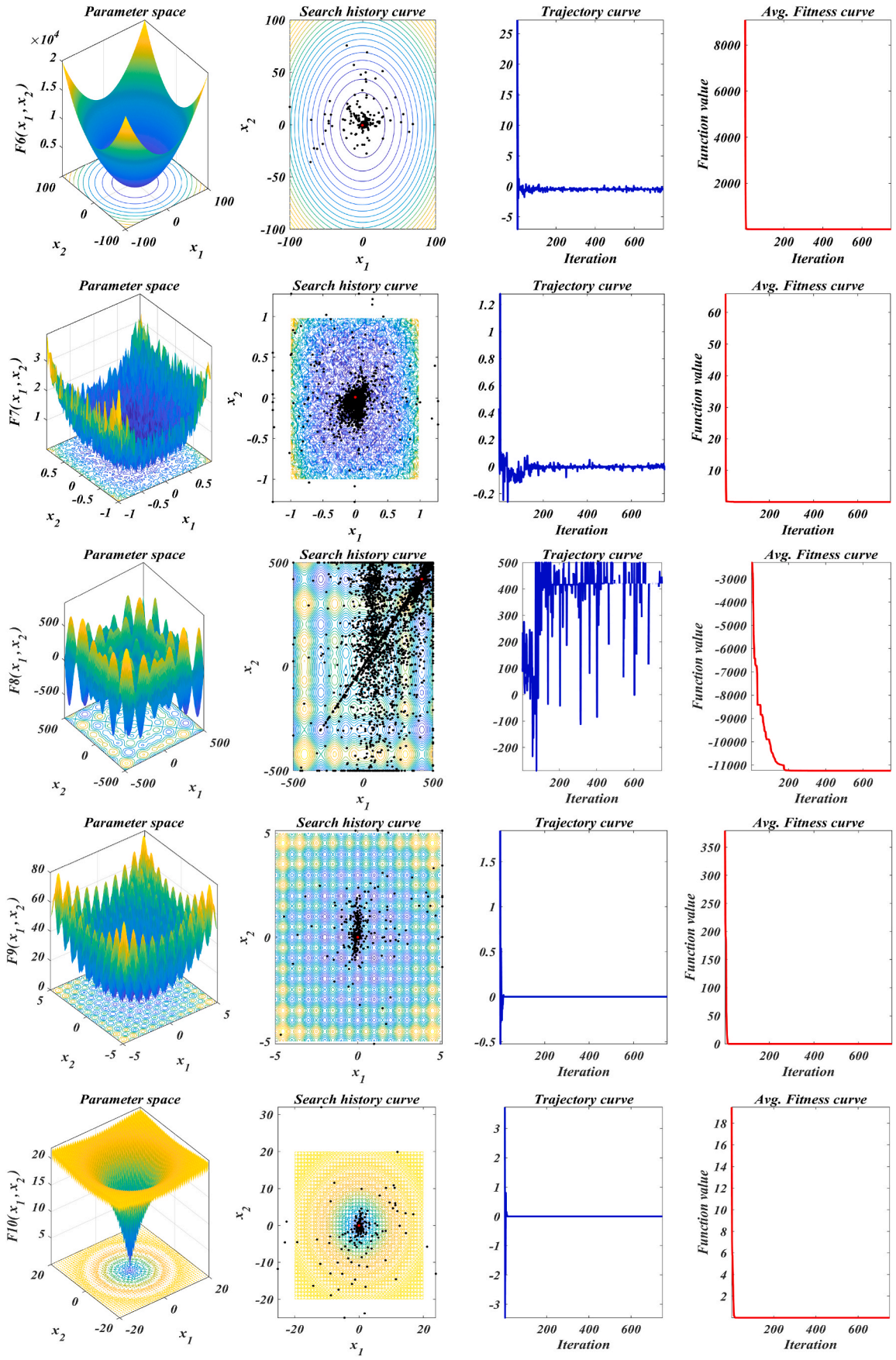
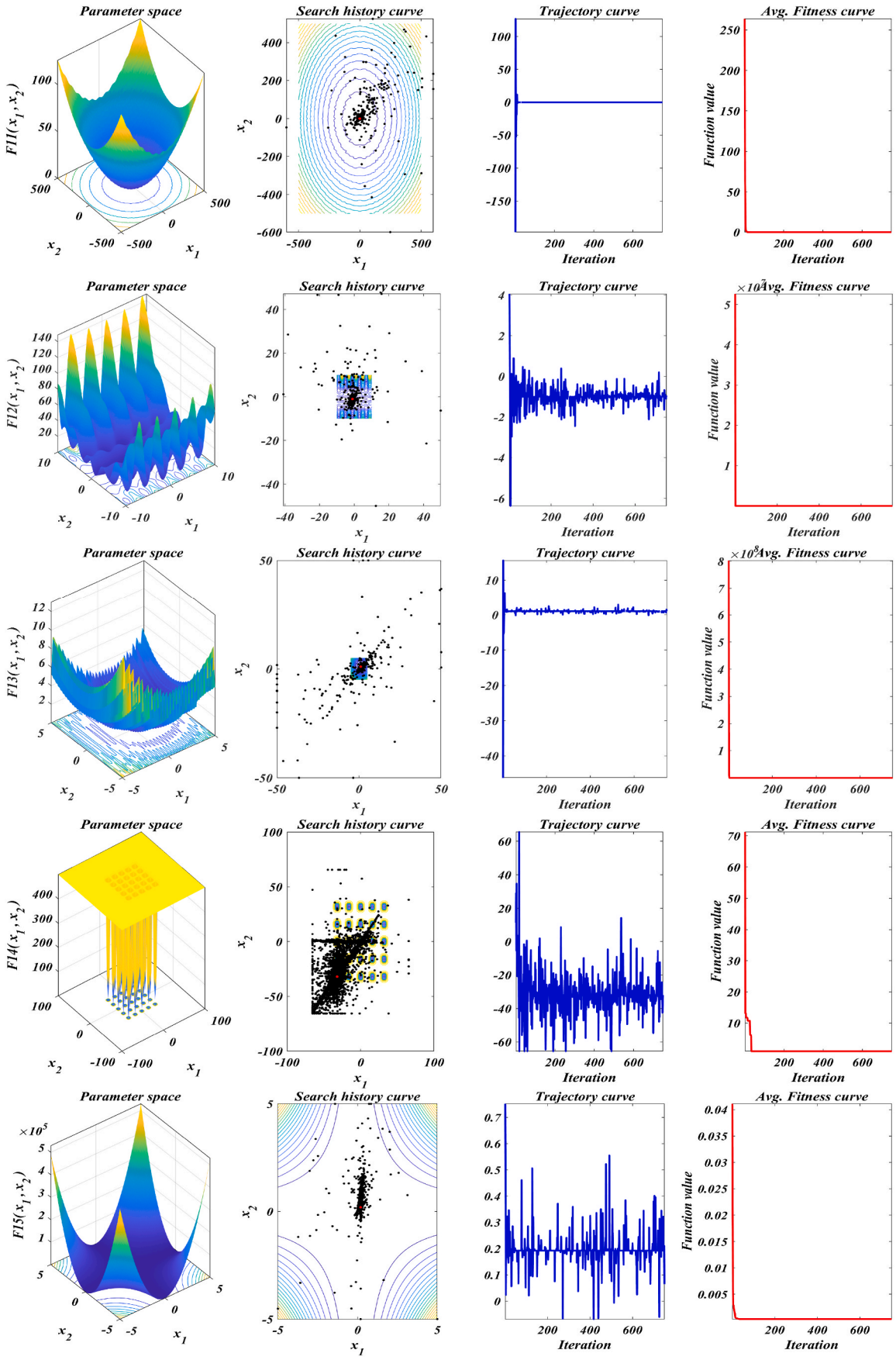


Fig. 5. (*continued*).



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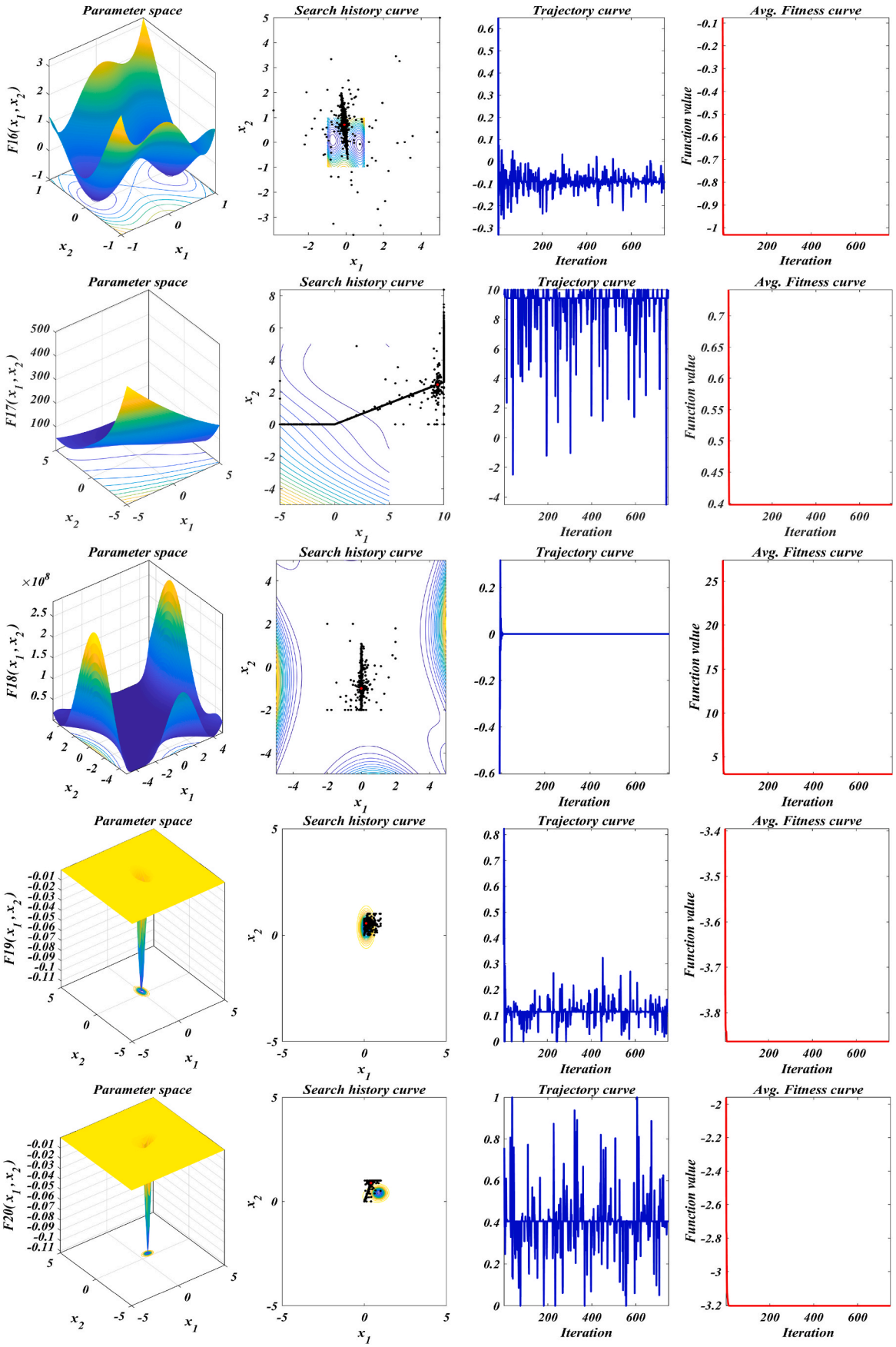

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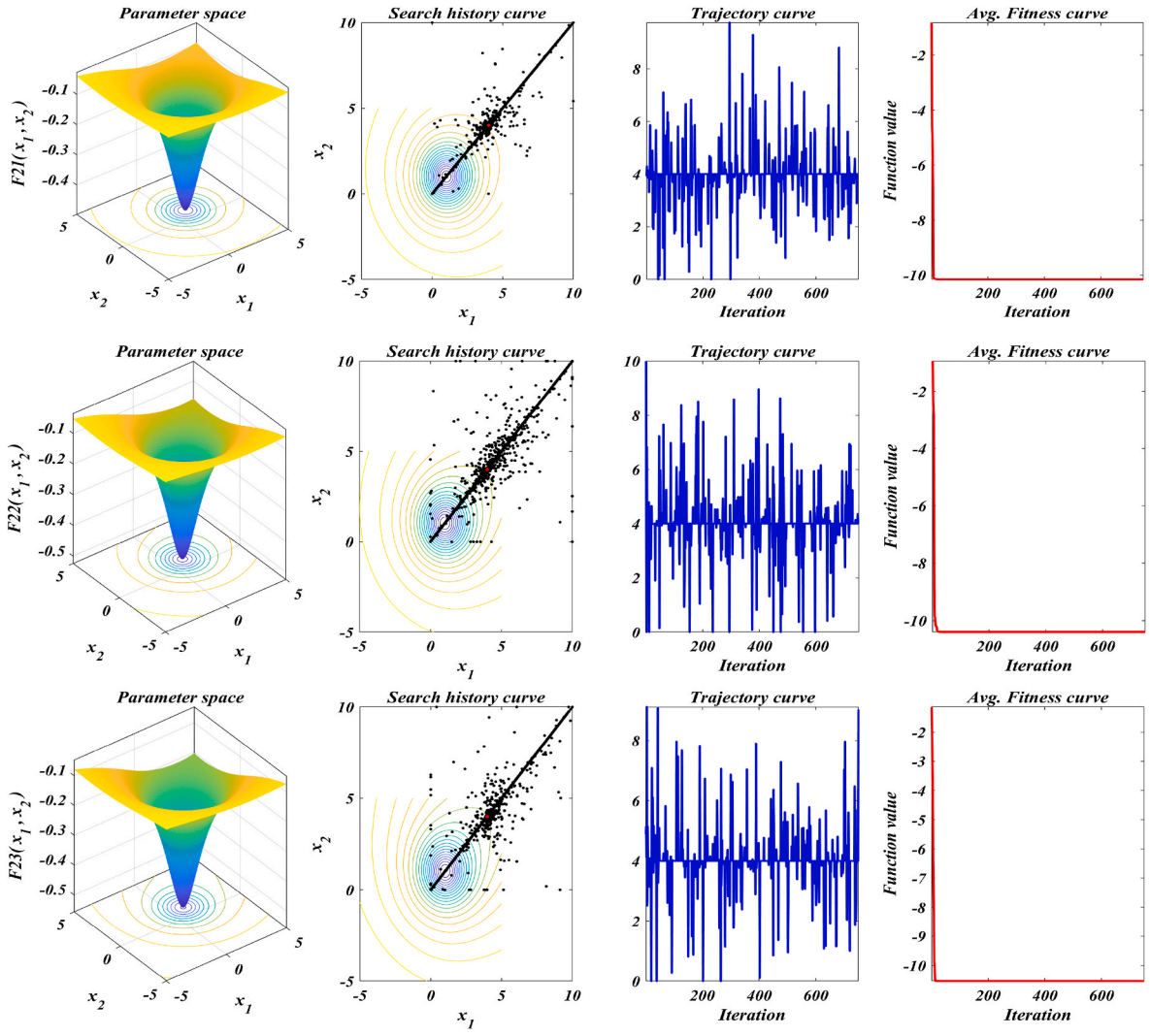

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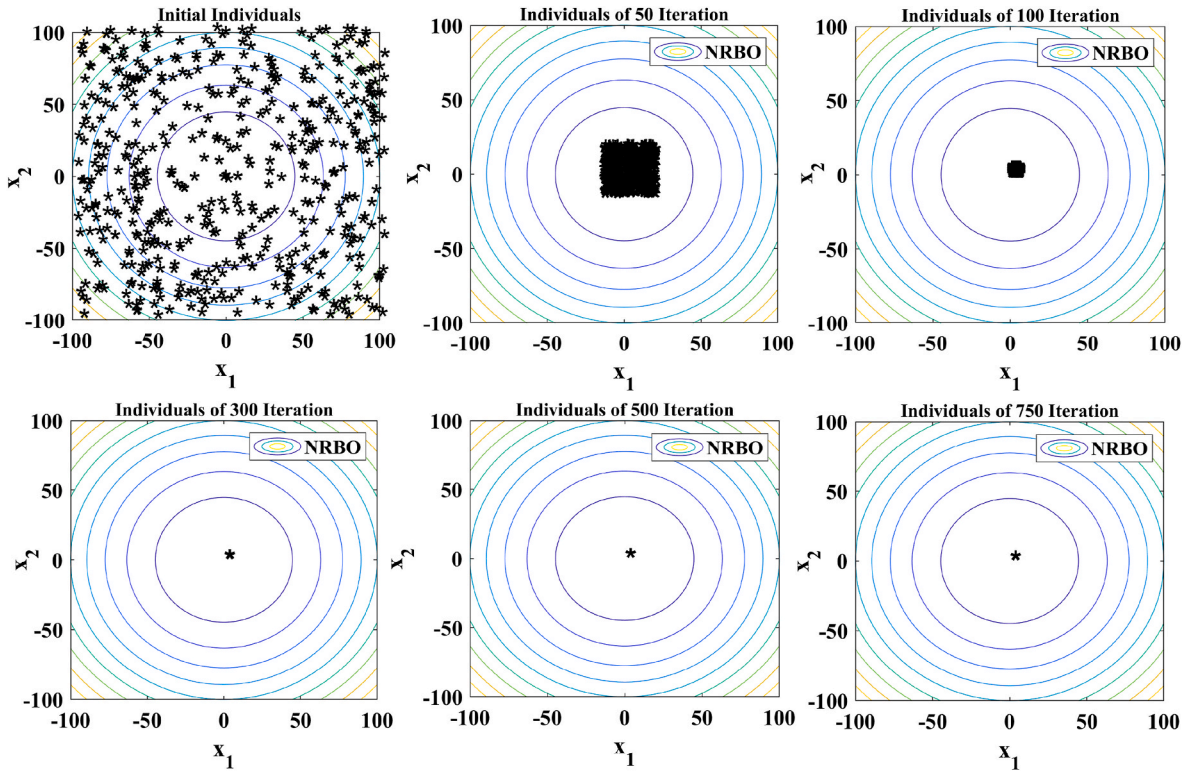

Fig. 6. Convergence behavior during different iteration levels.

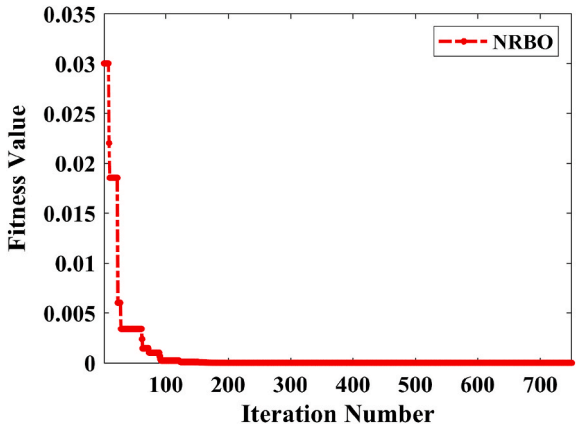

Fig. 7. Fitness evaluation of the best population.

convergence. We can now say that NRBO can handle the issues with premature convergence for unimodal functions. On the other hand, as shown in Fig. 6, randomly generated beginning search agents give NRBO a suitable level of variety. In addition, Fig. 7 displays the fitness value of the best population vs. iteration count.

3.4. Diversification and intensification analysis

The exploration and exploitation of the algorithm when handling the standard benchmark test functions are quantified using a dimension-wise diversity measure described in (Tilahun, 2019; Słowik, 2010) to analyze the search behaviours of NRBO quantitatively. The exploration (diversification) ratio, exploitation (intensification) ratio and balance between exploration and exploitation of the proposed algorithm are calculated as follows.

$$\text{Exploration in } \% = \frac{div^{IT}}{div_{max}} \times 100 \tag{33}$$

$$\text{Exploitation in } \% = \frac{\left|div^{IT} - div_{max}\right|}{div_{max}} \times 100 \tag{34}$$

$$\text{Balance in}\% = \frac{\frac{\left|div^{IT} - div_{max}\right|}{div_{max}} \times 100 + \frac{div^{IT}}{div_{max}} \times 100}{2} \tag{35}$$

where div^{IT} denotes the average population diversity of IT th iteration and div_{max} denotes the maximum value of population diversity in all iterations. The terminologies for div^{IT} and div_{max} are shown in Eq. (36) and Eq. (37), respectively.

$$div_j = \frac{1}{N_p}\left(\sum_{i=1}^{N_p} \text{median}\left(x^j\right) - x_{i,j}\right) \tag{36}$$

$$div^{IT} = \frac{1}{dim}\left(\sum_{j=1}^{dim} div_j\right) \tag{37}$$

where div_j represents the average population diversity of j th dimension and $x_{i,j}$ denotes i th population at j th dimension. The median of the j th dimension of the population with N_p individuals are represented by the expression median(x^j). Fig. 8 demonstrates the exploration and exploitation of the proposed NRBO while handling 23 traditional benchmark test functions.

Initially, the NRBO began with a high exploration rate on all 23 standard benchmark test functions. After a specific number of iterations, it increased its exploitation level and eventually turned into exploration. In most functions, such as F1, F4, F5, F7, F9, F10, F11, F12, F14, F16, F17, F19, F20, F21, and F23, the balance between exploration and exploitation is achieved in the first 150 iterations. In functions such as F2, F3, F6, F8, F13, F15, F18, and F22, the proposed NRBO achieves the balance in 250 iterations. Particularly, on F2, F3, F13, and F22, the

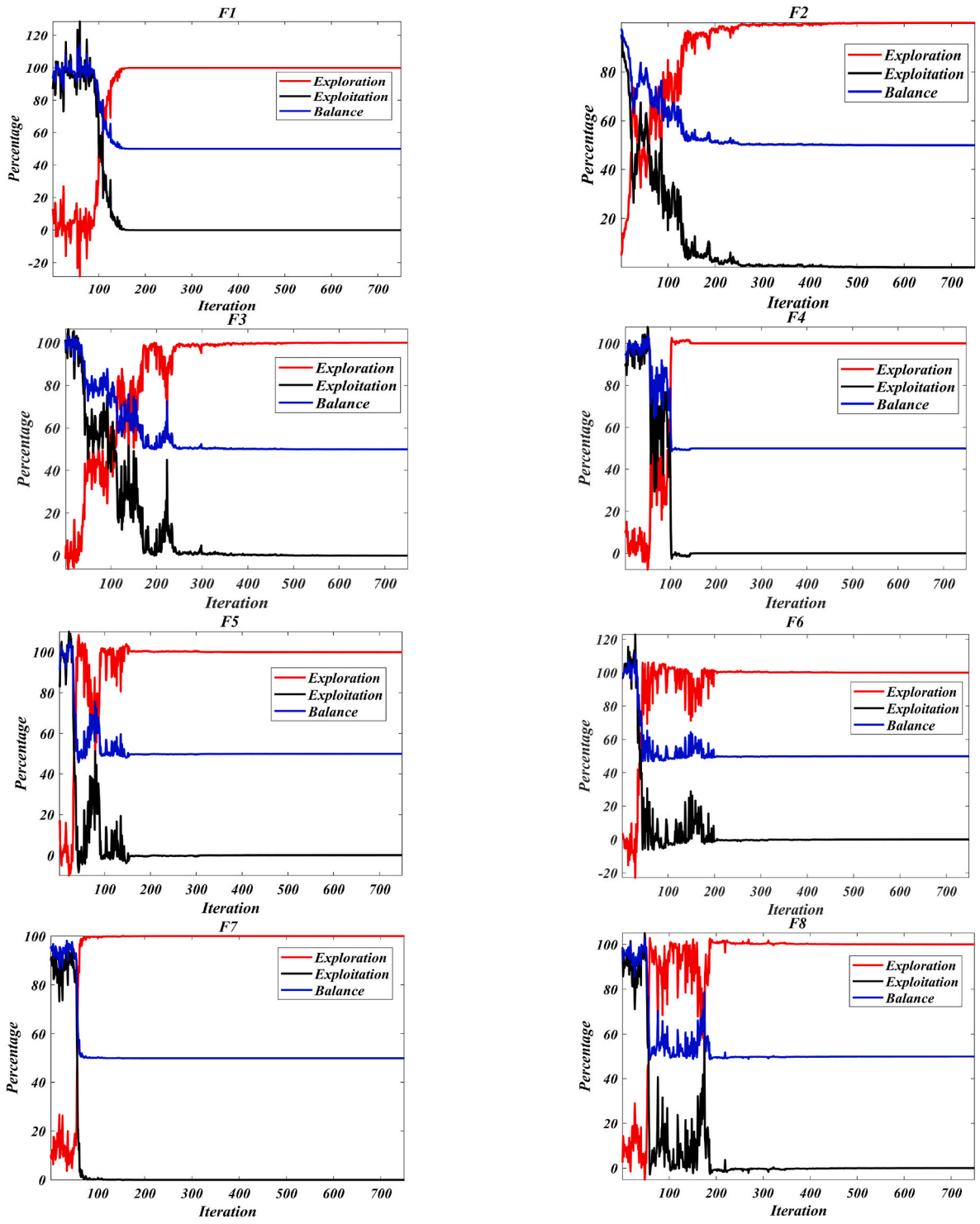

Fig. 8. Exploration and exploitation analysis of 23 standard benchmark problems.

exploration and exploitation ratios wavered and remained constant throughout the first half of iterations, while the algorithm's search strategy appeared to be more exploitative than exploratory during the second half. Above noteworthy, while viewing the trade-off between exploration and exploitation during the search phase, NRBO retained a rather dynamic search behaviour and outperformed such benchmark functions significantly. In general, the amount of exploration is more than that of the preliminary phase, and it diminishes as the number of iterations increases, whereas NRBO's exploitation ability increases.

3.5. Scalability analysis

The scalability of the proposed NRBO has been studied and evaluated in this subsection. The scalability assessment is conducted to evaluate the impact of several dimensions on the effectiveness of NRBO. It also indicates how an MA can maintain its strengths in large-dimensional exploration (Heidari et al., 2019; Ahmadianfar et al., 2022). In order to account for the random nature of NRBO and other selected algorithms, all algorithms have been examined 20 times, and statistical metrics, such as Min, Mean, and STD, have been gathered and reported

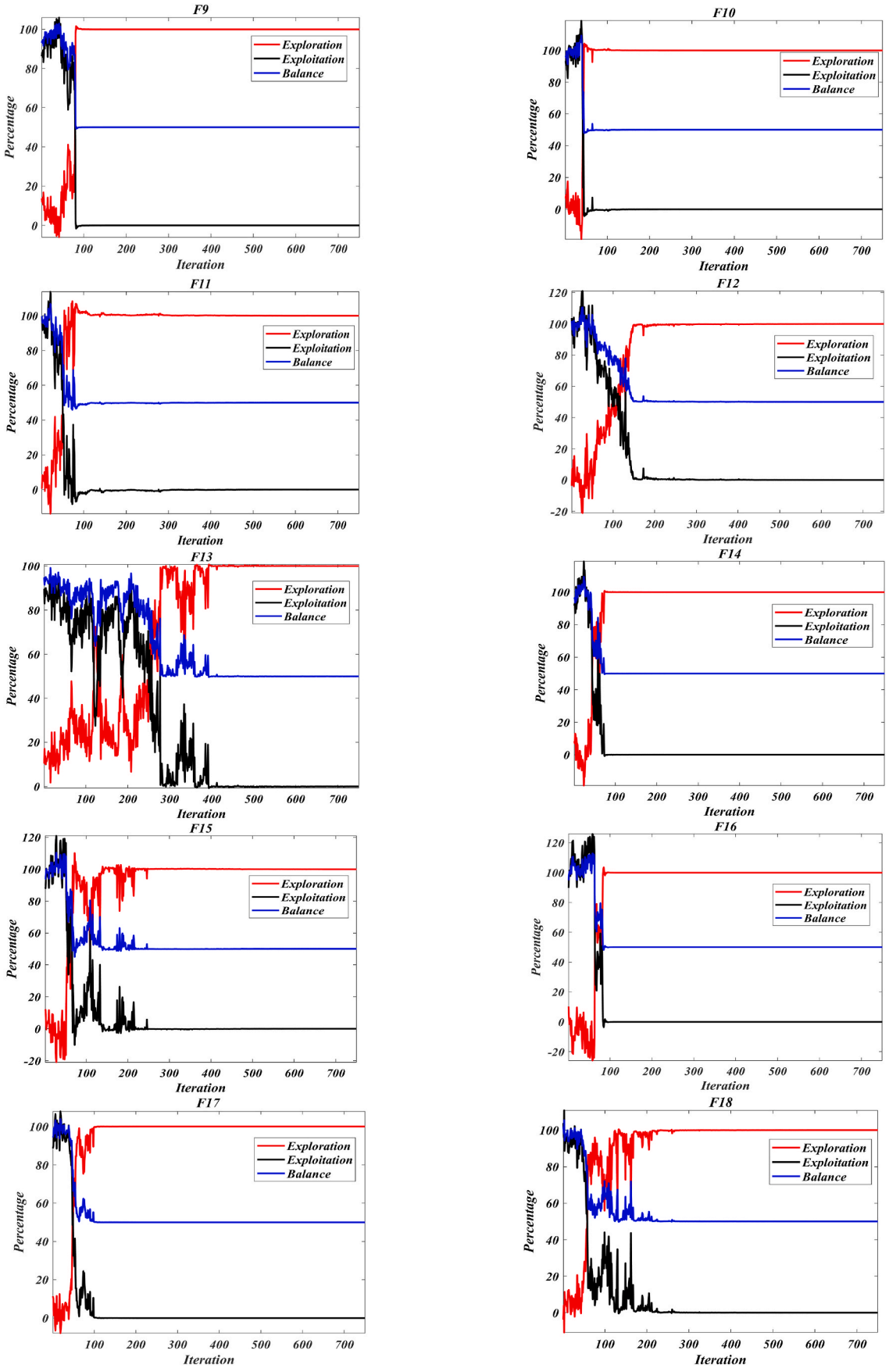

Fig. 8. (*continued*).

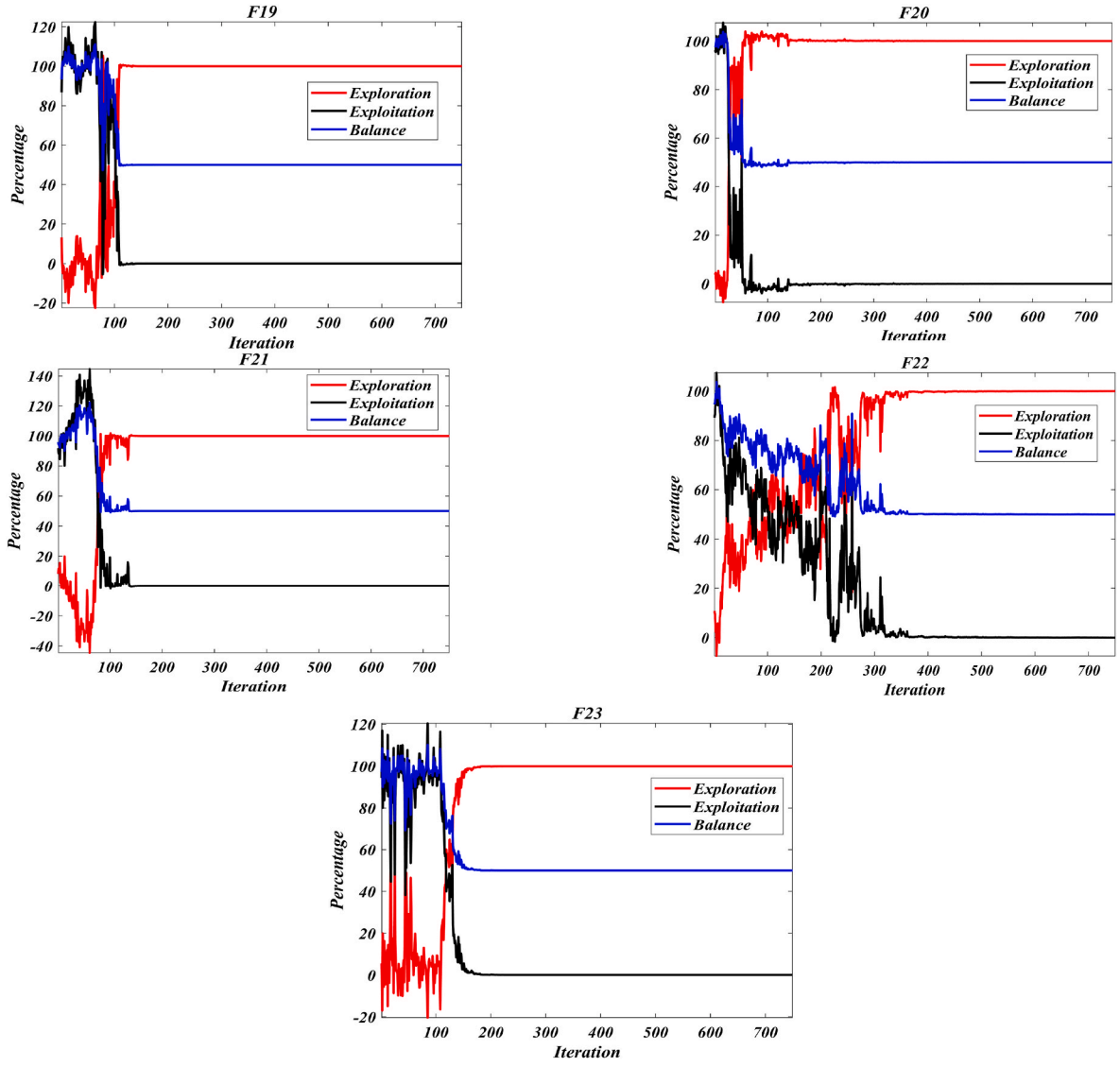

Fig. 8. (*continued*).

in Tables 2–5. For the investigations, 30, 100, 500, and 1000 dimensions are considered for the first 2 categories of functions (F1–F13). Each dimension's Min, Mean, and STD data were obtained and contrasted for all optimization algorithms with 30 population sizes and 750 iterations. The best results are highlighted with boldfaces in all tables.

In accordance with Table 2, the proposed NRBO yields the most successful outcomes. The obtained results by NRBO are much superior to those obtained by all selected algorithms when handling 30-dimensional benchmark functions. This is because when the NRBO is employed with NRSR and TAO, it performs significantly better than all algorithms on most benchmark functions. This has been demonstrated by the Friedman's Rank Test (FRT) values obtained and listed in Table 6, demonstrating that the proposed NRBO is statistically effective in most cases. When handling F1–F13 functions with 100 dimensions, NRBO can continue to perform better than other algorithms. The outcomes presented in Table 3 demonstrate that the NRBO outperforms all selected algorithms in nearly all tests performed. Table 4 shows that the NRBO has achieved superior outcomes in terms of Min, Mean, and STD than other algorithms in most situations with 500 dimensions. According to the FRT values in Tables 6 and it is clear that NRBO outperforms all other optimizers in all of the scenarios that have been evaluated. The proposed algorithm shows a considerable improvement when comparing the results obtained by the NRBO to those obtained by the other approaches. The employment of NRSR and TAO appears to favourably affect the performance of the NRBO, as demonstrated by these results. After seeing the effectiveness of the NRBR in lower-dimension problems, the test is extended to high-dimensional problems to evaluate its overall performance. It is revealed in Table 5 that the NRBO outperforms all other algorithms except for RKO in a few selected problems. It also outperforms other algorithms. The results of FRT are presented in Table 6 and indicate that NRBO has the potential to outperform other approaches. Importantly, NRBO has not only achieved a lesser global best in most functions with different dimensions than previous techniques but has also demonstrated superior results compared to RKO.

The results presented in Table 7 demonstrate that the proposed NRBO is highly competitive when handling F14–F23 benchmarks. On the other hand, the best optimal outcomes for F14–F23 can be obtained using all selected algorithms. Based on the findings, it can be concluded that whenever there is a good balance between the exploration and exploitation phases, such algorithms perform well when dealing with optimization problems. The NRBO takes advantage of TAO to speed up the exploration part of the process. Furthermore, incorporating NRSR into the NRBO balances the two phases. Based on the outcomes obtained by the NRBO when contrasted to other strategies, it has been demonstrated that the strategy used in the NRBO is efficient. This has been demonstrated by the FRT values obtained and listed in Table 8, demonstrating that the proposed NRBO is statistically significant in most cases.

Similar to traditional benchmarks, 29 CEC2017 problems are also considered to validate the proposed NRBO algorithm. The parameter settings of all selected algorithms are similar to the previous study. Due to the intrinsic difficulty of composite mathematical functions, it is extremely difficult for metaheuristic algorithms to handle them efficiently and effectively. A good balance between the exploration and exploitation stages of such challenges is required for their solution. This motivated the authors to select such problems to validate the proposed

Table 2
Results of F1–F13 benchmark functions (dim = 30).

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F1	Min	**0.00E + 00**	2.34E-201	1.44E-49	1.76E-04	2.27E-307	2.57E-164	6.00E-31	2.55E-73
	Mean	0.00E+00	2.04E-192	2.58E-48	1.34E-01	4.20E-278	1.23E-146	4.56E-19	2.53E-70
	STD	0.00E+00	0.00E+00	3.55E-48	3.09E-01	0.00E+00	4.78E-146	1.81E-18	4.69E-70
F2	Min	**5.95E-201**	1.12E-103	3.15E-29	4.00E-07	5.75E-167	1.17E-84	8.80E-16	1.01E-41
	Mean	1.67E-194	9.35E-98	1.72E-28	3.17E-04	8.04E-154	2.09E-75	3.71E-11	3.43E-40
	STD	0.00E+00	4.15E-97	1.08E-28	7.88E-04	2.86E-153	9.29E-75	9.28E-11	3.90E-40
F3	Min	**1.71E-296**	8.67E-171	1.02E-16	3.48E+02	5.01E-260	1.26E-142	1.04E-33	2.82E-24
	Mean	1.06E-262	7.84E-159	7.28E-13	4.25E+03	2.80E-239	9.93E-121	2.49E-19	4.78E-17
	STD	0.00E+00	3.17E-158	1.27E-12	3.86E+03	0.00E+00	3.35E-120	1.00E-18	2.06E-16
F4	Min	**2.56E-192**	1.00E-92	8.33E-13	7.76E+00	4.09E-147	1.92E-84	5.70E-16	1.15E-19
	Mean	3.13E-188	4.88E-88	6.89E-12	2.47E+01	8.87E-128	7.95E-73	3.61E-11	1.12E-17
	STD	0.00E+00	1.21E-87	6.30E-12	1.04E+01	3.11E-127	2.24E-72	6.79E-11	3.08E-17
F5	Min	16.1453	18.1323	25.5881	28.7732	22.2316	**0.0000**	28.0581	24.2907
	Mean	17.8648	20.4402	26.5891	981.4789	24.3769	0.0023	28.4745	24.5776
	STD	1.40E+00	1.28E+00	6.12E-01	3.41E+03	1.80E+00	4.24E-03	1.53E-01	1.61E-01
F6	Min	**2.02E-13**	1.26E-11	1.77E-05	4.13E+00	4.53E-10	7.80E-10	1.38E-01	5.05E-11
	Mean	3.47E-10	1.45E-08	3.95E-01	5.29E+00	6.85E-10	5.37E-05	4.38E-01	3.94E-09
	STD	5.84E-10	6.27E-08	3.13E-01	1.20E+00	1.76E-10	6.07E-05	1.99E-01	3.86E-09
F7	Min	2.74E-05	5.07E-05	2.36E-04	1.38E-03	4.70E-05	**7.77E-06**	9.13E-05	1.27E-04
	Mean	2.04E-04	4.08E-04	1.01E-03	3.51E-02	1.19E-04	6.53E-05	6.92E-04	5.88E-04
	STD	1.64E-04	2.72E-04	6.43E-04	2.38E-02	4.72E-05	7.10E-05	5.59E-04	3.05E-04
F8	Min	**−1.26E + 04**	−1.03E+04	−7.56E+03	−4.69E+03	−9.51E+03	−1.20E+04	−7.52E+03	−1.00E+04
	Mean	−1.26E+04	−9.05E+03	−6.00E+03	−3.88E+03	−8.56E+03	−1.06E+04	−5.85E+03	−8.89E+03
	STD	6.31E-01	7.28E+02	6.92E+02	2.56E+02	6.48E+02	8.71E+02	4.67E+02	5.89E+02
F9	Min	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	2.72E-05	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	9.07E-01	2.38E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	STD	0.00E+00	0.00E+00	2.09E+00	3.45E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F10	Min	**8.88E-16**	**8.88E-16**	1.51E-14	7.52E-03	**8.88E-16**	**8.88E-16**	1.87E-14	4.44E-15
	Mean	8.88E-16	8.88E-16	2.13E-14	1.49E+01	8.88E-16	8.88E-16	7.42E-11	5.68E-15
	STD	0.00E+00	0.00E+00	3.97E-15	8.86E+00	0.00E+00	0.00E+00	2.55E-10	1.74E-15
F11	Min	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	3.30E-03	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	4.60E-03	3.82E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	STD	0.00E+00	0.00E+00	7.59E-03	3.26E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F12	Min	**2.10E-14**	1.44E-12	1.31E-02	4.36E-01	1.20E-10	8.73E-08	2.25E-02	4.25E-12
	Mean	7.35E-10	2.06E-11	3.04E-02	4.64E+00	3.11E-10	3.72E-06	6.11E-02	1.36E-10
	STD	1.88E-09	3.36E-11	1.21E-02	5.19E+00	1.10E-10	4.05E-06	3.12E-02	2.95E-10
F13	Min	**1.04E-16**	2.38E-10	9.98E-02	2.20E+00	1.83E-09	2.00E-09	2.27E-01	2.79E-10
	Mean	2.57E-02	1.81E-02	4.70E-01	8.02E+03	8.05E-03	3.04E-05	5.66E-01	2.18E-02
	STD	3.17E-02	2.17E-02	2.31E-01	3.52E+04	8.33E-03	4.03E-05	2.82E-01	3.94E-02

Table 3
Results of F1–F13 benchmark functions (dim = 100).

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F1	Min	**0.00E + 00**	3.94E-186	5.96E-22	1.22E+03	1.02E-257	7.51E-162	1.40E-38	1.67E-48
	Mean	0.00E+00	4.34E-174	4.95E-21	8.70E+03	6.84E-233	7.93E-141	1.93E-21	3.25E-46
	STD	0.00E+00	0.00E+00	3.40E-21	5.27E+03	0.00E+00	2.35E-140	8.65E-21	5.17E-46
F2	Min	**4.30E-186**	1.53E-96	2.16E-13	4.94E-01	6.58E-147	3.01E-82	2.42E-21	7.84E-28
	Mean	7.29E-179	1.31E-89	5.28E-13	4.26E+00	1.34E-133	2.00E-73	2.16E-13	4.49E-27
	STD	0.00E+00	5.47E-89	2.00E-13	5.22E+00	5.95E-133	6.48E-73	6.18E-13	5.37E-27
F3	Min	**1.37E-236**	8.67E-158	6.46E-01	1.25E+05	1.25E-235	2.00E-134	2.21E-35	1.40E-06
	Mean	6.85E-208	3.78E-138	5.51E+01	2.27E+05	1.33E-202	9.86E-86	2.00E-24	1.17E-01
	STD	0.00E+00	1.69E-137	7.74E+01	6.29E+04	0.00E+00	4.41E-85	7.72E-24	4.50E-01
F4	Min	**4.75E-177**	9.10E-85	1.35E-03	7.95E+01	2.22E-123	1.31E-81	1.94E-20	1.30E-08
	Mean	1.39E-170	8.63E-79	1.09E-01	8.78E+01	2.09E-108	1.00E-71	8.83E-14	3.05E-06
	STD	0.00E+00	3.41E-78	2.07E-01	3.18E+00	6.72E-108	3.75E-71	1.94E-13	7.13E-06
F5	Min	8.95E+01	9.18E+01	9.61E+01	8.85E+06	9.36E+01	**9.26E-07**	9.86E+01	9.45E+01
	Mean	9.13E+01	9.44E+01	9.77E+01	7.02E+07	9.63E+01	1.75E-02	9.87E+01	9.57E+01
	STD	1.54E+00	2.07E+00	6.53E-01	4.52E+07	1.77E+00	2.70E-02	6.81E-02	1.03E+00
F6	Min	**3.75E-08**	1.47E-02	7.96E+00	1.33E+03	3.27E-07	6.85E-04	1.53E+01	1.25E+00
	Mean	1.87E-04	2.51E-02	9.32E+00	6.53E+03	1.57E-04	1.62E+00	1.85E+01	2.07E+00
	STD	2.99E-04	7.50E-03	9.35E-01	3.80E+03	3.95E-04	1.12E+00	1.47E+00	6.27E-01
F7	Min	6.85E-04	1.47E-02	7.96E+00	1.33E+03	3.27E-07	**3.75E-08**	1.53E+01	1.25E+00
	Mean	1.62E+00	2.51E-02	9.32E+00	6.53E+03	1.57E-04	1.87E-04	1.85E+01	2.07E+00
	STD	1.12E+00	7.50E-03	9.35E-01	3.80E+03	3.95E-04	2.99E-04	1.47E+00	6.27E-01
F8	Min	**−4.19E + 04**	−3.10E+04	−2.19E+04	−8.68E+03	−2.73E+04	**−4.19E + 04**	−1.36E+04	−2.92E+04
	Mean	−3.81E+04	−2.77E+04	−1.54E+04	−7.03E+03	−2.36E+04	−4.19E+04	−1.12E+04	−2.70E+04
	STD	3.12E+03	1.69E+03	3.60E+03	5.96E+02	2.14E+03	2.39E+00	1.13E+03	1.33E+03
F9	Min	**0.00E + 00**	**0.00E + 00**	3.41E-13	1.17E+02	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	3.26E+00	2.74E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	STD	0.00E+00	0.00E+00	3.95E+00	1.22E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F10	Min	**8.88E-16**	**8.88E-16**	2.61E-12	4.73E+00	**8.88E-16**	**8.88E-16**	8.88E-16	7.99E-15
	Mean	8.88E-16	8.88E-16	5.64E-12	1.80E+01	8.88E-16	8.88E-16	5.28E-14	1.05E-14
	STD	0.00E+00	0.00E+00	2.34E-12	5.42E+00	0.00E+00	0.00E+00	1.51E-13	3.48E-15
F11	Min	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	2.30E+01	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	3.22E-03	8.04E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	STD	0.00E+00	0.00E+00	9.03E-03	4.78E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F12	Min	**1.31E-08**	1.50E-04	1.78E-01	5.43E+07	1.64E-08	2.73E-06	3.55E-01	8.09E-03
	Mean	1.22E-06	4.75E-04	2.78E-01	2.25E+08	7.82E-08	1.29E-02	5.45E-01	1.67E-02
	STD	1.44E-06	3.22E-04	7.11E-02	1.08E+08	1.44E-07	8.64E-03	9.08E-02	6.76E-03
F13	Min	**4.03E-07**	3.97E-02	5.53E+00	1.19E+08	2.59E-06	4.40E-04	8.93E+00	2.75E+00
	Mean	8.26E-05	2.24E+00	6.41E+00	3.89E+08	1.82E-02	3.81E-01	9.88E+00	4.89E+00
	STD	1.17E-04	2.99E+00	3.44E-01	1.44E+08	2.01E-02	5.56E-01	2.81E-01	1.47E+00

algorithm. The obtained results by NRBO and other selected algorithms are listed in Table 9. The FRT values obtained for all 29 test functions are listed in Table 10.

When looking at the results described in Tables 9 and it can be seen that the proposed NRBO obtained the highest average fitness in 19 test functions (i.e., F25, F27, F28, F29, F30, F31, F32, F33, F35, F38, F39, F42, F43, F44, F45, F46, F47, F49, and F51); however, it did not outperform the GBO in other few test functions in terms of reaching the best mean values. The proposed NRBO attained optimal fitness values in 21 of the 29 functions tested but achieved satisfactory results in the remaining 8 functions. With respect to performance, the GBO came in second place and attained the best fitness value in 10 functions; nevertheless, in seven functions, its performance is similar to that of the proposed NRBO. The RKO produces the best results in six test functions, and in three functions, its performance is similar to the proposed NRBO. The EO produces the best results in five test functions, and in three functions, its performance is similar to the proposed NRBO. CSA also produces more competitive results than EO and other algorithms. Besides NRBO, GBO, EO, and CSA, all algorithms fail to handle the CEC2017 test functions. The reliability of the algorithm is judged by observing the STD values. In most test functions, the STD values obtained by the proposed NRBO are very low, i.e., the reliability of the NRBO is better than all other algorithms. Therefore, it is concluded from the preceding analysis that the NRBO is the superior algorithm in the CEC2017 benchmark functions, followed by GBO, RKO, CSA, GWO, EO, HHO, and SCA. Table 10 shows that, in most circumstances, NRBO produces superior and statistically significant results. Therefore, we can consider it a likely competitor for consideration as a current state-of-the-art algorithm.

In addition to the abovementioned discussions, 12 CEC2022 benchmarks are also considered to validate the proposed algorithm in this paper. Due to the fact that links between all variables in real-world problems are uncommon, CEC2022 separates parameters into subdimensions using a random distribution. In order to construct the rotation matrix for each set of subdimensions, Gram-Schmidt orthonormalization with a condition number equal to 1 or 2 is used in conjunction with standard normally distributed entries. The search range for all the problems is considered as [−100, 100], and the problem dimension is 20. The obtained results are presented in Table 11. The FRT values obtained for all CEC2022 functions are listed in Table 12.

When looking at the results described in Tables 11 and it can be seen that the proposed NRBO obtained the highest average fitness in seven test functions (i.e., F55, F56, F57, F59, F60, F61, and F64); however, it did not outperform the GBO in other few test functions in terms of reaching the best mean values. The proposed NRBO attained optimal fitness values in seven of the 12 functions tested but achieved satisfactory results in the remaining five functions. The GBO came in second place with respect to performance and attained the best fitness value in four functions. The RKO produces the best results in one test function. Other than NRBO, GBO, and RKO, all algorithms fail to handle the CEC2022 test functions. The reliability of the algorithm is judged by observing the STD values. In most test functions, the STD values obtained by the proposed NRBO are very low, i.e., the reliability of the NRBO is better than all other algorithms. Therefore, it is concluded from the preceding analysis that the NRBO is the superior algorithm in the CEC2022 benchmark functions, followed by GBO, RKO, GWO, CSA,

Table 4
Results of F1–F13 benchmark functions (dim = 500).

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F1	Min	**0.00E + 00**	1.64E-183	1.33E-08	3.55E+04	9.70E-245	2.98E-121	7.01E-33	7.76E-39
	Mean	0.00E+00	4.78E-166	4.97E-08	1.62E+05	7.64E-211	1.24E-108	3.08E-24	3.99E-37
	STD	0.00E+00	0.00E+00	2.10E-08	6.29E+04	0.00E+00	5.56E-108	9.54E-24	6.88E-37
F2	Min	**2.44E-185**	1.48E-92	1.36E-05	1.88E+01	8.27E-136	7.24E-81	7.04E-18	5.79E-23
	Mean	5.80E-177	2.02E-88	2.47E-05	9.09E+01	2.03E-125	2.29E-75	4.59E-13	4.18E-22
	STD	0.00E+00	5.13E-88	7.24E-06	4.43E+01	9.00E-125	5.15E-75	8.90E-13	3.14E-22
F3	Min	1.07E-184	2.88E-145	1.28E+05	3.82E+06	**6.61E-221**	1.52E+07	7.21E-37	3.56E+01
	Mean	1.89E-158	5.50E-133	1.99E+05	5.75E+06	6.00E-179	2.95E+07	1.09E-22	1.30E+04
	STD	8.45E-158	2.43E-132	5.94E+04	1.01E+06	0.00E+00	8.13E+06	2.79E-22	1.65E+04
F4	Min	**6.92E-170**	6.06E-76	5.19E+01	9.83E+01	5.22E-116	6.19E+00	1.81E-20	3.90E+01
	Mean	8.09E-160	1.54E-70	6.05E+01	9.89E+01	2.26E-104	8.00E+01	2.02E-14	7.24E+01
	STD	3.61E-159	5.74E-70	5.10E+00	2.94E-01	9.13E-104	2.31E+01	4.75E-14	2.09E+01
F5	Min	**4.92E + 02**	4.93E+02	4.97E+02	8.24E+08	4.91E+02	4.95E+02	4.99E+02	4.97E+02
	Mean	4.93E+02	4.95E+02	4.98E+02	1.59E+09	4.94E+02	4.96E+02	4.99E+02	4.97E+02
	STD	6.15E-01	7.33E-01	1.16E-01	3.69E+08	1.30E+00	4.26E-01	6.69E-02	2.71E-01
F6	Min	1.27E+01	9.05E+00	8.84E+01	5.78E+04	**9.88E-01**	1.45E+01	1.15E+02	7.30E+01
	Mean	2.07E+01	1.22E+01	9.26E+01	1.52E+05	1.37E+00	2.60E+01	1.19E+02	7.72E+01
	STD	5.87E+00	1.52E+00	1.81E+00	6.18E+04	1.73E-01	7.43E+00	1.41E+00	1.74E+00
F7	Min	**1.56E-06**	9.55E-05	1.03E-02	8.01E+03	1.26E-05	2.64E-05	1.44E-04	1.13E-03
	Mean	2.92E-04	9.47E-04	1.97E-02	1.22E+04	2.74E-04	2.95E-03	5.80E-04	2.45E-03
	STD	2.65E-04	9.74E-04	4.78E-03	2.62E+03	1.95E-04	1.98E-03	3.02E-04	1.01E-03
F8	Min	**−2.09E + 05**	−1.39E+05	−6.78E+04	−1.74E+04	−1.28E+05	**−2.09E + 05**	−2.94E+04	−9.63E+04
	Mean	−1.81E+05	−1.17E+05	−5.59E+04	−1.56E+04	−1.02E+05	−1.77E+05	−2.49E+04	−8.43E+04
	STD	2.16E+04	9.69E+03	1.10E+04	1.08E+03	1.88E+04	2.95E+04	1.91E+03	5.20E+03
F9	Min	**0.00E + 00**	**0.00E + 00**	1.11E+00	2.36E+02	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	1.66E+01	1.11E+03	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	STD	0.00E+00	0.00E+00	1.07E+01	6.16E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F10	Min	**8.88E-16**	**8.88E-16**	6.75E-06	1.03E+01	**8.88E-16**	**8.88E-16**	**8.88E-16**	1.15E-14
	Mean	8.88E-16	8.88E-16	1.04E-05	1.91E+01	8.88E-16	3.91E-15	3.23E-14	1.85E-14
	STD	0.00E+00	0.00E+00	2.34E-06	3.53E+00	0.00E+00	2.38E-15	6.90E-14	4.53E-15
F11	Min	**0.00E + 00**	**0.00E + 00**	2.08E-09	3.44E+02	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	1.69E-03	1.42E+03	0.00E+00	0.00E+00	0.00E+00	1.01E-03
	STD	0.00E+00	0.00E+00	7.56E-03	5.18E+02	0.00E+00	0.00E+00	0.00E+00	4.51E-03
F12	Min	1.26E-02	2.03E-02	6.75E-01	2.84E+09	**9.38E-04**	1.99E-02	9.87E-01	4.14E-01
	Mean	4.89E-02	2.84E-02	7.32E-01	4.88E+09	1.34E-03	6.55E-02	1.05E+00	4.60E-01
	STD	2.67E-02	4.75E-03	3.42E-02	1.02E+09	1.72E-04	3.50E-02	3.35E-02	2.55E-02
F13	Min	**1.98E + 00**	2.01E+01	4.53E+01	4.71E+09	2.57E+00	3.64E+00	5.00E+01	4.84E+01
	Mean	2.68E+00	4.63E+01	4.65E+01	8.25E+09	7.86E+00	1.26E+01	5.00E+01	4.90E+01
	STD	5.24E-01	6.20E+00	6.00E-01	1.99E+09	2.64E+00	4.66E+00	7.34E-03	2.88E-01

HHO, EO, and SCA. Table 12 shows that, in most circumstances, NRBO produces superior and statistically significant results. Therefore, we can consider it a likely competitor for consideration as a current state-of-the-art algorithm.

In addition, the performance of the proposed algorithm is also compared with CEC2022 winners, such as Evolutionary Algorithms with Eigen crossover (EA4eig) (Bujok and Kolenovsky, 2022), Non-Linear Population Size Reduction Success-History Adaptive Differential Evolution with Linear Bias Change (NL-SHADE-LBC) (Stanovov et al., 2022), Non-Linear Population Size Reduction Success-History Adaptive Differential Evolution with Rank-based Selective Pressure and Midpoint of the population (NL-SHADE-RSP-MID) (Biedrzycki et al., 2022), and Successful History-based Adaptive DE variants with Linear Population Size Reduction and Dynamic Perturbation (S-LSHADE-DP) (Le Van Cuong et al., 2022). The problems F53–F64 are considered with 10 dimensions. Each algorithm is executed 20 times individually, and the results are recorded in Table 13.

In order to rank the algorithms, as discussed earlier, FRT was conducted, and the average FRT scores are also listed in Table 13. Based on the average FRT values, it is found that the proposed algorithm is competitive with CEC2022 winners. With a 3.258 FRT average value, the proposed algorithm stood first among all CEC2022 winners, followed by S-LSHADE-DP, EA4eig, NL-SHADE-RSP-MID, and NL-SHADE-LBC. The improved performance of NRBO is because of the gradient- and population-based feature combinations.

3.6. Convergence analysis

It was determined and evaluated as the achieved optimum value to assess the convergence actions of the NRBO's best score metric, which had been acquired so far as possible (Ł, 2021; Davidović and Jakšić Krüger, 2018). This rate was computed and recorded for each iteration of all benchmark functions utilized. To compute this figure, we used all selected algorithms and validated them against the overall performance of the NRBO, which was gathered throughout 20 runs with 750 iterations. As illustrated in Fig. 9 (standard benchmarks from F1–F23), Fig. 10 (CEC2017 benchmarks), and Fig. 11 (CEC2022 benchmarks), in the argument over convergence rates, the NRBO, compared to other methods, has an acceptable convergence rate in the vast majority of circumstances.

The provided findings have proven the efficacy of applying TAO and NRSR, which has significantly improved the convergence speed and performance of the optimization algorithm, which is signified precisely by the suggested approach. However, in a few functions, NRBO attained optimal values almost identical to those produced by the GBO and RKO. Compared to other algorithms, the NRBO not only outperformed them in terms of convergence performance, but it also accomplished an optimal value in relatively few iterations and demonstrated an extremely fast convergence rate when contrasted to other approaches, as can be realized from the results shown in Figs. 9–11. While it is clear with other benchmarked algorithms, the local optimal solutions have been stacked with the algorithms from the beginning of the tests.

When the convergence graphs in Figs. 9–11, and the statistics are taken into consideration, the GBO is considered to be the second

Table 5
Results of F1–F13 benchmark functions (dim = 1000).

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F1	Min	**0.00E + 00**	1.86E-177	1.26E-05	1.49E+05	1.39E-241	1.11E-119	1.70E-34	2.17E-36
	Mean	0.00E+00	3.69E-164	4.80E-05	3.90E+05	6.98E-217	4.44E-107	8.40E-23	3.69E-35
	STD	0.00E+00	0.00E+00	2.03E-05	1.33E+05	0.00E+00	1.98E-106	3.47E-22	4.29E-35
F2	Min	**8.83E-183**	2.81E-92	1.85E-03	6.55E+04	7.30E-131	5.04E-84	1.70E-19	8.48E-22
	Mean	3.72E-174	3.80E-84	6.19E-03	6.55E+04	2.73E-122	1.84E-71	6.42E-11	2.48E-21
	STD	0.00E+00	1.70E-83	4.49E-03	6.55E+04	8.43E-122	8.23E-71	2.52E-10	1.41E-21
F3	Min	3.42E-168	5.21E-144	6.13E+05	1.85E+07	**1.97E-214**	3.41E+07	5.65E-30	4.15E+03
	Mean	1.09E-139	1.35E-129	9.65E+05	2.63E+07	1.91E-185	1.05E+08	7.48E-22	1.11E+05
	STD	4.87E-139	5.91E-129	1.94E+05	4.32E+06	0.00E+00	3.78E+07	3.31E-21	1.62E+05
F4	Min	**2.08E-166**	8.95E-74	7.26E+01	9.91E+01	1.79E-119	1.58E+01	6.87E-18	6.10E+01
	Mean	4.15E-156	4.68E-64	7.79E+01	9.96E+01	2.07E-103	7.53E+01	2.93E-13	8.22E+01
	STD	1.28E-155	2.07E-63	3.80E+00	1.65E-01	9.13E-103	2.76E+01	7.41E-13	1.19E+01
F5	Min	**9.88E + 02**	9.91E+02	9.97E+02	1.47E+09	**9.88E + 02**	9.92E+02	9.99E+02	9.97E+02
	Mean	9.90E+02	9.93E+02	9.97E+02	3.36E+09	9.91E+02	9.93E+02	9.99E+02	9.97E+02
	STD	7.12E-01	8.27E-01	9.23E-02	7.76E+08	1.17E+00	7.52E-01	7.56E-02	1.06E-01
F6	Min	4.43E+01	3.01E+01	2.02E+02	1.83E+05	**4.51E + 00**	2.98E+01	2.40E+02	1.90E+02
	Mean	6.65E+01	4.27E+01	2.06E+02	3.93E+05	5.40E+00	5.45E+01	2.44E+02	1.95E+02
	STD	1.37E+01	8.42E+00	2.38E+00	1.17E+05	3.79E-01	1.11E+01	1.50E+00	2.34E+00
F7	Min	**1.62E-05**	2.59E-04	2.72E-02	4.45E+04	8.50E-05	8.44E-05	1.25E-04	1.27E-03
	Mean	3.05E-04	7.76E-04	4.21E-02	6.43E+04	3.60E-04	2.66E-03	6.02E-04	3.43E-03
	STD	2.32E-04	6.05E-04	1.10E-02	1.27E+04	2.82E-04	2.97E-03	3.69E-04	1.19E-03
F8	Min	**−4.19E + 05**	−2.59E+05	−1.05E+05	−2.91E+04	−2.56E+05	−4.07E+05	−4.67E+04	−1.43E+05
	Mean	−3.76E+05	−2.05E+05	−9.62E+04	−2.25E+04	−2.00E+05	−3.54E+05	−3.70E+04	−1.28E+05
	STD	4.55E+04	2.66E+04	5.30E+03	2.23E+03	4.10E+04	4.73E+04	4.67E+03	6.84E+03
F9	Min	**0.00E + 00**	**0.00E + 00**	1.88E+01	6.97E+02	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	4.93E+01	2.02E+03	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	STD	0.00E+00	0.00E+00	1.84E+01	6.63E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F10	Min	**8.88E-16**	**8.88E-16**	1.65E-04	8.49E+00	**8.88E-16**	**8.88E-16**	**8.88E-16**	1.51E-14
	Mean	8.88E-16	8.88E-16	2.31E-04	1.86E+01	8.88E-16	4.26E-15	5.08E-14	2.08E-14
	STD	0.00E+00	0.00E+00	4.16E-05	4.01E+00	0.00E+00	2.70E-15	9.60E-14	4.37E-15
F11	Min	**0.00E + 00**	**0.00E + 00**	1.43E-06	1.62E+03	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**	**0.00E + 00**
	Mean	0.00E+00	0.00E+00	3.45E-06	3.73E+03	0.00E+00	0.00E+00	0.00E+00	1.05E-16
	STD	0.00E+00	0.00E+00	1.24E-06	1.40E+03	0.00E+00	0.00E+00	0.00E+00	2.48E-17
F12	Min	**2.31E-03**	4.06E-02	7.86E-01	8.01E+09	2.46E-02	1.82E-02	1.07E+00	6.73E-01
	Mean	2.67E-03	5.91E-02	8.39E-01	1.10E+10	6.72E-02	6.19E-02	1.11E+00	7.20E-01
	STD	1.74E-04	1.04E-02	2.27E-02	2.22E+09	2.85E-02	2.91E-02	1.71E-02	1.97E-02
F13	Min	1.16E+01	4.84E+01	9.60E+01	1.31E+10	**5.16E + 00**	1.45E+01	1.00E+02	9.90E+01
	Mean	1.84E+01	9.53E+01	9.92E+01	1.81E+10	6.05E+00	2.43E+01	1.00E+02	9.93E+01
	STD	4.57E+00	1.11E+01	1.61E+00	3.02E+09	6.42E-01	6.58E+00	8.02E-03	1.97E-01

strongest on a few test cases. Even though the convergence actions are fairly comparable to NRBO, the convergence behaviour tends to smooth out as the optimization process proceeds. This demonstrates that the TAO and NRSR in NRBO have been used to improve a pattern of this nature. The RKO is frequently ranked third among the best algorithms on a few test functions. RKO is among the best algorithms, and the supremacy of this algorithm is clear in the outcomes. Nevertheless, the results demonstrate that the proposed NRBO outperforms RKO when contrasted with its predecessor in every aspect. As illustrated in Figs. 9–11 and other statistical results, the CSA and GWO exhibit effectiveness close to that of the RKO. The results demonstrate that the effectiveness of CSA and GWO is extremely constant, with only a few instances of expedited or slowed convergence significantly. This is due to the algorithm's inability to adjust due to a lack of adaptive mechanisms.

The average performance of GWO is comparable only to that of RKO. As the number of iterations in the method increases, it demonstrates rapid convergence and minimal improvement. Compared to the GWO, the EO and HHO produce quite competitive results. The EO and HHO have a lower convergence rate and less accurate results. Overall, it is clear that the proposed NRBO outperforms all algorithms. The SCA received the lowest scores in the majority of case studies. According to a few studies, the fundamental cause for this could be the limited number of adaptive parameters to adjust intensification and diversification successfully. We are examining the findings in the preceding tables and Figs. 9–11 provides evidence in support of the notion. When using an algorithm to accelerate or slow down the rate of change of solutions to avoid optimal local solutions, monotonous convergence behaviour is unacceptable; therefore, monotonous convergence is avoided wherever possible.

3.7. Parameter tuning of the proposed algorithm

This sub-section is about selecting the appropriate parameter values of NRBO. For this purpose, the Taguchi method is employed (Taguchi, 1986). The control factors of the NRBO algorithm are the maximum number of iterations, the population size, and the parameter (*DF*). The NRBO requires tuning one parameter and has three levels. Thus, the L25 orthogonal array can be selected, fulfilling all the user requirements. The problem F1, with 5 dimensions, was chosen to evaluate the effects of the parameters of NRBO. The Relative Percent Deviation (RPD) employed as the performance indicator was calculated after 30 separate runs for each combination to discover how the outcomes varied (Ruiz and Allahverdi, 2007).

An analysis of variance (ANOVA) was carried out for the chosen problem to determine each variable's relative relevance in terms of its main impacts on the objective function. The Taguchi technique attempts to choose the optimal levels of variables with robustness as its main objective. The RPD readings should be considered as the second criterion if there is no noticeable difference between the S/N ratios (Mosadegh et al., 2012). The RPD values and S/N ratios derived at each set of parameters from the ANOVA test are shown in Table 14.

Table 15 lists the mean S/N ratio and RPD values obtained using the Taguchi design. According to the S/N ratios, DF has a meaningful effect over the objective value at the 5% significance level. Therefore, the parameter level of $DF(3)$ was selected based on RPD values and S/N

Table 6
FRT values obtained for F1–F13 benchmark functions of all dimensions.

Problem	*dim*	NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F1	30	**1.000**	3.000	6.000	8.000	2.000	4.000	7.000	5.000
F2		**1.000**	3.000	6.000	8.000	2.000	4.000	7.000	5.000
F3		**1.000**	3.000	7.000	8.000	2.000	4.000	5.150	5.850
F4		**1.000**	3.000	6.600	8.000	2.000	4.000	6.400	5.000
F5		2.050	3.000	5.800	8.000	4.600	**1.000**	6.950	4.600
F6		**1.650**	2.200	6.300	8.000	2.550	5.000	6.600	3.700
F7		2.950	4.350	6.200	8.000	2.550	**1.450**	5.300	5.200
F8		**1.000**	3.550	6.500	8.000	4.300	2.150	6.500	4.000
F9		**3.825**	**3.825**	5.100	7.950	**3.825**	**3.825**	**3.825**	**3.825**
F10		**2.500**	**2.500**	6.075	8.000	**2.500**	**2.500**	6.925	5.000
F11		**3.850**	**3.850**	4.950	7.950	**3.850**	**3.850**	**3.850**	**3.850**
F12		2.250	**1.550**	6.050	8.000	3.700	5.000	6.950	2.500
F13		**2.700**	3.250	6.450	8.000	3.050	3.150	6.550	2.850
F1	100	**1.000**	3.000	6.950	8.000	2.000	4.000	6.050	5.000
F2		**1.000**	3.000	6.900	8.000	2.000	4.000	6.100	5.000
F3		**1.300**	3.000	7.000	8.000	1.700	4.000	5.000	6.000
F4		**1.000**	3.000	7.000	8.000	2.000	4.000	5.000	6.000
F5		2.200	3.300	5.750	8.000	4.650	**1.000**	7.000	4.100
F6		**1.300**	3.050	6.000	8.000	4.300	1.750	7.000	4.600
F7		2.550	4.350	6.900	8.000	2.450	**1.500**	4.550	5.700
F8		**1.000**	3.550	6.200	7.900	4.800	2.000	6.900	3.650
F9		**3.500**	**3.500**	7.000	8.000	**3.500**	**3.500**	**3.500**	**3.500**
F10		**2.625**	**2.625**	7.000	8.000	**2.625**	**2.625**	4.900	5.600
F11		**3.925**	**3.925**	4.450	8.000	**3.925**	**3.925**	**3.925**	**3.925**
F12		**1.150**	3.050	6.050	8.000	4.300	1.850	6.950	4.650
F13		**1.100**	4.200	5.600	8.000	2.050	3.050	7.000	5.000
F1	500	**1.000**	3.000	7.000	8.000	2.000	4.000	6.000	5.000
F2		**1.000**	3.000	7.000	8.000	2.000	4.000	6.000	5.000
F3		2.000	3.000	6.000	7.000	**1.000**	8.000	4.000	5.000
F4		**1.000**	3.000	5.450	8.000	2.000	6.500	4.000	6.050
F5		**1.400**	2.850	5.950	8.000	1.950	3.800	7.000	5.050
F6		3.250	2.000	6.000	8.000	**1.000**	3.750	7.000	5.000
F7		**1.950**	3.350	7.000	8.000	2.100	4.950	3.400	5.250
F8		**1.450**	3.250	6.100	7.950	3.950	1.550	6.950	4.800
F9		**3.500**	**3.500**	7.000	8.000	**3.500**	**3.500**	**3.500**	**3.500**
F10		**2.350**	**2.350**	7.000	8.000	**2.350**	4.000	4.200	5.750
F11		**3.450**	**3.450**	6.950	8.000	**3.450**	**3.450**	**3.450**	3.800
F12		3.150	2.300	6.000	8.000	**1.000**	3.550	7.000	5.000
F13		**1.050**	4.850	4.150	8.000	2.050	2.900	7.000	6.000
F1	1000	**1.000**	3.000	7.000	8.000	2.000	4.000	6.000	5.000
F2		**1.000**	3.000	7.000	8.000	2.000	4.000	6.000	5.000
F3		2.050	2.950	6.000	7.050	**1.000**	7.950	4.000	5.000
F4		**1.000**	3.000	5.800	8.000	2.000	6.150	4.000	6.050
F5		**1.200**	3.350	5.300	8.000	1.850	3.600	7.000	5.700
F6		3.750	2.300	6.000	8.000	**1.000**	2.950	7.000	5.000
F7		**1.950**	3.450	7.000	8.000	2.300	4.400	3.200	5.700
F8		1.650	3.500	6.000	8.000	3.500	**1.350**	7.000	5.000
F9		**3.500**	**3.500**	7.000	8.000	**3.500**	**3.500**	**3.500**	**3.500**
F10		**2.300**	**2.300**	7.000	8.000	**2.300**	3.925	4.650	5.525
F11		**3.025**	**3.025**	7.000	8.000	**3.025**	**3.025**	**3.025**	5.875
F12		**1.000**	3.000	6.000	8.000	3.050	2.950	7.000	5.000
F13		**1.000**	4.200	5.450	8.000	2.250	2.750	6.750	5.600
Average		**1.950**	3.156	6.269	7.958	2.641	3.570	5.606	4.851
Overall		**1**	3	7	8	2	4	6	5

ratios in the second and third columns of Table 15, respectively. In other words, the parameter DF in NRBO was set to 0.6, respectively. The traditional way to select the algorithm parameters is provided in the supplementary material.

4. CEC2020 real-world engineering optimization problems

The performance of the suggested NRBO on CEC2020 real-world constrained optimization problems is defined in this section using 12 challenging problems (Kumar et al., 2020; Gurrola-Ramos et al., 2020). Reactor network design (RW1), process design problem (RW2), mechanical engineering problems (RW3–RW10), and power system problems (RW11-RW12) are among the challenging problems considered from the CEC2020 test suite. When dealing with any of the problems mentioned above, the decision vectors range from 2 to 126, and the number of inequality constraints and equality constraints varies from 0 to 116 and 0 to 11. The details of selected functions are listed in Table 16. All the selected algorithms are executed 30 times individually to get fair results, and statistically, all algorithms are compared and contrasted. For all algorithms, the population size is selected as 40, and the maximum number of iterations is selected as 1000.

4.1. Reactor network design problem (RW1)

The solution to the reactor network design is used to create a chain of two periodically Stirred Tank Reactors in tandem. The primary objective of this challenge is to maximize the concentration of the substance. Eq. (38) is the formulation of the problem.

$$\text{Maximize}, f1(\bar{x}) = x_4 \tag{38}$$

Subjected to the following constraints,

Table 7
Outcomes of F14–F23 benchmark test functions with different dimensions.

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F14	Min	**0.9980**	**0.9980**	**0.9980**	**0.9980**	**0.9980**	**0.9980**	**0.9980**	**0.9980**
	Mean	0.9980	2.3214	4.1792	1.7918	2.1885	1.0477	0.9980	0.9980
	STD	0.00E+00	3.01E+00	4.25E+00	9.97E-01	9.97E-01	2.22E-01	1.39E-09	5.09E-17
F15	Min	**3.07E-04**	**3.07E-04**	**3.07E-04**	3.13E-04	**3.07E-04**	3.08E-04	3.23E-04	**3.07E-04**
	Mean	3.07E-04	4.63E-04	6.37E-03	9.21E-04	6.74E-04	4.51E-04	4.20E-04	3.41E-03
	STD	2.41E-19	3.87E-04	9.40E-03	3.96E-04	4.60E-04	4.06E-04	7.10E-05	7.31E-03
F16	Min	**−1.0316**	**−1.0316**	**−1.0316**	**−1.0316**	**−1.0316**	**−1.0316**	**−1.0316**	**−1.0316**
	Mean	−1.0316	−1.0316	−1.0316	−1.0316	−1.0316	−1.0316	−1.0316	−1.0316
	STD	2.10E-16	2.28E-16	1.63E-08	4.06E-05	1.95E-13	8.26E-12	3.51E-07	1.97E-16
F17	Min	**0.3979**	**0.3979**	**0.3979**	0.3980	**0.3979**	**0.3979**	**0.3979**	**0.3979**
	Mean	0.3979	0.3979	0.3979	0.3988	0.3979	0.3979	0.3979	0.3979
	STD	0.00E+00	0.00E+00	7.75E-07	8.69E-04	9.01E-13	9.88E-07	2.08E-05	0.00E+00
F18	Min	**3.0000**	**3.0000**	**3.0000**	**3.0000**	**3.0000**	**3.0000**	**3.0000**	**3.0000**
	Mean	3.0000	3.0000	3.0000	3.0000	3.0000	3.0000	3.0001	3.0000
	STD	4.08E-16	5.58E-16	1.01E-05	3.49E-05	1.22E-13	1.82E-08	6.50E-05	2.06E-15
F19	Min	**−3.8628**	**−3.8628**	**−3.8628**	−3.8613	**−3.8628**	**−3.8628**	**−3.8628**	**−3.8628**
	Mean	−3.8628	−3.8628	−3.8615	−3.8550	−3.8628	−3.8621	−3.8626	−3.8628
	STD	2.28E-15	2.28E-15	2.56E-03	2.11E-03	1.86E-09	1.13E-03	7.25E-05	2.09E-15
F20	Min	**−3.3220**	**−3.3220**	**−3.3220**	−3.1258	**−3.3220**	−3.3054	−3.3206	**−3.3220**
	Mean	−3.2923	−3.2804	−3.2678	−2.9963	−3.2685	−3.1392	−3.3166	−3.2744
	STD	5.28E-02	5.82E-02	7.00E-02	1.76E-01	6.07E-02	1.12E-01	2.22E-03	5.98E-02
F21	Min	**−10.1532**	**−10.1532**	**−10.1531**	−8.6868	**−10.1532**	−9.7477	−10.1474	**−10.1532**
	Mean	−7.4830	−7.6042	−9.9001	−3.1277	−10.1532	−5.2893	−10.1072	−8.5071
	STD	1.12E-09	2.62E+00	1.13E+00	2.36E+00	2.79E+00	1.05E+00	4.50E-02	2.63E+00
F22	Min	**−10.4029**	**−10.4029**	−10.4028	−7.7861	**−10.4029**	−10.3876	−10.3974	**−10.4029**
	Mean	−7.0429	−8.5426	−10.1366	−3.8339	−10.4029	−5.6047	−10.3466	−9.4431
	STD	2.42E+00	2.60E+00	1.19E+00	2.22E+00	4.28E-10	1.59E+00	2.68E-02	3.20E+00
F23	Min	**−10.5364**	**−10.5364**	−10.5363	−8.7478	**−10.5364**	−10.3752	−10.5236	**−10.5364**
	Mean	−6.6221	−9.1844	−10.1301	−4.9349	−10.2660	−5.3898	−10.4717	−10.2014
	STD	3.42E+00	2.40E+00	1.81E+00	1.96E+00	1.21E+00	1.17E+00	3.60E-02	1.50E+00

Table 8
FRT values obtained for F14–F23 benchmark functions.

Problem	NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F14	**1.830**	3.330	6.300	6.950	5.600	4.600	5.500	1.900
F15	**1.600**	2.200	5.050	6.950	4.400	5.200	5.700	4.900
F16	2.025	**1.800**	6.000	8.000	4.350	4.650	7.000	2.175
F17	**2.150**	**2.150**	5.700	8.000	4.100	4.800	6.950	**2.150**
F18	**1.780**	1.875	6.600	6.700	4.200	4.800	7.700	2.350
F19	**1.775**	**1.775**	5.950	7.900	4.000	6.150	6.000	2.450
F20	**2.400**	2.775	4.750	7.750	4.100	6.800	4.150	3.275
F21	**2.700**	3.575	3.750	7.750	3.680	6.950	4.650	2.950
F22	**2.250**	3.025	4.000	7.400	2.950	6.850	4.950	4.580
F23	4.925	2.775	4.300	7.150	3.250	6.800	5.000	**1.800**
Average	**2.344**	2.528	5.240	7.455	4.063	5.760	5.760	2.853
Overall	**1**	2	5	8	4	6	7	3

$$h_1(\bar{x}) = k_1 x_5 x_2 + x_1 - 1 = 0$$

$$h_2(\bar{x}) = k_3 x_5 x_3 + x_3 + x_1 - 1 = 0$$

$$h_3(\bar{x}) = k_2 x_6 x_2 - x_1 + x_2 = 0$$

$$h_4(\bar{x}) = k_4 x_6 x_4 + x_2 - x_1 + x_4 - x_3 = 0$$

$$g_1(\bar{x}) = x_5^{0.5} + x_6^{0.5} \leq 4$$

The limits are,

$$0 \leq x_4, x_3, x_2, x_1 \leq 1,$$
$$0.00001 \leq x_6, x_5 \leq 16$$

where, $k_3 = 0.0391908$, $k_4 = 0.9k_3$, $k_1 = 0.09755988$, and $k_2 = 0.99k_1$.

The findings of the proposed NRBO and other algorithms are listed in Table 17. Table 17 shows that the NRBO outperformed all other approaches and had the maximum substance concentration. Table 18 includes statistics such as Min, Mean, STD, and Run-Time (RT). As a result, it is decided that the proposed NRBO is more reliable for the reactor network design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO also comes out on top when it comes to tackling the reactor network design problem. The performance of EO and GBO is almost similar to the proposed algorithm. But, with respect to RT, NRBO is a better algorithm.

4.2. Process design problem (RW2)

Challenges in chemical engineering (process design) can be characterized as a mixed-integer nonlinear constrained problem with constrained variables. Essentially, it is a minimization issue expressed in Eq. (39).

$$f2(\bar{x}) = 5.357854x_1^2 + 40792.141 - 37.29329x_4 + \quad 0.835689x_4x_3 \tag{39}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = -92 + a_3x_4x_2 + a_1 + a_2x_4x_3 - a_4x_4x_3 \leq 0$$

$$g_2(\bar{x}) = -110 + a_7x_4x_2 + a_5 + a_6x_5x_3 + a_8x_1^2 \leq 0$$

$$g_3(\bar{x}) = a_9 + a_{11}x_4x_1 + a_{10}x_4x_3 - 25 + a_{12}x_1x_2 \leq 0$$

Table 9
Results of F24–F52 benchmark functions with 10 problem dimensions.

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F24	Min	**123.76**	160.20	1.15E+04	5.11E+08	789.44	2.16E+05	2.73E+06	1385.08
	Mean	2860.62	2811.45	3.82E+06	8.84E+08	3266.34	8.09E+05	6.24E+06	5492.22
	STD	3516.44	2997.32	9.68E+06	2.77E+08	1395.05	9.37E+05	2.80E+06	3281.87
F25	Min	**300.00**	**300.00**	321.10	1024.90	**300.00**	302.02	303.26	**300.00**
	Mean	300.00	300.00	2065.60	1498.78	300.00	304.03	310.46	300.57
	STD	0.00	0.00	2829.46	484.30	0.00	1.89	5.65	1.47
F26	Min	403.65	**400.00**	407.43	438.86	400.04	403.27	405.15	401.07
	Mean	404.88	400.17	440.52	460.57	400.42	414.84	407.04	404.43
	STD	0.70	0.20	44.69	14.86	0.47	20.36	1.10	1.59
F27	Min	**503.98**	508.95	507.38	542.32	510.94	529.13	521.50	512.93
	Mean	513.42	518.47	516.61	553.27	526.86	544.51	529.10	535.41
	STD	7.51	8.61	11.45	6.68	9.26	12.71	4.30	14.89
F28	Min	**600.00**	**600.00**	600.09	611.03	606.56	611.14	601.86	605.46
	Mean	600.00	600.55	600.47	617.80	615.67	636.65	604.68	620.29
	STD	0.00	1.00	0.81	3.77	5.29	13.13	1.60	15.61
F29	Min	719.24	720.38	**707.06**	766.82	733.11	748.72	734.88	737.28
	Mean	724.35	735.94	727.66	777.51	750.91	787.05	740.37	757.69
	STD	5.51	9.13	12.25	6.87	17.65	27.72	4.27	25.89
F30	Min	**801.99**	808.95	804.00	835.45	821.89	808.18	808.89	807.96
	Mean	808.76	825.50	813.81	845.77	827.26	826.73	819.59	828.69
	STD	5.13	7.11	5.16	7.24	5.23	10.46	6.33	14.68
F31	Min	**900.00**	900.00	900.54	968.67	900.54	1245.61	901.81	917.73
	Mean	900.05	902.10	922.20	1024.58	963.62	1441.19	908.42	1061.55
	STD	0.14	3.39	42.33	55.01	51.25	183.59	5.65	128.19
F32	Min	**1027.02**	1362.53	1282.95	2351.73	1382.47	1596.34	1668.04	1390.15
	Mean	1532.93	1836.55	1600.53	2495.99	1618.94	2209.06	1856.34	1952.63
	STD	383.23	262.52	175.27	146.64	213.14	350.65	155.17	383.25
F33	Min	**1101.32**	1101.99	1111.69	1180.57	1112.95	1120.82	1115.96	1118.43
	Mean	1105.26	1113.23	1172.45	1241.26	1123.55	1181.83	1141.60	1172.28
	STD	4.82	11.33	63.58	60.19	4.81	58.43	12.98	45.53
F34	Min	6530.18	2750.76	1.01E+04	9.03E+06	1836.53	9998.19	60105.82	**1744.36**
	Mean	2.00E+04	1.79E+04	3.53E+05	1.99E+07	1.15E+04	4.19E+06	1.59E+05	2.05E+04
	STD	1.48E+04	1.26E+04	4.71E+05	9.40E+06	1.13E+04	3.85E+06	6.73E+04	21583.48
F35	Min	**1314.80**	1468.74	4176.06	4119.96	3410.69	3906.43	2859.22	1913.53
	Mean	6128.84	1831.02	1.35E+04	4.97E+04	7417.92	1.02E+04	5364.73	3989.82
	STD	7997.01	259.74	6299.23	4.35E+04	2109.06	7096.44	1877.11	3094.64
F36	Min	1453.78	**1432.97**	1454.71	1502.65	1431.48	1481.71	1489.76	1433.52
	Mean	1483.35	1473.82	2389.34	1976.89	1549.17	1581.37	1598.78	1485.81
	STD	31.09	29.78	1648.08	1117.28	204.78	76.61	111.19	39.43
F37	Min	1533.09	1507.15	1592.14	1952.73	**1504.09**	1836.16	1677.69	1559.62
	Mean	1638.22	1598.25	5314.45	2558.93	1543.78	6302.77	2178.17	1803.29
	STD	77.74	63.83	3457.07	799.43	35.55	2805.07	372.94	274.38
F38	Min	**1600.86**	1639.75	1604.60	1695.14	1607.62	1620.45	1649.37	1604.30
	Mean	1677.19	1725.20	1719.45	1761.18	1753.40	1869.65	1687.35	1816.69
	STD	58.82	56.76	133.31	64.24	94.78	226.07	21.57	101.52
F39	Min	**1716.94**	1719.84	1740.21	1765.39	1738.29	1740.32	1750.34	1737.49
	Mean	1743.72	1759.32	1772.82	1779.75	1758.55	1752.59	1765.68	1760.53
	STD	18.91	41.26	39.47	11.76	16.94	8.27	12.59	26.82
F40	Min	3070.77	**2039.47**	3615.76	3.44E+04	3270.98	2919.93	8724.36	7285.69
	Mean	1.47E+04	8793.41	2.69E+04	2.15E+05	1.86E+04	1.56E+04	1.46E+04	2.17E+04
	STD	1.36E+04	7712.51	1.42E+04	1.83E+05	9928.98	1.34E+04	7862.53	1.08E+04
F41	Min	**1906.51**	**1906.51**	1968.62	2159.60	1910.49	2540.58	1978.95	1911.18
	Mean	1975.62	1954.70	9476.20	5127.85	5191.35	6384.96	2558.77	1960.45
	STD	128.07	51.21	7041.39	4419.52	5218.63	8530.77	450.22	41.85
F42	Min	**2005.60**	2012.49	2021.46	2053.50	2042.27	2081.68	2047.55	2024.36
	Mean	2022.49	2077.74	2092.37	2109.93	2089.96	2180.51	2072.45	2147.94
	STD	26.65	68.34	64.25	38.38	49.25	73.98	17.27	87.98
F43	Min	**2200.00**	**2200.00**	2201.35	2206.33	**2200.00**	2202.89	2202.93	2315.01
	Mean	2299.03	2235.56	2308.76	2225.79	2238.13	2311.03	2203.78	2332.62
	STD	35.10	56.70	39.41	44.33	60.59	73.21	0.99	17.73
F44	Min	**2242.66**	2243.39	2301.48	2361.64	2303.18	2310.02	2300.00	2257.33
	Mean	2293.61	2292.97	2311.30	2385.52	2306.88	2316.79	2393.77	2298.70
	STD	0.34	20.92	11.27	21.64	3.94	5.44	29.13	14.61
F45	Min	2607.82	2614.55	2608.48	2648.68	2610.02	2622.34	**2366.35**	2618.10
	Mean	2616.52	2627.32	2620.75	2657.08	2625.77	2647.03	2601.07	2639.21
	STD	5.32	12.07	9.41	4.44	9.15	22.38	82.53	30.00
F46	Min	**2500.00**	**2500.00**	2735.42	2617.00	2737.29	2501.20	2517.72	**2500.00**
	Mean	2696.65	2706.51	2746.63	2769.89	2751.94	2800.98	2719.77	2716.55
	STD	103.78	109.57	11.21	55.01	10.41	111.23	82.28	115.28
F47	Min	**2897.74**	2897.89	2902.71	2953.03	2899.61	2898.31	2898.75	2898.00
	Mean	2931.41	2930.75	2932.11	2971.79	2927.50	2941.53	2907.71	2930.69
	STD	23.09	40.54	17.61	23.46	24.03	15.26	14.79	40.44
F48	Min	2815.83	2800.00	2900.20	3034.39	**2600.04**	2829.14	2912.93	2800.00
	Mean	2931.73	2947.69	3045.62	3091.18	2955.61	3562.67	2921.79	3313.31

(*continued on next page*)

Table 9 (*continued*)

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
	STD	54.33	111.83	287.91	26.26	248.26	595.10	8.91	466.02
F49	Min	**3089.52**	**3089.52**	3091.11	3101.13	3090.75	3099.54	3097.01	**3089.52**
	Mean	3091.65	3108.06	3098.06	3102.53	3093.40	3152.14	3104.37	3117.11
	STD	1.49	28.26	7.53	1.27	2.27	39.22	5.89	26.97
F50	Min	**3100.00**	**3100.00**	3217.15	3242.93	**3100.00**	3217.84	3136.30	**3100.00**
	Mean	3312.66	3244.15	3387.89	3327.60	3339.51	3451.79	3169.70	3387.96
	STD	147.17	143.60	100.20	88.99	120.06	127.29	47.95	174.97
F51	Min	**3135.25**	3158.12	3164.18	3171.00	3149.85	3195.07	3152.39	3175.96
	Mean	3185.43	3226.69	3211.25	3241.31	3213.56	3351.98	3178.66	3297.78
	STD	38.84	39.58	45.04	30.08	41.11	107.35	23.70	89.57
F52	Min	1.08E+04	4008.18	9192.12	1.54E+05	**3779.54**	2.31E+04	1.07E+04	6167.03
	Mean	3.83E+05	2.60E+05	1.03E+06	8.66E+05	4742.44	1.99E+06	8.54E+05	2.38E+06
	STD	4.86E+05	4.46E+05	1.18E+06	4.30E+05	1.37E+03	2.31E+06	1.39E+06	3.33E+06

Table 10
FRT values obtained for F24–F52 benchmark functions.

Problem	NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F24	**2.1**	**2.1**	5.4	8	2.5	5.7	6.9	3.3
F25	2.4	**1**	7.4	7.6	2.6	4.9	6	4.1
F26	3.9	**1.3**	6.8	7.6	1.7	5.6	5.5	3.6
F27	**1.8**	2.8	2.7	7.6	4.1	6.4	4.8	5.8
F28	**1**	2.5	2.6	6.3	6	7.4	4.2	6
F29	**1.6**	3.1	2	7.1	5	7.3	4.3	5.6
F30	**1.4**	5	2.5	7.8	5.3	5.1	3.6	5.3
F31	**1.3**	2.1	3.6	6.1	5.1	8	3.5	6.3
F32	**2.4**	4.4	2.7	7.6	3	6.4	4.4	5.1
F33	**1.2**	2.2	5.3	7.2	3.5	5.9	4.7	6
F34	3	3	5.5	8	1.9	6.4	5.5	**2.7**
F35	2.8	**1.5**	6.5	7.6	5.2	5.3	4	3.1
F36	3.2	**2.5**	5.3	6.6	2.9	6.1	6.4	3
F37	2.7	2.4	6.6	6.2	**1.5**	7.1	5.4	4.1
F38	3	4.6	3.6	5	4.8	6.2	**2.7**	6.1
F39	**2.5**	3.7	5.1	6.9	4.1	3.9	5.8	4
F40	3.4	**2.4**	5.4	7.8	4.6	3.7	3.8	4.9
F41	**2.3**	2.4	6.4	6.3	4	6.5	5.2	2.9
F42	**1.5**	3.6	4.3	5.1	4.6	7	4	5.9
F43	4.6	**2.4**	5.8	4.3	2.9	6.7	2.7	6.6
F44	**1.6**	2.7	4.8	8	4.9	6.5	4.3	3.2
F45	**2**	3.6	3.2	7.5	4.2	6.6	4.4	4.5
F46	**2.7**	4	3.4	6.3	3.8	7.3	4.1	4.4
F47	3.9	3.5	4.9	7.8	4.3	4.7	**3.3**	3.6
F48	**3.1**	**3.1**	4.1	6	4.1	7	**3.1**	5.5
F49	**1.6**	4.2	3.7	5.7	2.6	7.1	5.7	5.4
F50	3.6	2.9	5.4	5.1	4.9	6.8	**2.1**	5.2
F51	2.7	4.9	3.4	5.5	3.9	6.9	**2.3**	6.4
F52	4.1	**2.6**	5.5	6	1.2	6	4.8	5.8
Average	**2.53**	2.98	4.62	6.71	3.77	6.22	4.40	4.77
Overall	**1**	2	5	8	3	7	4	6

The limits are,

$$27 \le x_3, x_1, x_2 \le 45$$
$$x_4 \in \{78, 79, \ldots, 102\}$$
$$x_5 \in \{33, 34, \ldots., 45\}$$

The values of a_1-a_{12} are as follows. $a_1 = 85.334407$, $a_2 = 0.0056858$, $a_3 = 0.0006262$, $a_4 = 0.0022053$, $a_5 = 80.51249$, $a_6 = 0.0071317$, $a_7 = 0.0029955$, $a_8 = 0.0021813$, $a_9 = 9.300961$, $a_{10} = 0.0047026$, $a_{11} = 0.0012547$, and $a_{12} = 0.0019085$.

The findings of the proposed NRBO and other algorithms are listed in Table 19. Table 19 shows that the NRBO and all other algorithms performed well, and all algorithms computed the minimum objective value. Table 20 includes statistical information such as Min, Mean, STD, and R). As a result, it is decided that the proposed NRBO and all other algorithms are more reliable for the process design problem. Furthermore, FRT values derived by all algorithms are presented. All algorithms are equally ranked based on FRT values. But, with respect to RT, SCA is a better algorithm. However, the proposed NRBO also equally performs with minimal RT than GBO, RKO, HHO, CSA, and EO.

4.3. Weight minimization of a speed reducer problem (RW3)

A small aircraft engine speed reduction is being designed, and this study is a part of it. The following is the structure of the optimization problem that resulted.

$$\begin{aligned} f3(\bar{x}) \quad &= 0.7854x_2^2x_1\left(14.9334x_3 - 43.0934 + 3.3333x_3^2\right) \\ &+0.7854\left(x_5x_7^2 + x_4x_6^2\right) - 1.508x_1\left(x_7^2 + x_6^2\right) + 7.477\left(x_7^3 + x_6^3\right) \end{aligned} \tag{40}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = -x_1x_2^2x_3 + 27 \le 0$$

$$g_2(\bar{x}) = -x_1x_2^2x_3^2 + 397.5 \le 0$$

$$g_3(\bar{x}) = -x_2x_6^4x_3x_4^{-3} + 1.93 \le 0$$

$$g_4(\bar{x}) = -x_2x_7^4x_3x_5^{-3} + 1.93 \le 0$$

$$g_5(\bar{x}) = 10x_6^{-3}\sqrt{16.91 \times 10^6 + \left(745x_4x_2^{-1}x_3^{-1}\right)^2} - 1100 \quad \le 0$$

$$g_6(\bar{x}) = 10x_7^{-3}\sqrt{157.5 \times 10^6 + \left(745x_5x_2^{-1}x_3^{-1}\right)^2} - 850 \quad \le 0$$

$$g_7(\bar{x}) = x_2x_3 - 40 \le 0$$

$$g_8(\bar{x}) = -x_1x_2^{-1} + 5 \le 0$$

$$g_9(\bar{x}) = x_1x_2^{-1} - 12 \le 0$$

$$g_{10}(\bar{x}) = 1.5x_6 - x_4 + 1.9 \le 0$$

$$g_{11}(\bar{x}) = 1.1x_7 - x_5 + 1.9 \le 0$$

The limits are,

$$0.7 \le x_2 \le 0.8, 17 \le x_3 \le 28, 2.6 \le x_1 \le 3.6$$

$$5 \le x_7 \le 5.5, 7.3 \le x_5, x_4 \le 8.3, 2.9 \le x_6 \le 3.9$$

The findings of the proposed NRBO and other algorithms are listed in Table 21. Table 21 shows that the NRBO outperformed all other approaches except EO and explored minimum weight. Table 22 includes statistics such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the weight minimization of a speed reducer problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO and EO come out on top with a 1.8 FRT value when it comes to tackling the weight minimization of a speed reducer problem. The performance of EO and GBO is similar to the proposed algorithm. But, with respect to RT, NRBO is a better algorithm.

4.4. Tension/compression spring design (Case1) problem (RW4)

The prime objective of this challenge is to reduce the overall weight

Table 11
Results of F53–F64 CEC2022 benchmark functions with 20 dimensions.

Problem		NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F53	Min	430.18	300.60	7369.46	10583.38	**300.00**	2031.32	9942.28	12977.67
	Mean	745.28	317.24	12948.91	14277.99	300.01	5186.86	23199.85	21043.67
	STD	290.89	30.75	5066.55	2297.12	0.00	3352.76	7534.57	9762.55
F54	Min	407.61	**406.33**	446.85	630.11	403.36	450.14	447.89	455.64
	Mean	454.54	445.41	492.88	726.38	450.97	489.40	475.17	557.86
	STD	18.66	19.64	49.72	77.86	19.81	30.87	17.66	58.52
F55	Min	**600.00**	601.32	601.49	628.82	620.26	636.71	613.31	632.44
	Mean	600.01	606.96	604.64	643.63	632.19	657.06	622.04	662.57
	STD	0.01	5.23	2.67	8.46	9.16	9.63	4.83	18.84
F56	Min	**818.91**	843.78	827.00	922.39	847.76	854.55	870.58	855.25
	Mean	837.75	873.28	860.39	948.20	875.16	881.82	888.80	918.28
	STD	13.40	19.53	33.14	14.09	10.19	15.62	11.66	34.45
F57	Min	**900.00**	947.41	913.51	1598.61	1270.81	2288.37	1101.21	1749.21
	Mean	903.00	1363.47	1078.04	2327.50	1768.74	2817.72	1457.81	3167.08
	STD	3.58	387.60	190.26	649.13	252.01	243.67	311.71	1090.53
F58	Min	1971.04	**1823.38**	3245.82	2.01E+07	1934.88	32456.73	2.16E+05	2.71E+03
	Mean	5466.35	7007.17	3.12E+06	1.19E+08	3941.00	1.09E+05	6.98E+05	3.85E+05
	STD	3558.93	6029.80	1.22E+07	8.52E+07	2845.19	4.50E+04	4.87E+05	1.05E+06
F59	Min	**2025.35**	2029.14	2026.99	2101.27	2045.58	2085.63	2066.15	2099.47
	Mean	2047.05	2096.10	2082.58	2136.60	2093.77	2198.98	2093.06	2207.23
	STD	17.75	31.72	41.86	20.47	21.02	72.32	14.12	77.83
F60	Min	2222.86	**2220.93**	2225.97	2243.11	2223.92	2231.69	2230.22	2230.33
	Mean	2229.65	2235.52	2277.93	2269.19	2233.68	2284.96	2237.73	2284.47
	STD	4.45	26.35	59.42	22.15	27.32	87.22	3.70	69.45
F61	Min	**2480.78**	2480.78	2481.55	2535.51	2480.79	2484.89	2486.47	2489.85
	Mean	2480.78	2480.78	2510.87	2590.06	2480.84	2501.08	2502.78	2547.49
	STD	0.00	0.00	26.04	33.15	0.05	19.74	16.09	34.00
F62	Min	**2500.36**	2500.53	2500.51	2522.35	2500.68	2501.46	2500.55	2500.72
	Mean	2969.46	3583.80	3266.03	2600.33	2566.59	3885.58	2814.43	3844.18
	STD	600.13	992.10	685.02	118.52	202.52	800.77	886.61	1436.66
F63	Min	2900.00	**2600.00**	2989.46	3680.02	2600.30	2657.98	3148.72	2821.18
	Mean	2920.00	2895.00	3395.97	4687.00	2910.28	3147.10	3313.86	3637.14
	STD	41.04	75.92	428.07	536.73	85.15	620.78	105.30	886.59
F64	Min	**2935.19**	2955.06	2945.20	3013.73	2949.81	2997.72	2960.18	2961.74
	Mean	2948.08	3004.60	2959.77	3056.07	2964.05	3120.62	3044.97	3031.57
	STD	7.80	51.60	11.58	26.81	11.06	128.16	74.87	54.59

Table 12
FRT values obtained for F53–F64 CEC2022 benchmark functions.

Problem	NRBO	GBO	GWO	SCA	RKO	HHO	CSA	EO
F53	3.000	2.000	5.350	6.000	**1.000**	4.200	7.350	7.100
F54	2.600	**1.650**	4.850	7.900	2.800	5.100	4.450	6.650
F55	**1.000**	2.600	2.500	6.250	5.050	7.300	4.000	7.300
F56	**1.400**	3.800	2.800	7.800	3.850	4.600	5.200	6.550
F57	**1.000**	3.450	2.450	5.950	4.900	7.300	3.750	7.200
F58	2.150	2.300	4.500	8.000	**1.900**	5.100	6.750	5.300
F59	**1.200**	4.000	3.350	5.800	3.600	7.100	3.650	7.300
F60	2.650	2.950	4.850	6.750	**2.550**	5.500	4.400	6.350
F61	1.850	**1.150**	5.300	7.800	3.000	4.850	5.250	6.800
F62	3.700	5.100	4.850	4.500	3.050	6.350	**2.850**	5.600
F63	2.250	**1.400**	5.800	7.800	2.900	4.200	5.750	5.900
F64	**1.150**	4.400	2.400	6.750	2.900	7.000	5.950	5.450
Average	**1.996**	2.900	4.083	6.775	3.125	5.717	4.946	6.458
Overall	**1.000**	2.000	4.000	8.000	3.000	6.000	5.000	7.000

of a tension or compression spring to the lowest possible value. In this problem, there are four constraints, and to compute the weight, three parameters must be taken into consideration: the diameter of the wire (x_1), the mean diameter of the coil (x_2), and the number of active coils (x_3). Eq. (41) is a definition of the problem.

$$f4(\bar{x}) = x_1^2 x_2 (2 + x_3) \tag{41}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = 1 - \frac{x_2^3 x_3}{71785 x_1^4} \leq 0$$

$$g_2(\bar{x}) = \frac{4x_2^2 - x_1 x_2}{12566(x_2 x_1^3 - x_1^4)} + \frac{1}{5108 x_1^2} - 1 \leq 0$$

$$g_3(\bar{x}) = 1 - \frac{140.45 x_1}{x_2^2 x_3} \leq 0$$

$$g_4(\bar{x}) = \frac{x_1 + x_2}{1.5} - 1 \leq 0$$

The limits are,

$$0.05 \leq x_1 \leq 2.00$$
$$0.25 \leq x_2 \leq 1.30$$
$$2.00 \leq x_3 \leq 15.0$$

The findings of the proposed NRBO and other algorithms are listed in Table 23. Table 23 shows that the NRBO outperformed all other approaches and explored the minimum tension or compression spring

Table 13
Performance comparison of NRBO with CEC2022 winners.

Function		NRBO	EA4eig	NL-SHADE-LBC	NL-SHADE-RSP-MID	S-LSHADE-DP
F53	Mean	300	300	300	300.124	300
	STD	0.00E+00	1.34E-09	0.00E+00	0.00E+00	0.00E+00
F54	Mean	400.235	401.4618	400.133	400	400
	STD	4.27E-02	1.95395	7.16E-01	0.00E+00	0.00E+00
F55	Mean	600	600	600	600	600
	STD	0.00E+00	9.98E-10	0.00E+00	0.00E+00	0.00E+00
F56	Mean	800	801.2603	801.3	800	804.72
	STD	9.42E-04	1.04298	7.78E-01	4.55E+00	1.33E+00
F57	Mean	900	900	900	901.69	900
	STD	0.00E+00	1.62E-09	0.00E+00	3.88E+00	0.00E+00
F58	Mean	1800.004	1800.017	1800.124	1800.167	1800.26
	STD	5.58E-02	3.57E-02	1.25E-01	2.45E−01	1.27E-01
F59	Mean	2000	2000	2000	2000	2000
	STD	0.00E+00	1.17E-09	0.00E+00	0.00E+00	0.00E+00
F60	Mean	2200.071	2200.071	2200.046	2200.238	2200.189
	STD	9.58E-01	6.81E-02	3.80E-02	2.78E−01	2.73E-01
F61	Mean	2380	2485.502	2529	2529	2491
	STD	0.00E+00	5.78E-14	5.68E-14	0.00E+00	8.54E+01
F62	Mean	2442.587	2500.157	2500	2404.53	2400.013
	STD	4.28E-02	3.60E-02	2.95E-02	1.83E+01	6.35E-02
F63	Mean	2600	2600	2600	2600	2600
	STD	0.00E+00	1.06E-09	0.00E+00	6.50E−10	0.00E+00
F64	Mean	2805.234	2847.378	2865	2865	2862
	STD	3.02E-02	3.90145	4.04E-01	9.72E−01	1.77E+00
Average FRT Values		**3.258**	5.0	5.83	5.78	4.16
Rank		**1**	3	5	4	2

weight. Table 23 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the tension/compression spring design (case 1) problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO comes out on top with an FRT value of 1 when it comes to tackling the weight minimization of a tension/compression spring design (case 1) problem. The Min values explored by all algorithms except SCA are the same; however, based on Mean and STD values, the NRBO is better than all algorithms. The performance of RKO and GBO is similar to the proposed algorithm. But, with respect to RT, NRBO is a better algorithm.

4.5. Pressure vessel design problem (RW5)

The primary objective of this challenge is to reduce the price of welding, the amount of material used, and the shape of the vessel. It is necessary to satisfy four constraints, and four parameters are required to compute the fitness function: the shell thickness (x_1), the head thickness (x_2), the inner radius (x_3), and the total length of the vessel without the head (x_4). Eq. (42) represents the fitness function.

$$f5(\bar{x}) = 1.7781 z_2 x_3^2 + 0.6224 z_1 x_3 x_4 + 3.1661 z_1^2 x_4 \quad + 19.84 z_1^2 x_3 \tag{42}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = 0.00954 x_3 \le z_2$$

$$g_2(\bar{x}) = 0.0193 x_3 \le z_1$$

$$g_3(\bar{x}) = x_4 \le 240$$

$$g_4(\bar{x}) = -\pi x_3^2 x_4 - \frac{4}{3}\pi x_3^3 \le -1296000$$

Where:

$$z_1 = 0.0625 x_1,$$
$$z_2 = 0.0625 x_2$$

The limits are,

$$10 \le x_4, x_3 \le 200$$
$$1 \le x_2, x_1 \le 99 (\text{integer variables})$$

The findings of the proposed NRBO and other algorithms are listed in Table 24. Table 24 shows that the NRBO outperformed all other approaches and explored the minimum cost of welding, material weight, and proper shape. Table 24 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the pressure vessel design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO comes out on top with an FRT value of 1 when it comes to tackling the pressure vessel design problem. The Min value explored by NRBO, GBO, and RKO are the same; however, based on Mean and STD values, the NRBO is better than all algorithms. The performance of RKO, GBO, and EO is similar to the proposed algorithm. But, with respect to the RT and metrics, NRBO is a better algorithm.

4.6. Welded beam design problem (RW6)

The primary objective is to construct a welded beam as economically as possible. Five constraints and four parameters complicate the development of a welded beam. Eq. (43) is a mathematical representation of the problem.

$$f6(\bar{x}) = 0.04811 x_3 x_4 (x_2 + 14) + 1.10471 x_1^2 x_2 \tag{43}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = x_1 - x_4 \le 0$$

$$g_2(\bar{x}) = \delta(\bar{x}) - \delta_{max} \le 0$$

$$g_3(\bar{x}) = P \le P_c(\bar{x})$$

$$g_4(\bar{x}) = \tau_{max} \ge \tau(\bar{x})$$

$$g_5(\bar{x}) = \sigma(\bar{x}) - \sigma_{max} \le 0$$

Where:

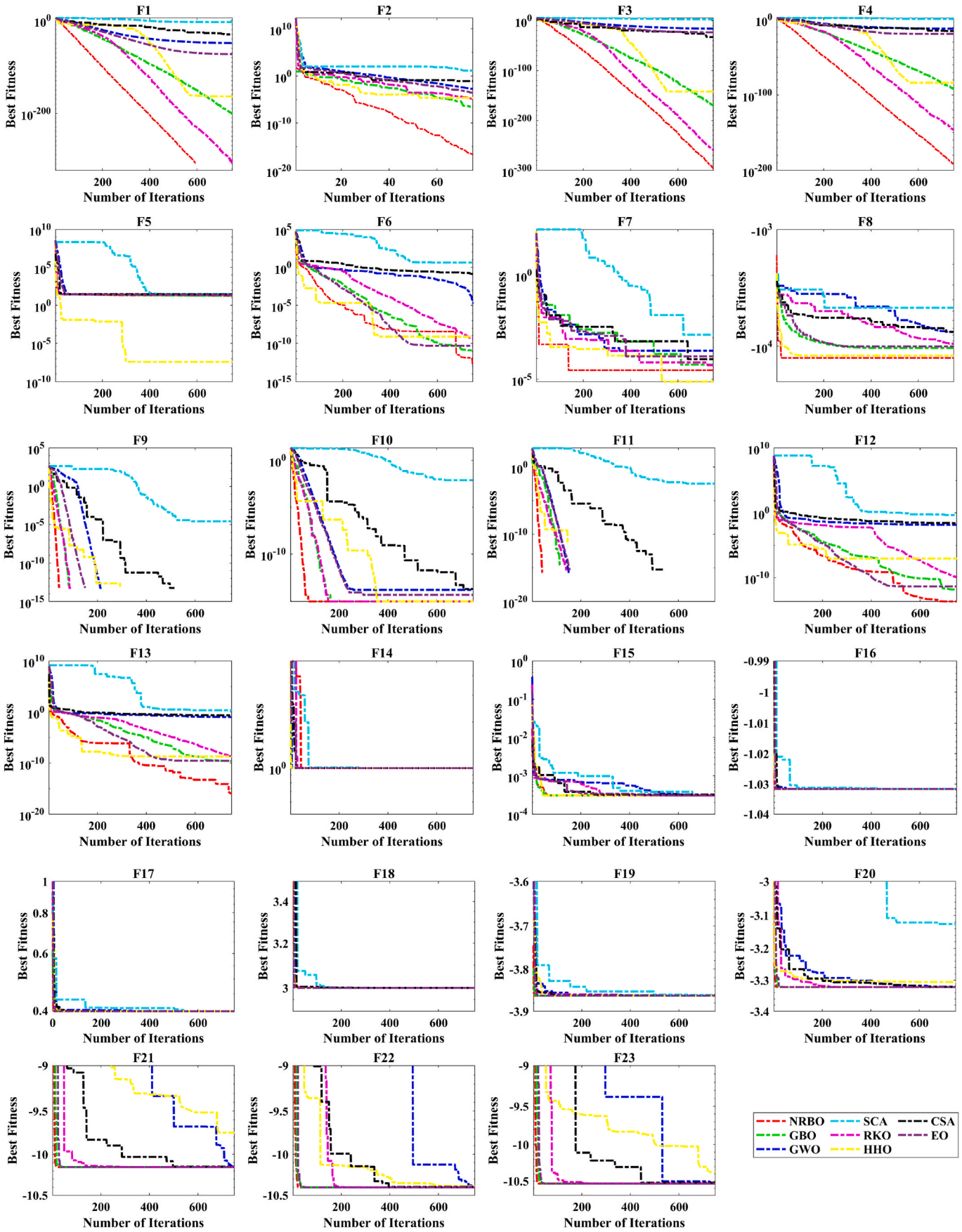

Fig. 9. Convergence curves obtained by all algorithms on F1–F23 benchmark functions.

$$\tau = \sqrt{\tau'^2 + \tau''^2 + 2\tau'\tau''\frac{x_2}{2R}},\ \tau'' = \frac{RM}{J},\ \tau' = \frac{P}{\sqrt{2}x_2x_1},$$

$$M = p\left(\frac{x_2}{2} + L\right),$$

$$R = \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2},\ J = 2\left(\left(\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2\right)\sqrt{2}x_1x_2\right),$$

$$\sigma(\overline{x}) = \frac{6PL}{x_4x_3^2},\ \delta(\overline{x}) = \frac{6PL^3}{Ex_3^2X_4},\ P_c(\overline{x}) = \frac{4.013Ex_3x_4^3}{6L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right)$$

$$L = 14\text{in},\ P = 6000\text{lb},\ E = 30.10^6\text{psi},\ \sigma_{max} = 30,000\text{psi},$$

$$\tau_{max} = 13,600\text{psi},\ G = 12.10^6\text{psi},\ \delta_{max} = 0.25\text{in}$$

The limits are,

$$0.1 \le x_3, x_2 \le 10$$
$$0.1 \le x_4 \le 2$$
$$0.125 \le x_1 \le 2$$

The findings of the proposed NRBO and other algorithms are listed in Table 25. Table 25 shows that the NRBO outperformed all other approaches and explored the minimum cost. Table 25 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the welded beam design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO comes out on top with an FRT value of 1.8 when it comes to tackling the welded beam design problem. The Min value explored by NRBO, GBO, and EO are the same; however, based on Mean and STD values, the NRBO is better than all algorithms. The performance of GBO and EO is similar to that of NRBO. But, with respect to RT and other metrics, NRBO is a better algorithm.

4.7. Three-bar truss design problem (RW7)

An accidented confined space constrains the solution space of this optimization problem, which is derived from civil engineering. The primary objective of this challenge is to reduce the weight of the bar

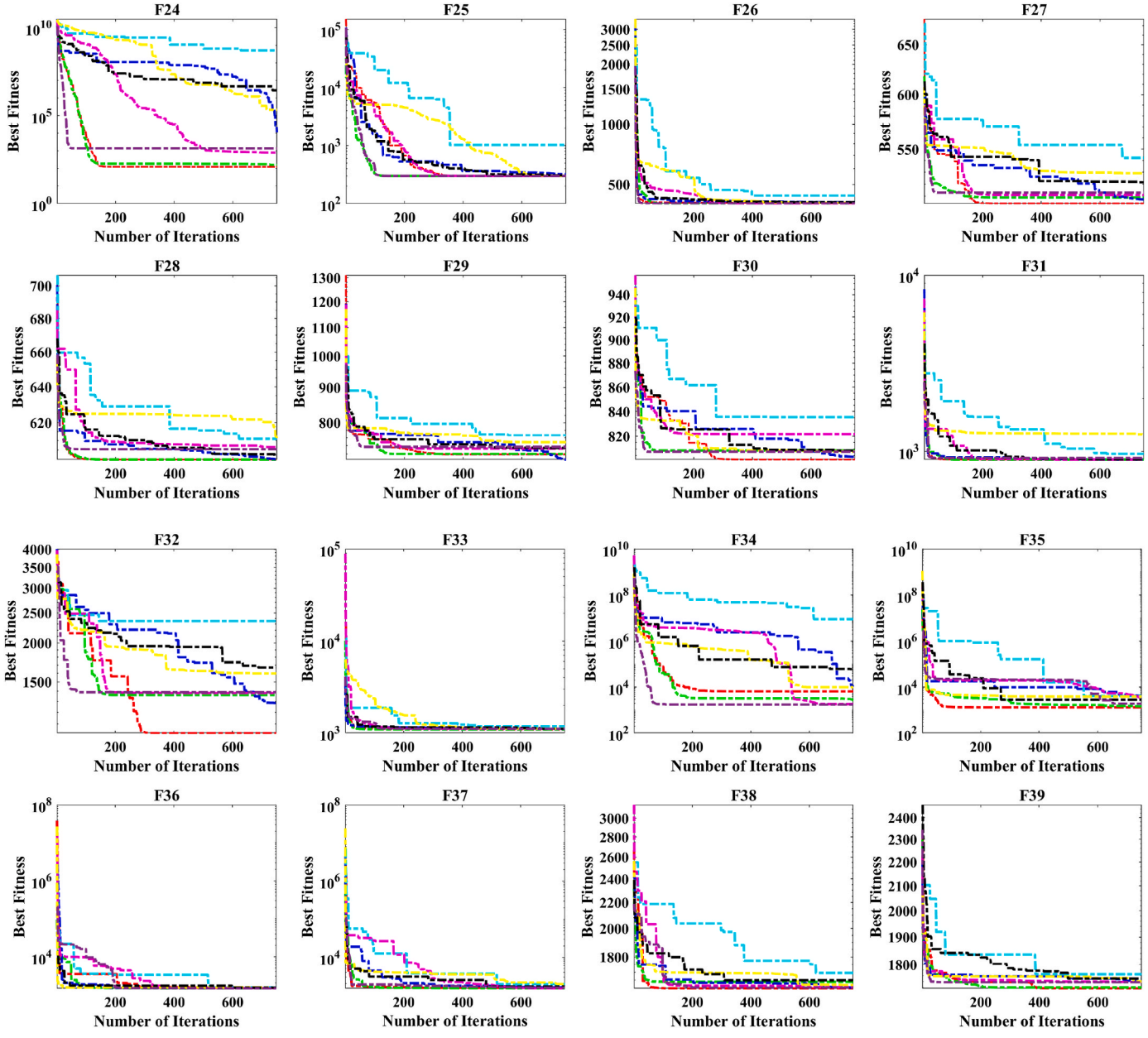

Fig. 10. Convergence curves obtained by all algorithms on F24–F52 CEC2017 benchmark functions.

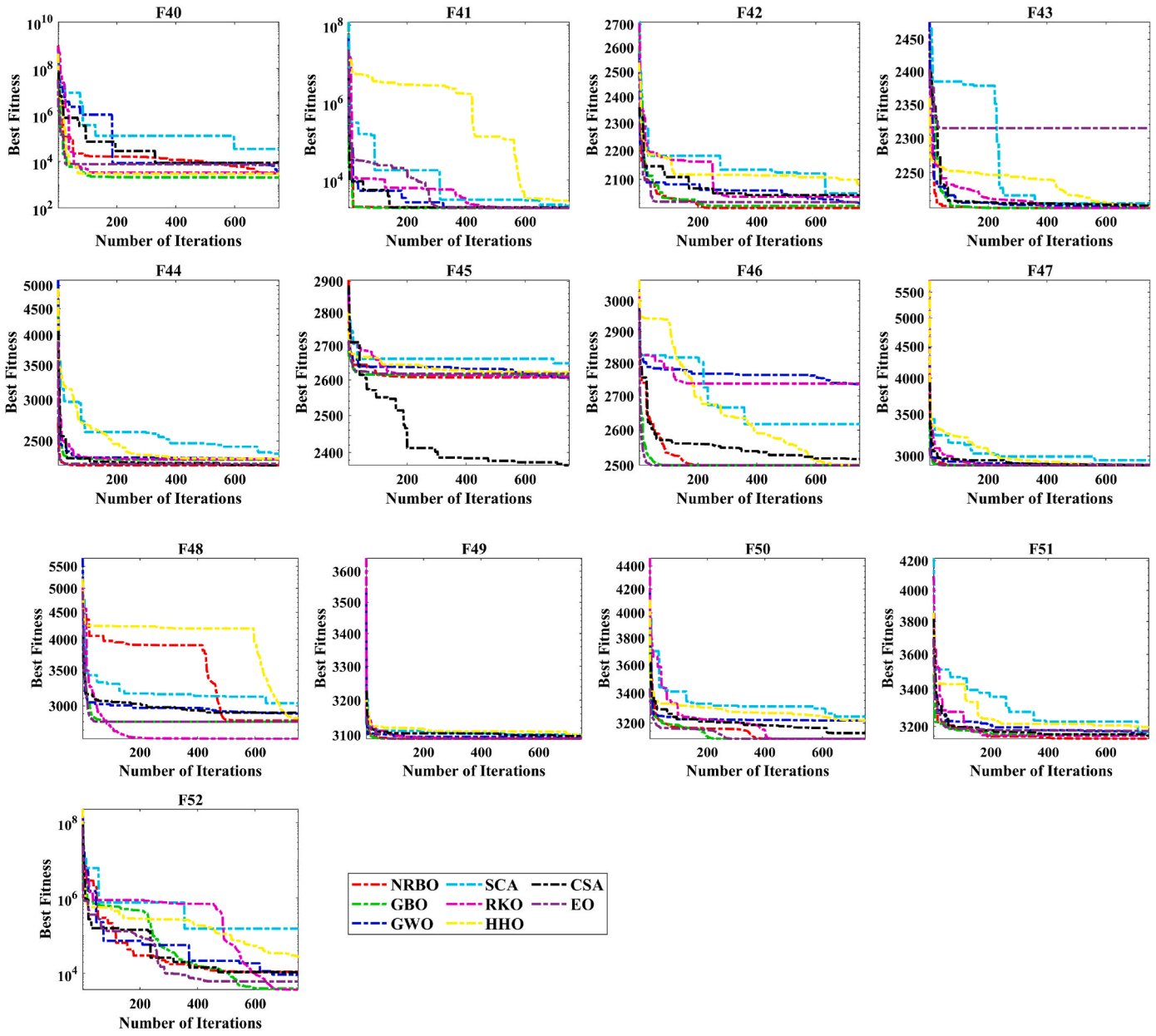

Fig. 10. (*continued*).

constructions to the bare minimum. Constrained by the stress limitations of each bar, this problem's constraints are as follows: The resulting problem has a linear objective function with three non-linear constraints, which makes it a linear optimization problem. Eq. (44) is a mathematical description of this problem.

$$f7(\bar{x}) = l\left(x_2 + 2\sqrt{2}\,x_1\right) \tag{44}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = \frac{x_2}{2x_2x_1 + \sqrt{2}x_1^2}P - \sigma \le 0$$

$$g_2(\bar{x}) = \frac{x_2 + \sqrt{2}x_1}{2x_2x_1 + \sqrt{2}x_1^2}P - \sigma \le 0$$

$$g_3(\bar{x}) = \frac{1}{x_1 + \sqrt{2}x_2}P - \sigma \le 0$$

Where:

$l = 100,\ P = 2,\ \text{and}\ \sigma = 2$

The limits are,

$0 \le x_1, x_2 \le 1$

The findings of the proposed NRBO and other algorithms are listed in Table 26. Table 26 shows that the NRBO outperformed all other approaches and explored the minimum bar weight. Table 26 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the 3-bar truss design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO and GBO come out on top with an FRT value of 1.9 when it comes to tackling the 3-bar truss design problem. The Min value explored by all algorithms is the same; however, based on Mean and STD values, the NRBO is better than all algorithms. And, with respect to the RT and other metrics, NRBO is a better algorithm.

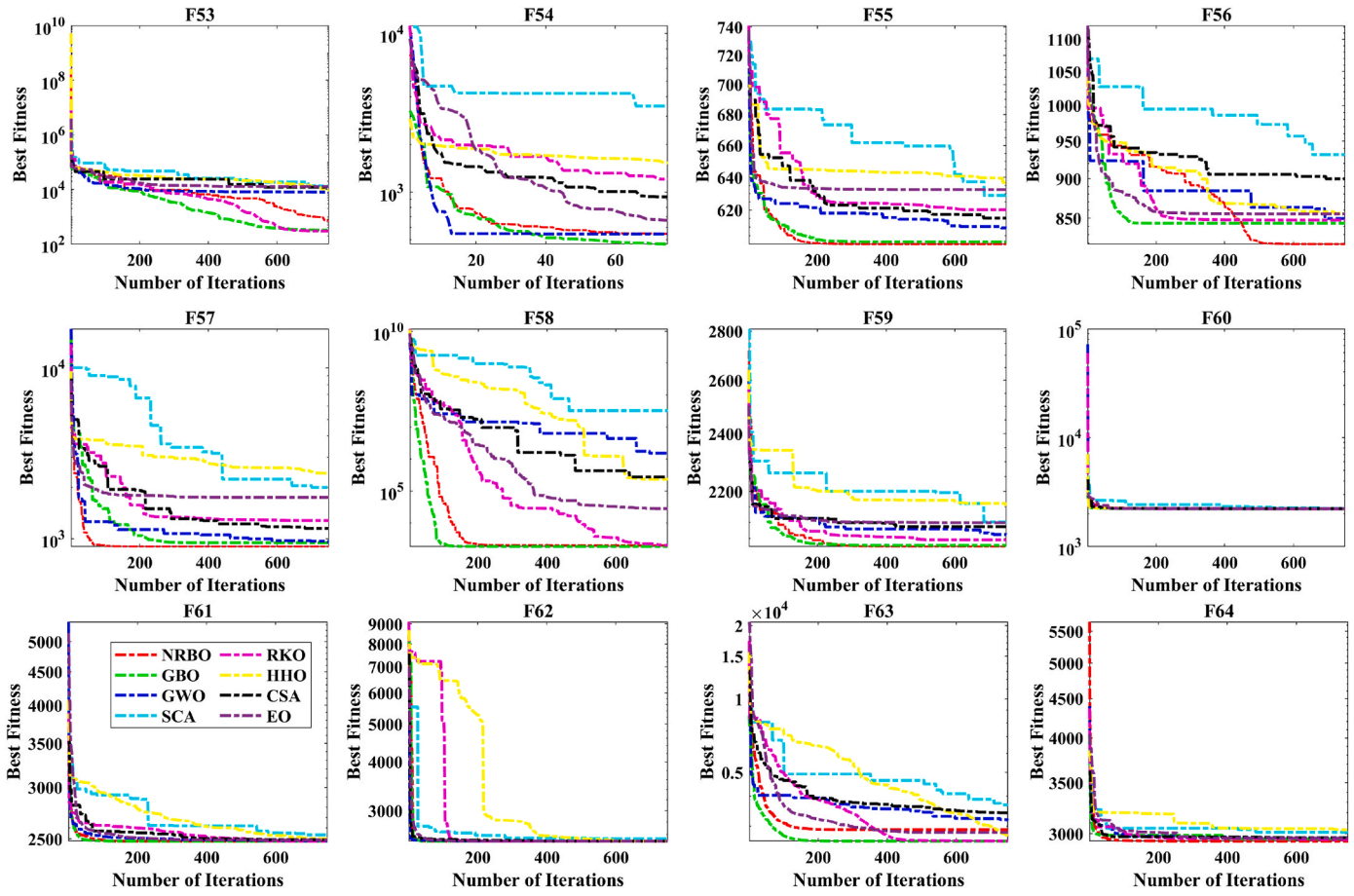

Fig. 11. Convergence curves obtained by all algorithms on F53–F64 CEC2022 benchmark functions.

Table 14
ANOVA for S/N ratios and RPD values.

Empty Cell	Parameter	Degree of freedom	Mean of squares	Sum of squares	p-value	F-distribution
ANOVA results for S/N ratios	**τ**	4	1.498	2.157	0.000[a]	242.578
	Error	17	0.003	0.124		
	Total	21		2.281		
ANOVA results for RPD values	**DF**	4	11.985	23.586	0.004[a]	159.52
	Error	17	0.069	1.257		
	Total	21		24.843		

[a] At the 5% significance level, the variable has a significant impact.

Table 15
Results obtained using Taguchi design.

Parameter level	Mean S/N ratios	Mean RPD
DF(1)	−42.68	14.29
DF(2)	−40.73	13.48
DF(3)[a]	−38.85	8.15

[a] Represents the best level.

4.8. Multiple disk clutch brake design problem (RW8)

The primary objective is to reduce the mass of a multiple-disc clutch brake as much as possible. A total of five integer model parameters are employed, which are the inner radius (x_1), the outer radius (x_2), the disc thickness (x_3), the force of actuators (x_4), and the number of frictional surfaces (x_5). It has nine non-linear constraints that must be satisfied. Eq. (45) is a description of the problem.

$$f8(\bar{x}) = \pi\left(x_2^2 - x_1^2\right)x_3(x_5+1)\rho \tag{45}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = -p_{max} + p_{rz} \le 0$$

$$g_2(\bar{x}) = p_{rz}V_{sr} - V_{sr,max}p_{max} \le 0$$

$$g_3(\bar{x}) = \triangle R + x_1 - x_2 \le 0$$

$$g_4(\bar{x}) = -L_{max} + (x_5+1)(x_3+\delta) \le 0$$

$$g_5(\bar{x}) = sM_s - M_h \le 0$$

$$g_6(\bar{x}) = T \ge 0$$

$$g_7(\bar{x}) = T - T_{max} \le 0$$

Where:

Table 16
Details of all selected real-world constrained functions from CEC2020 (Kumar et al., 2020).

Problem	Name of the problem	*dim*	Equality constraints (*h*)	Inequality constraints (*g*)	f_{min}
RW1	Reactor network design	6	1	4	−3.8826E-01
RW2	Process design Problem	5	3	0	2.6887E+04
RW3	Weight minimization of a speed reducer	7	11	0	2.9944E+03
RW4	Tension/ compression spring design (case 1)	3	3	0	1.2665E-02
RW5	Pressure vessel design	4	4	0	5.8853E+03
RW6	Welded beam design	4	5	0	1.6702
RW7	3-bar truss design problem	2	3	0	2.6390E+02
RW8	Multiple disk clutch brake design problems	5	7	0	2.3524E-01
RW9	Hydro-static thrust bearing design problem	4	7	0	1.6254E+03
RW10	Gear train design Problem	4	1	1	0
RW11	Optimal power flow (minimization of active power loss)	126	0	116	2.1963E-02
RW12	Optimal power flow (minimization of fuel cost)	126	0	116	2.7766

Table 17
Obtained results by all algorithms on RW1.

Algorithm	x_1	x_2	x_3	x_4	x_5	x_6
NRBO	0.9991	0.3925	0.0032	0.3914	1.000E-05	15.9755
GBO	0.9991	0.3925	0.0032	0.3914	1.000E-05	15.9755
GWO	1.0000	0.4012	0.0000	0.3891	4.945E-03	15.4289
SCA	1.0000	1.0000	0.0000	0.0000	1.015E-05	0.0000
RKO	0.9985	0.3959	0.0043	0.3917	1.100E-03	15.7365
HHO	0.8780	0.5477	0.1289	0.3693	1.972E+00	6.3796
CSA	1.0000	0.3943	0.0000	0.3914	1.000E-05	15.8857
EO	0.9991	0.3930	0.0031	0.3913	5.067E-05	15.9439

Table 18
Statistical results of all algorithms on RW1.

Algorithm	Min	Mean	STD	RT	FRT
NRBO	−0.3898	−0.3837	1.26E-02	0.66	2.2
GBO	−0.3898	−0.3827	6.53E-03	1.98	3.6
GWO	−0.3884	−0.2992	1.65E-01	1.18	4.2
SCA	0.0000	0.0000	0.00E+00	0.65	7.6
RKO	−0.3896	−0.3046	1.67E-01	2.76	4.6
HHO	−0.2719	0.2257	5.79E-01	1.72	7.2
CSA	−0.3878	−0.3735	2.62E-02	4.92	4
EO	−0.3898	−0.3850	6.57E-03	0.71	2.6

Table 19
Obtained results by all algorithms on RW2.

Algorithm	x_1	x_2	x_3	x_4	x_5
NRBO	45.00	27.00	45.00	80.08	36.20
GBO	45.00	27.00	45.00	79.69	43.18
GWO	45.00	27.00	45.00	79.97	33.37
SCA	45.00	27.00	45.00	80.27	33.52
RKO	45.00	27.00	45.00	79.71	32.51
HHO	45.00	27.00	45.00	79.92	33.46
CSA	45.00	27.00	45.00	80.01	45.49
EO	45.00	27.00	45.00	80.39	32.51

Table 20
Statistical results of all algorithms on RW2.

Algorithm	Min	Mean	STD	RT	FRT
NRBO	24201.83	24201.83	0	0.66	4.5
GBO	24201.83	24201.83	0	1.89	4.5
GWO	24201.83	24201.83	0	0.62	4.5
SCA	24201.83	24201.83	0	0.61	4.5
RKO	24201.83	24201.83	0	2.79	4.5
HHO	24201.83	24201.83	0	1.49	4.5
CSA	24201.83	24201.83	0	4.83	4.5
EO	24201.83	24201.83	0	1.14	4.5

Table 21
Obtained results by all algorithms on RW3.

Algorithm	x_1	x_2	x_3	x_4	x_5	x_6	x_7
NRBO	3.230	0.700	17.000	7.300	7.641	3.351	5.287
GBO	3.230	0.700	17.000	7.300	7.641	3.351	5.287
GWO	3.230	0.700	17.000	7.419	7.720	3.352	5.289
SCA	3.276	0.700	17.000	8.289	8.300	3.472	5.321
RKO	3.231	0.700	17.000	7.300	7.599	3.351	5.287
HHO	3.248	0.700	17.000	7.329	7.706	3.351	5.287
CSA	3.217	0.700	17.005	7.300	7.982	3.365	5.287
EO	3.230	0.700	17.000	7.300	7.641	3.351	5.287

Table 22
Statistical results of all algorithms on RW3.

Algorithm	Min	Mean	STD	RT	FRT
NRBO	2911.20	2911.20	0.00E+00	0.82	1.80
GBO	2911.20	2911.20	4.55E-13	2.16	2.40
GWO	2914.93	2918.04	3.43E+00	0.79	5.00
SCA	2998.25	3024.43	1.70E+01	0.75	7.80
RKO	2911.47	2914.91	5.08E+00	3.03	4.20
HHO	2915.03	2991.36	6.73E+01	1.94	6.60
CSA	2923.98	2940.10	1.59E+01	5.50	6.40
EO	2911.20	2911.20	0.00E+00	1.33	1.80

$$M_h = \frac{2}{3}\mu x_4 x_5 \frac{x_2^3 - x_1^3}{x_2^2 - x_1^2}\text{N.mm},\ \omega = \frac{\frac{\pi n}{30}}{\text{s}}\text{rad},\ A = \pi\left(x_2^2 - x_1^2\right)\text{mm}^2,\ p_{rz} = \frac{x_4}{A}\text{N}\Big/\text{mm}^2,\ V_{sr} = \frac{\frac{\pi R_{sr} n}{30}}{\text{s}}\text{mm},\ R_{sr} = \frac{2}{3}\frac{x_2^3 - x_1^3}{x_2^2 x_1^2}\text{mm},\ T = \frac{I_z \omega}{M_h + M_f},\ \triangle R = 20\text{mm},\ L_{max} = 30\text{mm},\ \mu = 0.6,\ V_{sr,max} = \frac{10\text{m}}{\text{s}},\ \delta = 0.5\text{mm},\ s = 1.5,\ T_{max} = 15\text{s},\ n = 250\text{rpm},\ I_z = 55\text{Kg.m}^2,\ M_s = 40\text{Nm},\ M_f = 3\text{Nm},\ \text{and}\ \ p_{max} = 1$$

The limits are,

$$60 \le x_1 \le 80, 90 \le x_2 \le 110, 1 \le x_3 \le 3$$

$$0 \le x_4 \le 1000, 2 \le x_5 \le 9$$

The findings of the proposed NRBO and other algorithms are listed in

Table 23
Obtained results by all algorithms on RW4, including statistical results.

Algorithm	x_1	x_2	x_3	Min	Mean	STD	RT	FRT
NRBO	0.0517	0.3575	11.2382	0.0127	0.0127	2.108E-06	0.689	1
GBO	0.0520	0.3644	10.8515	0.0127	0.0127	1.599E-05	1.938	2.8
GWO	0.0510	0.3414	12.2526	0.0127	0.0127	1.654E-05	0.644	4.4
SCA	0.0500	0.3165	14.1580	0.0128	0.0129	1.130E-04	0.631	7.2
RKO	0.0523	0.3717	10.4586	0.0127	0.0127	2.239E-05	2.797	3.6
HHO	0.0525	0.3760	10.2395	0.0127	0.0132	8.697E-04	1.600	5
CSA	0.0523	0.3720	10.4500	0.0127	0.0128	5.759E-05	4.972	5.6
EO	0.0528	0.3831	9.8918	0.0127	0.0138	2.165E-03	1.247	6.4

Table 24
Obtained results by all algorithms on RW5, including statistical results.

Algorithm	x_1	x_2	x_3	x_4	Min	Mean	STD	RT	FRT
NRBO	0.555	0.977	40.480	197.775	592.332	592.332	2.32E-11	0.752	1
GBO	0.891	1.343	40.480	197.781	592.332	592.332	7.60E-07	1.956	2
GWO	0.838	0.770	40.500	197.502	592.333	592.342	7.90E-03	0.659	5
SCA	0.510	1.380	40.321	200.000	592.374	592.611	2.48E-01	0.641	6.8
RKO	0.514	0.559	40.476	197.829	592.332	592.337	7.35E-03	2.925	3.6
HHO	0.617	0.510	42.462	172.178	593.766	628.976	3.10E+01	1.575	8
CSA	0.929	0.831	40.320	199.997	592.349	592.403	4.73E-02	4.900	6
EO	0.510	0.510	40.457	198.099	592.333	592.334	1.68E-03	1.250	3.6

Table 25
Obtained results by all algorithms on RW6, including statistical results.

Algorithm	x_1	x_2	x_3	x_4	Min	Mean	STD	RT	FRT
NRBO	0.2022	3.2748	9.1920	0.1988	1.6686	1.6686	9.17E-06	0.74	1.8
GBO	0.2021	3.2773	9.1920	0.1988	1.6686	1.6686	1.34E-05	1.23	2.2
GWO	0.2031	3.2598	9.1930	0.1989	1.6692	1.6696	6.44E-04	0.68	4
SCA	0.2174	3.1001	9.4208	0.1988	1.7544	1.7898	2.81E-02	0.67	7.2
RKO	0.1980	3.3537	9.1920	0.1988	1.6711	1.7156	5.94E-02	2.88	5.4
HHO	0.1701	3.9343	9.3251	0.1982	1.7207	1.8353	9.79E-02	1.65	7.8
CSA	0.2030	3.3206	9.2301	0.1988	1.6827	1.6957	8.25E-03	4.94	5.6
EO	0.2021	3.2769	9.1920	0.1988	1.6686	1.6686	9.58E-05	2.03	2

Table 26
Obtained results by all algorithms on RW7, including statistical results.

Algorithm	x_1	x_2	Min	Mean	STD	RT	FRT
NRBO	0.6783	0.3511	242.851	242.851	4.02E-14	0.67	1.9
GBO	0.6783	0.3511	242.851	242.851	4.02E-14	1.86	1.9
GWO	0.6783	0.3511	242.851	242.851	6.65E-07	0.61	5.8
SCA	0.6784	0.3516	242.851	242.852	1.22E-03	0.59	8
RKO	0.6783	0.3511	242.851	242.851	2.57E-11	2.89	4
HHO	0.6783	0.3511	242.851	242.851	5.06E-07	1.50	5.2
CSA	0.6785	0.3507	242.851	242.851	6.56E-05	4.82	7
EO	0.6783	0.3511	242.851	242.851	3.18E-14	1.15	2.2

Table 27
Obtained results by all algorithms on RW8, including statistical results.

Algorithm	x_1	x_2	x_3	x_4	x_5	Min	Mean	STD	RT	FRT
NRBO	70.00	90.00	1.00	438.82	2.00	0.2352	0.2352	3.10E-17	0.73	2.5
GBO	70.00	90.00	1.00	225.40	2.00	0.2352	0.2352	3.10E-17	2.01	2.5
GWO	70.00	90.00	1.00	280.11	2.00	0.2352	0.2353	1.22E-05	0.71	6
SCA	69.96	90.00	1.00	386.78	2.00	0.2356	0.2371	1.61E-03	0.68	7.4
RKO	70.00	90.00	1.00	286.81	2.00	0.2352	0.2352	1.84E-08	2.77	5
HHO	70.00	90.00	1.00	1000.00	2.00	0.2352	0.2352	3.10E-17	1.78	2.5
CSA	70.00	90.00	1.00	967.75	2.00	0.2353	0.2927	5.22E-02	4.93	7.6
EO	70.00	90.00	1.00	7.08	2.00	0.2352	0.2352	3.10E-17	1.23	2.5

Table 28
Obtained results by all algorithms on RW9, including statistical results.

Algorithm	x_1	x_2	x_3	x_4	Min	Mean	STD	RT	FRT
NRBO	5.734	5.738	9.17E-06	1.0000	124.931	419.897	330.258	0.763	2.6
GBO	8.613	8.614	9.62E-06	1.0011	140.649	263.599	87.438	2.088	3
GWO	8.734	8.738	9.08E-06	2.3313	358.821	502.834	114.399	0.706	5.2
SCA	6.818	6.834	6.78E-06	1.0000	478.913	1935.846	824.406	0.694	7.4
RKO	10.934	10.934	9.28E-06	1.0000	138.216	399.762	295.696	2.459	3.4
HHO	15.999	16.000	1.00E-05	16.0000	527.930	1547.874	1348.601	2.538	7
CSA	9.098	9.098	9.16E-06	1.0000	164.076	296.569	126.308	4.844	3.2
EO	7.693	7.693	9.45E-06	1.0133	210.532	354.630	139.743	1.241	4.2

Table 29
Obtained results by all algorithms on RW10, including statistical results.

Algorithm	x_1	x_2	x_3	x_4	Min	Mean	STD	RT	FRT
NRBO	17.44	14.85	57.73	31.09	0.00E+00	0.00E+00	0.00E+00	0.92	1.9
GBO	12.04	43.02	60.00	59.83	2.65E-25	5.60E-18	7.62E-18	1.52	4.2
GWO	24.96	14.10	41.60	58.65	3.11E-15	4.35E-13	4.97E-13	0.48	6.8
SCA	12.00	21.75	60.00	30.15	2.92E-11	7.54E-10	9.76E-10	0.47	8
RKO	22.07	19.67	53.67	56.05	0.00E+00	2.30E-15	5.03E-15	0.49	4.9
HHO	13.20	38.63	59.44	59.44	0.00E+00	0.00E+00	0.00E+00	11.60	1.9
CSA	13.53	15.17	36.15	39.37	1.73E-17	3.72E-15	4.39E-15	3.73	5.8
EO	15.26	34.03	60.00	60.00	0.00E+00	1.18E-27	2.63E-27	0.67	2.5

Table 27. Table 27 shows that the NRBO outperformed GWO, SCA, and CSA and explored the minimum weight of multiple disk clutch brakes. Table 27 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the multiple disk clutch brake design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO, GBO, HHO, and EO come out on top with an FRT value of 2.5 when it comes to tackling the multiple disk clutch brake design problem. The Min value explored by NRBO, GBO, RKO, and EO are the same; however, based on Mean and STD values, the NRBO is better than all algorithms. The performance of GBO, HHO, and EO is similar to that of NRBO. But, with respect to RT and other metrics, NRBO is a better algorithm.

4.9. Hydro-static thrust bearing design problem (RW9)

The primary objective is to minimize the amount of bearing energy lost while utilizing four design parameters. Such design parameters are the oil viscosity, the bearing radius R, the flow rate Q, and the recess radius R_o. There are seven non-linear constraints related to inlet oil pressure, load-carrying capacity, and oil film thickness. Eq. (46) is a description of the problem.

$$f9(\bar{x}) = \frac{QP_0}{0.7} + E_f \tag{46}$$

Subjected to the following constraints,

$$g_1(\bar{x}) = 1000 - P_0 \leq 0$$

$$g_2(\bar{x}) = W - 101000 \leq 0$$

$$g_3(\bar{x}) = 5000 - \frac{W}{\pi\left(R^2 - R_0^2\right)} \leq 0$$

$$g_4(\bar{x}) = 50 - P_0 \leq 0$$

$$g_5(\bar{x}) = 0.001 - \frac{0.0307}{386.4P_0}\left(\frac{Q}{2\pi Rh}\right) \leq 0$$

$$g_6(\bar{x}) = R - R_0 \leq 0$$

$$g_7(\bar{x}) = h - 0.001 \leq 0$$

Where:

$$W = \frac{\pi P_0}{2}\frac{R^2 - R_0^2}{ln\left(\frac{R}{R_0}\right)},\ P_0 = \frac{6\mu Q}{\pi h^3} ln\left(\frac{R}{R_0}\right)$$

$$E_f = 9336Q \times 0.0307 \times 0.5 \triangle T,\ \triangle T = 2\left(10^P - 559.7\right)$$

$$P = \frac{log_{10}log_{10}\left(8.122 \times 10^6\mu + 0.8\right) + 3.55}{10.04}$$

$$h = \left(\frac{2\pi \times 750}{60}\right)^2 \frac{2\pi\mu}{E_f}\left(\frac{R^4}{4} - \frac{R_0^4}{4}\right)$$

The limits are,

$$1 \leq R \leq 16, 1 \leq R_0 \leq 16$$

$$1 \times 10^{-6} \leq \mu \leq 16 \times 10^{-6}, 1 \leq Q \leq 16$$

Table 30
Statistical results of all algorithms on RW11.

Algorithm	Min	Mean	STD	RT	FRT
NRBO	81.26	98.15	21.79	12.90	1
GBO	5173.90	17979.89	14048.99	18.60	6.4
GWO	692.48	977.73	245.09	7.11	3
SCA	127240.36	148578.23	13101.75	7.78	8
RKO	1920.28	2517.90	574.97	8.70	4
HHO	4767.40	5556.70	708.33	18.91	5.4
CSA	5094.03	8331.72	3499.19	36.14	6.2
EO	90.51	123.80	20.60	7.63	2

Table 31
Statistical results of all algorithms on RW12.

Algorithm	Min	Mean	STD	RT	FRT
NRBO	86.40	99.38	8.66	12.54	1.2
GBO	1777.10	18765.64	14062.93	18.68	6.6
GWO	533.96	1069.23	414.52	8.06	3.2
SCA	113850.54	187998.90	87307.01	7.98	8
RKO	88.72	1705.26	994.03	8.51	3.4
HHO	5740.73	6517.75	653.44	19.12	5.4
CSA	5226.84	7374.89	1989.86	38.31	6
EO	101.43	159.07	77.18	8.19	2.2

The findings of the proposed NRBO and other algorithms are listed in Table 28. Table 28 shows that the NRBO outperformed all other approaches and explored the minimum power loss. Table 28 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the hydro-static thrust bearing design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO comes out on top with an FRT value of 2.6 when it comes to tackling the hydrostatic thrust-bearing design problem. The Min, Mean, and STD values explored by NRBO are lesser than any other algorithms, and therefore, the NRBO is better than all algorithms, including RT values.

4.10. Gear train design problem (RW10)

The primary objective is to reduce the ratio of gears used in the layout of the composite gear train to the smallest possible value. The gear train ratio is defined as the relationship between angular velocities at both the output and the input shafts in a gear train. The compound gear train is constructed from two sets of gearwheels to achieve the appropriate overall gear ratio. The total ratio of gears is presented in Eq. (47).

$$i_{tot} = \frac{\omega_o}{\omega_i} = \frac{z_d z_b}{z_a z_f} \tag{47}$$

Where z denotes the total number of teeth on each gear wheel, while the parameters ω_i and ω_o denote the relative rotational speeds of the input and output shafts, respectively. The primary aim of this phase is to determine the total number of teeth for each gear wheel in order to construct an optimal ratio of gears that is nearer to the desired ratio ($i_{trg} = 1/6.931$) than possible. The highest number of teeth required for each gearwheel is 60, with the minimum number of teeth being 12. Eq. (48) presents the mathematical formulation of this problem.

$$f10(\bar{x}) = \left(i_{trg} - i_{tot}\right)^2 = \left(\frac{1}{6.931} - \frac{x_1 x_2}{x_3 x_4}\right)^2 \tag{48}$$

Subjected to the following constraints,

$$g_{1-4}(\bar{x}) = 12 - x_i \le 0,$$
$$g_{5-8}(\bar{x}) = (60 - \bar{x}) \le 0$$

The findings of the proposed NRBO and other algorithms are listed in Table 29. Table 29 shows that the NRBO outperformed all other approaches and explored the minimum gear ratio. Table 29 also includes statistical information such as Min, Mean, STD, and RT. As a result, it is decided that the proposed NRBO is more reliable for the gear train design problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO and HHO come out on top with an FRT value of 1.9 when it comes to tackling the gear train design problem. The Min value explored by NRBO, RKO, HHO, and EO are the same; however, based on Mean and STD values, the NRBO and HHO are better than all algorithms. The performance of RKO and EO is similar to that of NRBO. But, with respect to RT and other metrics, NRBO is a better algorithm.

4.11. Optimal Power Flow (minimization of active power loss) problem (RW11)

Optimal Power Flow (OPF) continues to be a popular research area among academics and industry professionals. To simplify, the OPF can be seen as a single-objective COP of emission and voltage deviation and minimizing fuel cost and transmission loss, with limitations based on

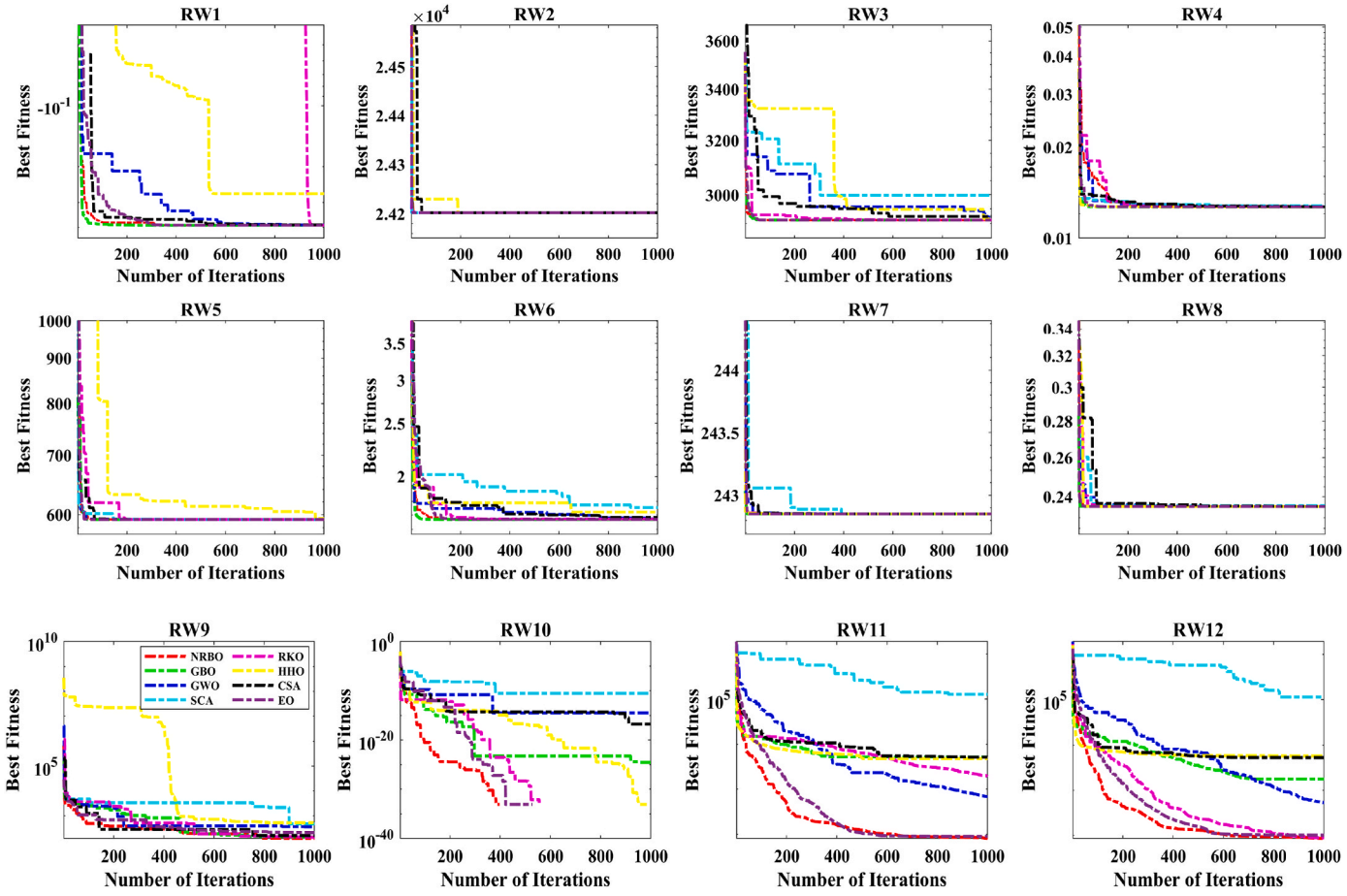

Fig. 12. Convergence curves of all algorithms for real-world CEC2020 problems.

line capacity and generator potential, power flow balance, and bus voltage that must be met. This situation examines the minimization of active power losses as an objective function, and the problem can be expressed in Eq. (49).

$$f11 = \sum_{i=1}^{N} P_{g,i} - P_{l,i} \tag{49}$$

Subjected to the following constraints,

$$h_i = P_{g,i} - P_{l,i} - V_i \sum_{j=1}^{N} V_j \left(G_{ij} \cos\left(\delta_i - \delta_j\right) + B_{ij} \sin\left(\delta_i - \delta_j\right)\right) = 0,$$

$$h_{N+i} = Q_{g,i} - Q_{l,i} - V_i \sum_{j=1}^{N} V_j \left(G_{ij} \sin\left(\delta_i - \delta_j\right) - B_{ij} \cos\left(\delta_i - \delta_j\right)\right) = 0$$

The limits are,

$$V_{min} \le V_i \le V_{max}, \delta_{min} \le \delta_i \le \delta_{max}, P_{min} \le P_{g,i} \le P_{max}, \text{and } Q_{min} \le Q_{g,i} \le Q_{max}$$

where $i = 1, 2, \ldots, N$, $P_i (= P_{g,i} - P_{l,i})$, $Q_i (= Q_{g,i} - Q_{l,i})$ denote the active and reactive power injected at the i^{th} bus, $Y_{bus,i,j} (= G_{ij} - 1B_{ij})$ is the ij^{th} element of the admittance matrix, $V_i = V_i \angle^{\delta_i}$ denote the bus voltage at the i^{th} bus, and N denote the total number of buses.

Due to many problem dimensions, this paper does not list the optimized decision vector of RW11. The findings, such as Min, Mean, and STD of the proposed NRBO and other algorithms, are listed in Table 30. Table 30 shows that the NRBO outperformed all other approaches and explored the minimum active power loss. As a result, it is decided that the proposed NRBO is more reliable for the OPF problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO comes out on top with an FRT value of 1 when it comes to tackling the OPF problem. The Min, Mean, and STD values explored by NRBO are lesser than any other algorithms, and therefore, the NRBO is better than all algorithms, including RT values.

4.12. Optimal Power Flow (minimization of fuel cost) problem (RW12)

In this situation, the objective function is the reduction of fuel costs to the lowest possible level. This problem can also be expressed in Eq. (50).

$$f12 = \sum_{i=1}^{N} \left(a_i + b_i P_{g,i} + c_i P_{g,i}^2\right) \tag{50}$$

Subjected to the following constraints,

$$h_i = P_{g,i} - P_{l,i} - V_i \sum_{j=1}^{N} V_j \left(G_{ij} \cos\left(\delta_i - \delta_j\right) + B_{ij} \sin\left(\delta_i - \delta_j\right)\right) = 0,$$

$$h_{N+i} = Q_{g,i} - Q_{l,i} - V_i \sum_{j=1}^{N} V_j \left(G_{ij} \sin\left(\delta_i - \delta_j\right) - B_{ij} \cos\left(\delta_i - \delta_j\right)\right) = 0$$

The limits are,

$$V_{min} \le V_i \le V_{max}, \delta_{min} \le \delta_i \le \delta_{max}, P_{min} \le P_{g,i} \le P_{max}, \text{and } Q_{min} \le Q_{g,i} \le Q_{max}$$

where $i = 1, 2, \ldots, N$, $P_i (= P_{g,i} - P_{l,i})$, $Q_i (= Q_{g,i} - Q_{l,i})$ denote the active and reactive power injected at the i^{th} bus, $Y_{bus,i,j} (= G_{ij} - 1B_{ij})$ is the ij^{th} element of the admittance matrix, $V_i = V_i \angle^{\delta_i}$ denote the bus voltage at the i^{th} bus, and N denote the total number of buses.

Due to many problem dimensions, the optimized decision vector of RW12 is also not listed in this paper. The findings, such as Min, Mean, and STD of the proposed NRBO and other algorithms, are listed in Table 31. Table 31 shows that the NRBO outperformed all other approaches and explored the minimum fuel cost. As a result, it is decided that the proposed NRBO is more reliable for the OPF problem. Furthermore, FRT values derived by all algorithms are presented. The proposed NRBO comes out on top with an FRT value of 1.2 when it comes to tackling the OPF problem. The Min, Mean, and STD values explored by NRBO are lesser than any other algorithms, and therefore, the NRBO is better than all algorithms, including RT values.

In order to assess the convergence rate of the proposed NRBO and all other selected algorithms, the convergence curves of all problems are illustrated in Fig. 12. The provided findings have proven the efficacy of applying TAO and NRSR, which has resulted in a significant improvement in the convergence speed and the performance of the optimization algorithm, as discussed earlier. However, in a few functions, NRBO attained optimal values almost identical to those produced by the GBO and EO. Compared to other algorithms, the NRBO not only outperformed them in terms of convergence curve performance, but it also attained an optimal value in relatively few iterations and demonstrated an extremely fast convergence rate when contrasted to other approaches, as can be observed from the results shown in Fig. 12.

When the convergence graphs in Fig. 12 and the statistics are taken into consideration, the GBO is considered to be the second strongest in a few test cases. The EO is frequently ranked third among the best algorithms on a few test functions. The RKO, HHO, CSA, and GWO exhibit competitive results compared to the proposed algorithm. The SCA received the lowest scores in the majority of case studies. According to a few studies, the fundamental cause for this could be the limited number of adaptive parameters to adjust intensification and diversification successfully. According to the experimental findings, the proposed NRBO outperformed the other selected algorithms in real-world CEC2020-constrained optimization tasks. The proposed NRBO also has increased searchability to obtain a good balance between exploration and exploitation using TAO and NRSR.

5. Application of Internet of Vehicles (IoV) routing

Specifically, an IoV routing optimization model was designed and executed in order to assess the effectiveness of the proposed NRBO. Fig. 13 depicts an Internet of Vehicles environment with two highlighted cars, including one that serves as the source of Internet traffic and the other that serves as the target to which the data must be transferred (Dehkordi et al., 2021). If we look at this from the perspective of the origin vehicle, there are numerous alternatives for forming an optimal path to the target vehicle that may include numerous crossings. The quality-of-service values vary depending on traffic, and these values are closely connected with various performance parameters. Fig. 13 depicts the purpose of this challenge, which is to exploit the likelihood of connectivity and connect the quality of service of the possible channels from the starting point to the endpoint as much as possible. The probability of route link (P_c) and packet reception (P_r) of every channel are taken into account using the maximization approach, which is subject to the constraint of route delay (D). The greater the values of P_c and P_r, the greater the maximum fitness value. In a realistic situation, each communication vehicle has a varying chance of connecting to the network and receiving packets. As a result of taking into account the various delays provided by every accessible route from the sender to the receiver, choosing the proper mixture of such variables among all accessible delays given by cars in a road section as the best path with the highest possible fitness function value is a non-polynomial problem.

To handle the optimization challenge of identifying the most efficient and associated path in software-defined network-based vehicular communication, the proposed NRBO has been combined with other benchmark optimization methods to allow IoV applications. The urban road links in the vehicular situation are represented as a graphical model $G(i, s)$, in which i indicates the intersection (IS) and s denotes the road section between two ISs. Hence, every best route x comprises a set of ISs and a set of road sections. As per the assumptions made above, the fitness function of the IoV routing problem can be stated in Eq. (51).

$$\text{Maximize} : f(x) = \delta_1 \times P_c(x) + \delta_2 \times P_r(x) + \delta_3 \times D(x) \times \frac{1}{1+T} \tag{51}$$

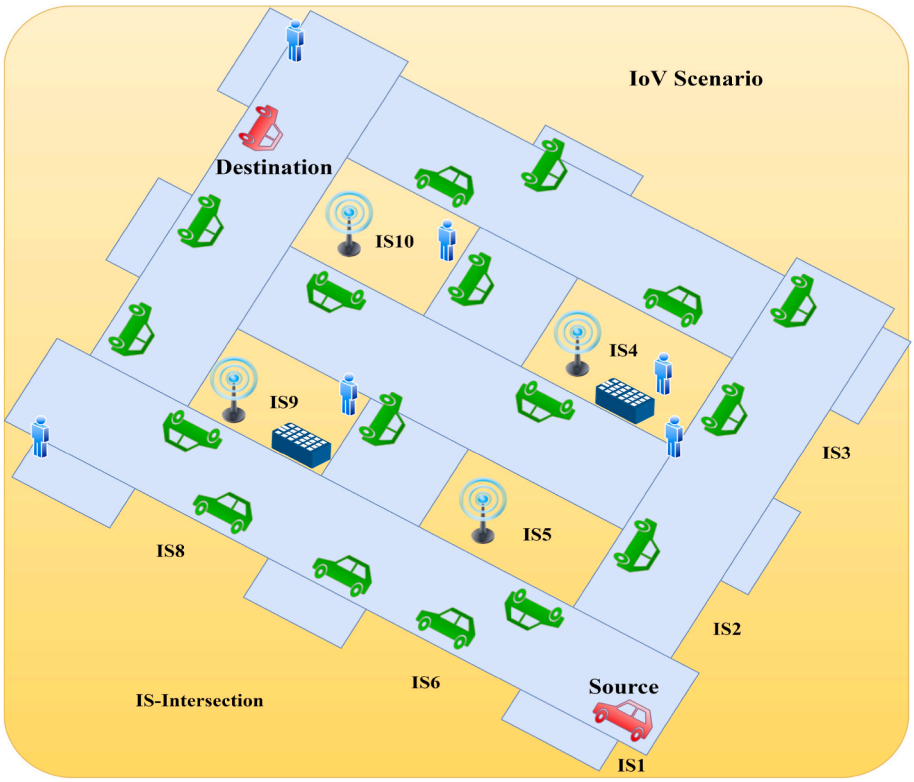

Fig. 13. IoV routing scenario.

$$P_c(x) = \prod_{j=1}^{N} P_c(e_j) \tag{52}$$

$$P_r(x) = \prod_{j=1}^{N} P_r(e_j) \tag{53}$$

$$D(x) = \frac{\sum_{j=1}^{N} D_t(e_j) - \sum_{j=1}^{N} D(e_j)}{\sum_{j=1}^{N} D_t(e_j)} \tag{54}$$

where $D_t(e_j)$ denotes the threshold value for the delay, and T denotes the time variation of the graphical model of the network map ($G(i,s)$).

$$D_t(e_j) = \frac{\sum_{j=1}^{N} D(e_j)}{2} \tag{55}$$

Subjected to: $D(e_j) \leq D_t(e_j)$.

$$T = \frac{\sum_{j=1}^{N} D(e_j)}{\text{size } (G(i,s))} \tag{56}$$

The IoV problem was handled by using a collection of sophisticated algorithmic approaches. It was decided to use the NRBO and other selected algorithms to discover the most efficient path from the source to the target vehicles. Importantly, for each IoV scenario formed, initial random solutions were generated for each target variable. Such values would be recorded for every route solution, and each algorithm would look for the best route to provide the highest possible fitness value. Eq. (51) shows how to compute and aggregate the probability of connectivity for each existing route of a highway section to the endpoint vehicle, as well as the probability of packet reception and route delay. Consequently, to arrive at a more accurate judgment, the delayed metric was collected and contrasted with each specified road segment threshold, where the threshold is the mean delay encountered by a specific route set. In addition, a time variance was also incorporated to strengthen the consistency of the selected best path.

To tackle the specified IoV optimization problem, a comprehensive experiment has been conducted using the proposed NRBO and other selected algorithms. Fig. 14 illustrates the convergence rates of the best value obtained throughout 100 iterations for different graphical model sizes (N), such as 100 × 100, 200 × 200, 500 × 500, and 1000 × 1000 and different dimensions (2 or 4). The attained results resulted from a mean of 20 runs for each optimization procedure. Each algorithm's capacity is tested to optimize various dimensions ($dim = 2$ or 4) of the IoV routing problem. Fig. 14(a, c, e, g) shows the convergence curves for problem dimension of 2 (P_c and P_r) with different model sizes, such as 100 × 100, 200 × 200, 500 × 500, and 1000 × 1000. Fig. 14(b, d, f, h) shows the convergence curves for the problem dimension of 4 (P_c, P_r, D and T) with different model sizes, such as 100 × 100, 200 × 200, 500 × 500, and 1000 × 1000. Table 32 lists all the statistical measures, such as best, Mean, and STD values of all selected algorithms.

Over an average of 20 runs, the authors discovered that the NRBO could achieve good convergence when contrasted to the other algorithms; however, when the graph dimensions were set to 100 and 200, the STD value of all algorithms was slightly higher 2-dimension problem was considered. However, the proposed NRBO achieved a better STD when the graph dimensions were set to 100, 200, 500, and 1000. It also underlines the fact that using P_c and P_r variables, it was impossible to accurately represent the fitness value to obtain its best maximum value. As a result, the problem was also experimented with four dimensions in the second trial. These better overall STD metrics of the effectiveness of all algorithms have been obtained; however, whenever the problem dimension was increased to four, the NRBO outperformed all other algorithms in terms of performance with a low STD value. When NRBO's performance is compared with other algorithms with high dimensions, it is concluded to outperform them. This conclusion is supported by the STD values shown in Table 32. Perhaps this is due to the TAO and NRSR

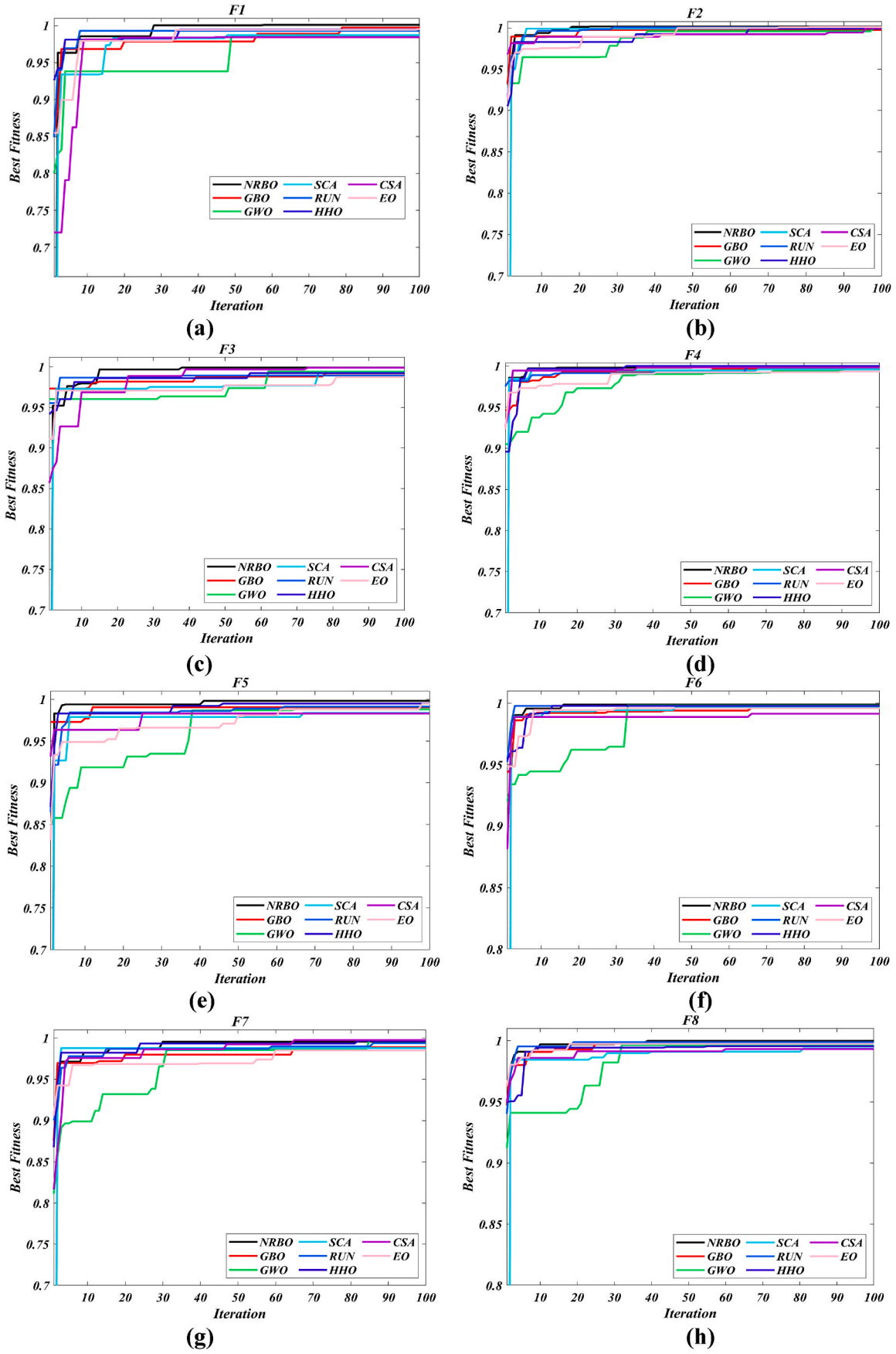

Fig. 14. Convergence curves of all testing scenarios.

Table 32
Obtained results for different dimensions and sizes of the graphical model (N).

Algorithm	Best	Mean	STD
$dim = 2, N = 100 \times 100$			
NRBO	**1**	0.9998	0.0014
GBO	0.9975	0.9884	0.0057
GWO	0.9917	0.9909	0.0048
SCA	0.9890	0.9864	0.0030
RKO	**1**	0.9945	0.0045
HHO	0.9962	0.9912	0.0074
CSA	0.9946	0.9899	0.0053
EO	0.9957	0.9924	0.0042
$dim = 2, N = 200 \times 200$			
NRBO	**0.9947**	0.9985	0.0020
GBO	0.9865	0.9931	0.0038
GWO	0.9799	0.9899	0.0058
SCA	0.9865	0.9917	0.0036
RKO	0.9892	0.9951	0.0033
HHO	0.9779	0.9917	0.0065
CSA	0.9846	0.9918	0.0046
EO	0.9870	0.9932	0.0046
$dim = 2, N = 500 \times 500$			
NRBO	**0.9943**	0.9971	0.0015
GBO	0.9863	0.9918	0.0042
GWO	0.9770	0.9867	0.0052
SCA	0.9789	0.9881	0.0062
RKO	0.9913	0.9946	0.0029
HHO	0.9819	0.9905	0.0041
CSA	0.9831	0.9923	0.0041
EO	0.9880	0.9929	0.0035
$dim = 2, N = 1000 \times 1000$			
NRBO	**0.9951**	0.9963	9.48e-04
GBO	0.9858	0.9917	0.0043
GWO	0.9794	0.9900	0.0068
SCA	0.9808	0.9887	0.0048
RKO	0.9851	0.9918	0.0031
HHO	0.9854	0.9932	0.0040
CSA	0.9851	0.9916	0.0044
EO	0.9849	0.9918	0.0047
$dim = 4, N = 100 \times 100$			
NRBO	**1**	1	5.52e-04
GBO	0.9997	0.9968	0.0040
GWO	0.9999	0.9960	0.0044
SCA	0.9959	0.9970	0.0041
RKO	0.9992	0.9945	0.0016
HHO	**1**	1	0.0012
CSA	**1**	0.9990	0.0028
EO	0.9991	0.9984	0.0020
$dim = 4, N = 200 \times 200$			
NRBO	**0.9985**	1.0001	7.99e-04
GBO	0.9945	0.9984	0.0019
GWO	0.9924	0.9968	0.0026
SCA	0.9935	0.9961	0.0021
RKO	0.9976	0.9991	0.0010
HHO	0.9962	0.9990	0.0014
CSA	0.9952	0.9976	0.0014
EO	0.9933	0.9969	0.0022
$dim = 4, N = 500 \times 500$			
NRBO	**0.9981**	0.9991	6.15e-04
GBO	0.9940	0.9969	0.0014
GWO	0.9901	0.9951	0.0031
SCA	0.9898	0.9951	0.0034
RKO	0.9951	0.9975	0.0015
HHO	0.9919	0.9959	0.0022
CSA	0.9915	0.9969	0.0025
EO	0.9938	0.9956	0.0017
$dim = 4, N = 1000 \times 1000$			
NRBO	**0.9981**	0.9992	6.86e-04
GBO	0.9917	0.9963	0.0024
GWO	0.9927	0.9954	0.0022
SCA	0.9903	0.9965	0.0027
RKO	0.9935	0.9970	0.0015
HHO	0.9845	0.9946	0.0044
CSA	0.9922	0.9964	0.0022
EO	0.9911	0.9948	0.0028

operator that the algorithm has acquired. According to the Mean values, it was obvious that the NRBO could outperform the other algorithms in terms of convergence potential, as evidenced by its capacity to acquire the best result among the other algorithms. In contrast, when it comes to reliability analysis, the NRBO has succeeded in achieving better STD values, and the NRBO is capable of dealing with optimization problems of relatively large dimensions and graphical model size, as in the IoV routing problem, is reflected in this result.

6. Further discussions

6.1. Performance of NRBO in deep reinforced learning (DRL) agent training

To further analyze the performance of NRBO, DRL agent training is studied. For the performance of NRBO for training DRL agents, the continuous mountain car environment from the classic control problem in the OpenAI Gym is considered (Brockman et al., 2016). The problem involves a car along a one-dimensional track stationed between two peaks. The objective is to drive the car to the top of the mountain on the right, but its engine is not powerful enough to do it in a single pass. Therefore, the car needs to build momentum by driving back and forth. The reward is a function of the energy expended to achieve the objective. Specifically, the reward is inversely related to the energy expended. Furthermore, this reward formulation is the most commonly used formulation for the mountain car problem. Based on the continuous mountain car environment, a six black-box optimization problems testbed is generated with dimensions ranging from 34 to 8834 using six proven continuous DRL agents. Table 33 shows the dimensionality of the resulting problems for each DRL agent. Therefore, the effectiveness and robustness of NRBO can be sufficiently evaluated by using the base DRL Specifically, PPO (Schulman et al., 2017), Trust Region Policy Optimization (TRPO) (Schulman et al., 2015), Advantage Actor-Critic (A2C) (Mnih et al., 2016), DDPG (Lillicrap et al., 2015b), Actor-Critic using Kronecker-factored Trust Region (ACKTR) (Wu et al., 2017) and Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor (SAC) (Haarnoja et al., 2018) over 25 independent runs (see Table 34).

As shown in Table 34, for the high-dimension problems (i.e., DDPG and SAC), NRBO shows an average of 10–100x better sample complexity than the base DRL agents, while for low dimension problems (i.e., A2C, PPO2, ACKTR, and TRPO), NRBO achieved up to 5–10000x better sample complexity than their associated DRL agents. From the results of the final scores shown in Tables 34 and it is evident that for the low-dimension problems (i.e., A2C, PPO2, ACKTR, and TRPO), NRBO shows a significantly better final score compared to those of their associated DRL agents. On the other hand, for the high-dimension problems (i.e., DDPG and SAC), although the final score of NRBO is still better, those of their corresponding DRL agents are comparable.

6.2. Summary

According to findings and FRT analysis, the suggested NRBO generated the best "Mean" and "Best" scores for 42 out of 64 functions, or 65.6% success on unconstraint benchmark tests. Conversely, NRBO generated the best "Mean" and "Best" scores for the 12 RW challenges out of 12, which denotes a 100% success rate on real-world design challenges. The results acquired for the 11 functions are satisfactory; however, NRBO could not give the best "Best" values for 22 functions. The suggested method performs on par with CEC2022 winners, and

Table 33
DRL agents and the respective problem dimension.

RL agents	A2C	PPO2	ACKTR	TRPO	SAC	DDPG
Problem dimension	34	67	68	134	4482	8834

Table 34
A ratio of DRL timesteps to NRBO timesteps is needed to reach 100 percent of each DRL agent's learning progress.

Algorithms	% DRL final score	DRL score	**NRBO score**	DRL timesteps	NRBO timesteps	DRL Timesteps/NRBO timesteps
A2C	100%	92.28	97.55	7.89E+06	6.00E+02	13150.00
ACKTR	100%	92.58	97.57	3.10E+05	6.00E+02	516.66
TRPO	100%	92.36	97.53	1.22E+05	6.00E+02	203.33
PPO2	100%	92.50	97.51	8.99E+05	6.00E+02	1498.33
SAC	100%	92.42	92.57	6.30E+04	5.10E+03	12.35
DDPG	100%	90.93	92.38	3.40E+05	7.30E+03	46.575

according to the observation of the STD values, NRBO has the best stability out of the other seven algorithms. On most test functions, NRBO can generate results with the lowest STD values of any method. We can conclude that the NRBO search method is very stable and can generate relatively consistent results.

Fig. 15 visually compares the total "average FRT value" for each algorithm. Fig. 15 also simply sums the "average FRT value" of every algorithm over 78 functions. It is also clearly and precisely seen that NRBO (8.82) surpasses all of the other algorithms, such as GBO (11.564), GWO (20.212), SCA (28.898), RKO (13.599), HHO (21.267), CSA (20.712), and EO (18.932) in case of the FRT value over all the unconstraint and real-world benchmark set. So, we can conclude that NRBO can produce satisfactory results due to overall less FRT value. Additionally, Fig. 16 which is formed by the percentage success rate of all 78 functions under both the "Best" and "Mean" values for each algorithm. The superiority of NRBO (73.80%) over the other algorithms, such as GBO (9.52%), GWO(2.38%), SCA(2.38%), RKO(7.14%), HHO (3.57%), CSA(1.19%), & EO(2.38%) can be precisely and clearly seen in Fig. 16.

6.3. Benefits and limitations of the proposed algorithm

In the proposed NRBO algorithm, initial points (i.e., initial population) are randomly generated. Each population has a search direction determined by the information acquired from previous results. The optimization process is continued by updating the search directions until the convergence criterion is met. The NRBO provides great strength to find the global optima. Thus, one of the unique features of this study combines the concept of the gradient-based methods with the population-based methods for creating a powerful and efficient NRBO algorithm to overcome traditional algorithms' drawbacks and require

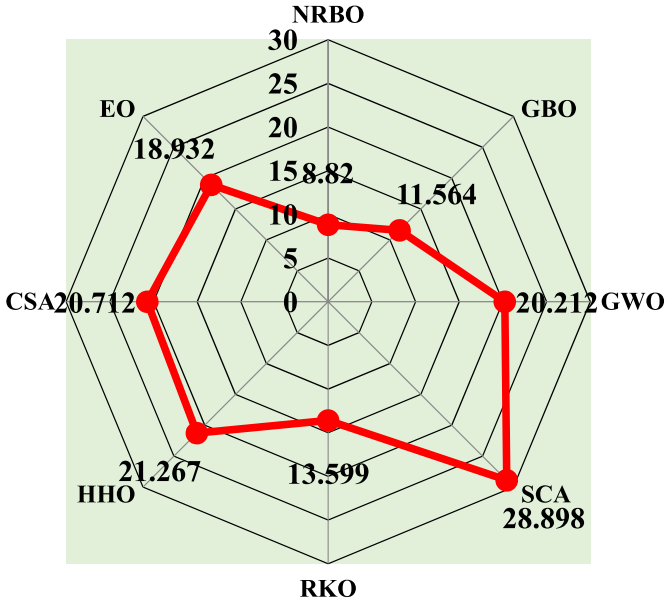

Fig. 15. Average FRT values over selected 78 functions.

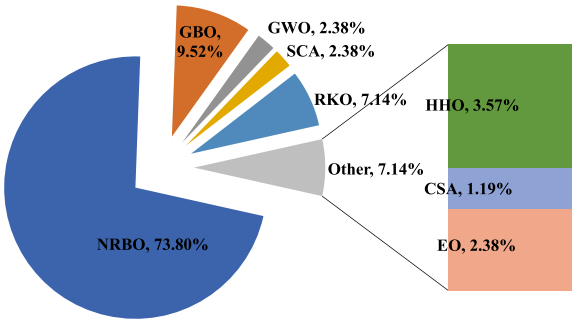

Fig. 16. Success rate of all algorithms over 78 functions.

very few iterations for a better solution. From earlier discussions, it can be seen that NRBO has a strong ability to manage various engineering design problems with various equality/inequality constraints and can find the best solution for such difficulties. The NRBO is a strong, useful, and appealing metaheuristic algorithm for solving global optimization problems with/without constraints, it can be said. The proposed NRBO's main benefits include simplicity in implementation, the discovery of superior solutions to the tested problems, prompt response, lower probability of slow convergence, enhanced population diversity, the balance between exploitation and exploration, smooth convergence curves, more efficient and targeted delivery of new solutions, a limited number of control parameters, etc.

There are some limitations in this study. First, the parameter setting needs more analysis to obtain a better configuration. Second, the performance of the proposed algorithm should be measured on higher dimensional functions, such as 5000 and 10000. Third, the first zone, which represents the reference area, can be replaced with one of the remaining zones to observe the effect of the change on the solution qualities. Lastly, since the proposed algorithm can tend to be stuck in local minima for problems that include many constraints, the different improvement mechanisms, such as the opposition-based, chaos-based, Levy-flight-based, adaptive parameters, etc., can be integrated with NRBO to enhance its performance. Apart from this, the proposal of discrete or multi-objective versions of the developed algorithm could also be a significant contribution. Additionally, minor flaws can be addressed, including the miss self-learning process, the impact of fitness values on the mechanisms for updating solutions, and a novel operator between search techniques.

7. Conclusions and future scope

The NRBO is a new metaheuristic population-based method proposed in this paper. The proposed NRBO is basically developed using the Newton-Raphson method to investigate the search space's best positions and find the best solution. The NRSR and TAO were developed and integrated with the NRBO to make exploration and exploitation more convenient in the NRBO environment. When utilized as a search engine in the NRSR, Newton-Raphson's method helped to boost the exploration and exploitation phases, and the TAO helped to prevent the use of the local best solution. The effectiveness of the NRBO was assessed utilizing

a total of 64 test functions from standard benchmarks, CEC2017 and CEC2022, which included unimodal, multimodal, hybrid, and composite test functions. Using unimodal functions and convergence behaviour, the exploitation ability of the proposed NRBO was demonstrated along with the usage of NRSR and TAO. With the support of multimodal functions, the exploration ability of the proposed NRBO was demonstrated. The findings of both the hybrid and composite functions revealed that the GBO effectively balanced exploration and exploitation by utilizing the TAO and NRSR, which was another important finding. The balance analysis, scalability analysis, sensitivity analysis, and convergence analysis were made to prove the effectiveness of the algorithm. It was found that the proposed NRBO outperformed seven recently proposed metaheuristic algorithms, which included the GBO, GWO, SCA, HHO, RKO, CSA, and EO. The Friedman ranking test was used to compare the performance of the algorithms, and the results were analyzed. Consequently, the NRBO produced a highly positive performance and outscored the other algorithms in most test functions, suggesting that the NRBO may be used as an alternate optimization technique to handle various numerical problems.

In addition, to assess the suitability of the proposed algorithm in handling real-world constrained problems, the NRBO is validated using 12 real-world CEC2020 engineering-constrained optimization problems. The performance obtained by the NRBO was outstanding compared to all other selected algorithms. The results prove that the proposed NRBO can handle the problems with large dimensions and many equality and inequality constraints. The proposed NRBO was also used at the end, and its results were evaluated to identify an optimized solution for a difficult challenge in the IoV routing problem, which was solved successfully. Locating the optimal path for an IoV application using NRBO yielded good results. Further, the performance of NRBO in DRL agent training is also studied and analyzed. Compared to the current state-of-the-art methods, the proposed algorithm produced significantly better results, which is solid evidence of its superiority in this domain.

It is important to note that the proposed strategy has future scopes that should be considered. First and foremost, the NRBO necessitates the development of a specific means of coping with different constraint-handling mechanisms in problems with a high degree of constraint. Second, this approach should be modified at the algorithmic level to address mixed-integer (binary, discrete, and integer) optimization problems. The third point is to construct a multi-objective and many-objective version of NRBO to handle multiple objectives. The possible applications of NRBO are photovoltaic panel parameter estimation, charge scheduling in electric vehicles, microgrid management, unit commitment, economic load and emission dispatch, energy forecasting, demand forecasting, motor design, controller tuning, load frequency control, image segmentation, image enhancement, time-series forecasting, regression problems, feature selection, feature extraction, etc. can be tested in future. Finally, a hyper-heuristic mechanism may be necessary to determine the best regulating parameters for the suggested algorithm.

CRediT authorship contribution statement

Ravichandran Sowmya: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Validation, Writing – original draft, Writing – review & editing. **Manoharan Premkumar:** Conceptualization, Methodology, Software, Data curation, Visualization, Validation, Supervision, Project administration, Formal analysis, Writing – original draft, Writing – review & editing. **Pradeep Jangir:** Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.engappai.2023.107532.

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